

Slippery Fish:

Enforcing Regulation when Agents Learn and Adapt

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Abstract

Curbing undesired behaviors through regulation gets complicated when agents can adapt to circumvent enforcement. We test a model of enforcement with learning and adaptation, by auditing vendors selling illegal fish in Chile in a randomized controlled trial, and tracking them daily using mystery shoppers. Conducting audits on a predictable schedule and (counter-intuitively) at high frequency is less effective as agents learn to take advantage of loopholes. We observe the specific defensive actions vendors adopt to circumvent fines, and their pattern of adoption over time is consistent with the model of learning. The experiment also examines whether a demand-side consumer information campaign can either complement or substitute for monitoring and enforcement. The information campaign proves almost as cost-effective at curbing illegal sales, and the Chilean government subsequently chose to scale up the campaign.

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1 Introduction

Correcting market failures and improving economic efficiency often require curbing undesirable behaviors of market agents who act to maximize their own private benefits. Examples span actions that affect the natural environment, such as deforestation, pollution, or resource exploitation (Stavins, 2011; Duflo et al., 2013, 2018; Hansman et al., 2018); actions that affect community health such as open defecation or drunk driving (Banerjee et al., 2017); or actions that undermine government performance such as corruption or tax evasion (Carrillo et al., 2017). Informing decision-makers about the ramifications of their actions, and enforcing regulations are the most direct strategies to deter such behavior. Enforcement requires both strong state capacity and sophisticated policing, because its design needs to be robust enough to deter cheating even after agents learn how to ‘game’ the new system.

Targeted agents adapt to new rules, finding loopholes that allow them to continue maximizing private benefits at the expense of others.¹ In many instances, it is therefore insufficient to evaluate the effectiveness of enforcement activities based on their immediate, short-run effects (Fudenberg and Levine, 2020). Comprehensive analysis requires tracking the (sometimes unanticipated) strategies that targeted agents may deploy to circumvent the regulation as they adjust to the new regime.

We develop a model of enforcement paired with an experimental design and data collection strategy that delineate how agents learn about the patterns of, and loopholes in, enforcement. We study adaptation along two different margins: (i) the agent learns audit patterns and schedules over time and focuses illegal activities on days when she is not likely to be inspected, and (ii) the agent devises defensive strategies to avoid paying fines even when she is audited. Relative to existing deterrence and audit design models (e.g., Becker, 1968; Lazear, 2006; Ball and Knoepfle, 2023), our framework allows agents to learn about enforcement patterns and

¹For example, Carrillo, Pomeranz, and Singhal (2017) show that when the Ecuadorian tax authority improves the quality of their information on firm revenues, the firms react by raising their estimates of costs in line with the revised revenue estimates, to keep total tax payments unchanged. Blattman et al. (2017) shows that intensive policing pushes crime “around the corner”, with null impacts on overall violent crimes. Health officials adapt to undermine a monitoring scheme to punish delinquent nurses in Banerjee et al. (2008), making an initially-effective program completely ineffective in 18 months.

imperfections over time, which yields new predictions about how inspection frequency and predictability interact with strategic adaptation. Our Bayesian model of learning also yields predictions on which specific designs of enforcement strategy are more robust to agents’ subversive adaptation efforts. We test these predictions by conducting a large-scale randomized controlled trial (RCT) in which government monitors penalize vendors that sell illegal fish in Chile, while we surreptitiously monitor vendors’ reactions to that enforcement by deploying “mystery shoppers” in fish markets. We cross-randomize the frequency of enforcement visits and the (un)predictability of schedules to test theoretical predictions about the optimal audit policy design.

This experiment is set in Chile, where the government instituted a ban on fishing and sale of the critically endangered Pacific hake fish (*merluza*) during September each year, when the fish reproduce. Catching hake during this period is especially ecologically harmful. We randomized the fish markets where the government sent monitors to levy penalties on vendors illegally selling fish. We also randomized a consumer information campaign designed to educate consumers about the environmental risks associated with over-fishing of hake, and discourage consumption during the September ban period. Information campaigns, attempts to change social norms (Chetty et al., 2014; Guiteras et al., 2015), marketing that appeals to people’s sense of fairness (Hainmueller et al., 2015), or encouraging third-party reporting (Naritomi, 2019), may be more cost-effective in settings where it is difficult to enforce rules. Our consumer information campaign not only serves as a useful benchmark because the “private politics” of activism and boycotts have emerged as leading alternatives to public regulation (Egorov and Harstad, 2017), but our 2x2 experimental design with enforcement, information, both, and neither can test for complementarities between these demand- and supply-side strategies. We develop a theoretical model that specifies the conditions under which the information campaign would complement or substitute for the audit strategy. This provides generalizable guidance for other regulatory settings.

Since we are tracking illegal activities, we measure outcomes using “mystery shoppers” – trained surveyors who look like typical shoppers – sent to each market to pose as buyers and

(try to) purchase hake fish during the ban. We linked the daily reports from mystery shoppers to the enforcement logbook recorded by government inspectors to test our model’s specific predictions on the dynamics of learning and adaptation in response to variable patterns of enforcement visits that different vendors experienced.

Additionally, we conducted consumer surveys to record changes in demand for hake and other substitutes, and consumer knowledge about the hake ban. We mapped all spatial and market relationships between vendors and fishers to study spill-overs across markets. Finally, we surveyed the fishers who supply to these markets to explore whether interventions implemented “downstream” (at the point of sale from vendors to consumers) traveled “upstream” the supply chain of fish. Our sample covers all major markets in Chile where the majority of hake is caught, which allows us to report on equilibrium outcomes, such as changes in fishers’ activities, or availability and prices of hake substitutes. This produces a more comprehensive evaluation of the full range of effects up and down the supply chain.

We first specify a model of learning to develop a more precise understanding of how regulated agents learn about the audit system, adapt, and develop defensive strategies. The model also specifies the conditions for substitutability or complementarity between the enforcement and information campaigns. We then conduct a program evaluation of the government’s audit and information campaigns to test the model’s predictions. We observe the variation in vendors’ behaviors week to week, and mystery shoppers recorded the precise defensive strategies vendors introduced to circumvent enforcement. For example, many did not display the hake openly during the ban, but were willing to sell illegal fish hidden from plain view. We also introduce experimental variations in the design of the audit system to test which strategies are more robust to such subversive adaptation. Not surprisingly, we see that fish vendors find it more difficult to adapt when monitoring visits are unpredictable. Audits on a predictable schedule become less and less effective over time, as vendors learn monitoring schedules and shift sales away from targeted days and markets. We also tried increasing monitoring frequency to better contain hake sales, but this strategy backfired. Somewhat surprisingly, auditing was found to be more effective when conducted at *lower* frequency. Our model explains that increased

frequency allows fish vendors to learn about the flaws in the system more quickly and adopt more successful defensive strategies. This generates a new insight: even when enforcement is very cheap to conduct, the auditor can do better by holding back some effort.

Our findings shed light on a larger theoretical literature in Law and Economics on adaptation and subversive reactions to regulations (Glaeser and Shleifer, 2003; Becker, 1968; Eeckhout et al., 2010; Lazear, 2006). Also related is the literature on *gaming* incentive schemes where agents adapt to undermine the intent of the regulator (Ederer et al., 2018; Oyer, 1998; Gravelle et al., 2010). That literature suggests that introducing unpredictability and opacity to incentives can mitigate gaming by the agent and improve payoffs for the regulator. Both our model and Okat (2016) predicts that random and less frequent enforcement hinders or delays agents’ learning about the weaknesses of the auditing process.

Our results call into question any enforcement mechanism that economic theory deems “most efficient” without grappling with the (potentially unanticipated) behavioral responses by regulated agents. We conduct this evaluation under the real-world complexities of implementing a large government enforcement program at scale, and contribute to the empirical literature on the effects of monitoring and penalties (Boning et al., 2018; Shimshack and Ward, 2005; Gray and Shimshack, 2011; Hansen, 2015; Pomeranz, 2015; Johnson et al., 2019). Banerjee et al. (2017)’s policing intervention to curb drunk driving in India, where they randomized fixed vs rotational checkpoints and monitoring frequency is closely related. Beyond regulation and enforcement, our results also show that an easier-to-implement consumer information campaign is almost as effective in curbing the illegal activity as direct monitoring.² Our model explains that the information campaign acts as a substitute for enforcement because many fish vendors were already close to the margin for compliance, as evidenced by the behavior of vendors in our control group. The model elaborates the conditions under which demand- and supply-side strategies to curb undesirable behaviors could complement each other in other regulatory settings.

²Like our consumer information campaign, many other papers have evaluated indirect strategies in pursuit of social goals, in environments where enforcement is expensive or difficult (Johnson, 2016; Jin and Leslie, 2003; Reinikka and Svensson, 2005; Alm et al., 2009; Shimeles et al., 2017; Kollmuss and Agyeman, 2002).

The context of our experiment is policy-relevant. [FAO \(2014\)](#) estimates that 31.4% of the world’s fish stocks were over-exploited to biologically unsustainable levels in 2013, up from 10% in 1974. [Costello et al. \(2012\)](#) reports that over-exploitation is worse in small-scale fisheries, like the one we study, and such fisheries represent the majority of the global catch. Illegal fishing accounts for US\$10-23 billion worth of fish each year. Fishing bans of the type we study in Chile are in effect in many countries around the world, including China, Fiji, India, Ghana, Bangladesh, Peru and Myanmar. Some of these other policies are extremely similar in structure to the Chile hake ban, such as a 22-day ban on selling Hilsa fish in Bangladesh during the fish’s reproduction period, and a 60-day ban on silverfish in Peru.³ Our results add deeper understanding and direction to a policy issue which is pertinent globally.

This paper is organized as follows: Section 2 describes the context, section 3 develops the theory of learning and presents a formal model to guide our empirical strategies, section 4 describes experimental design, section 5 the data collection, section 6 presents the empirical strategy and results, and section 7 tests for learning and adaptation. Sections 8 and 9 document spillovers and market equilibrium effects and the cost-effectiveness of the program. Lastly, section 10 concludes the paper.

2 Background

2.1 Context

With around 4,000 miles of coastline, Chile is one of the top ten fish producers in the world ([FAO, 2014](#)). However, as in many other low and middle-income countries, the marine ecosystems have been threatened by over-fishing. The Pacific Hake is the fish low and middle-income Chileans consume most, and also one of the most important sources of protein for this population. The domestic hake market is served entirely by the domestic supply. Imports and exports of hake are quite uncommon. In an effort to protect the hake population, the Chilean

³See <http://www.newagebd.net/article/52220/22-day-ban-on-hilsa-fishing-from-oct-7> and <https://elcomercio.pe/economia/peru/produce-establece-veda-nacional-pejerrey-60-dias-noticia-543012>

National Marine Authority (*Sernapesca*) and the central government have enacted various policies including restrictive fishing quotas and a one-month ban on fishing and selling hake during the fish’s September reproduction cycle. Due to difficulties in enforcing the ban, the hake population is now critically threatened, with 72% of species rated as overexploited or collapsed ([Subpesca, 2015](#)).

2.2 Supply Chain of Illegal Fish

2.2.1 *Caletas*: Coastal Villages where Artisanal Fishers Bring in their Catch

The majority of people engaged in fishing are small-scale and artisanal fishers. Small-scale fishers contribute almost 40% of the national fishing volume, and up to 75% of the hake fish market. Artisanal fishers operate out of hundreds of fishing villages called *Caletas* dotting the coastline.

Each *caleta* contains between 10 and 100 fishing boats. Around 76% of the caletas are located in rural areas along the extended Pacific coast, and are highly spatially dispersed ([Subpesca, 2013](#)). Their geographic dispersion, informality, and the small-scale of operations of each individual fisher make it difficult for the government to monitor their activities. The absence of alternative income-generating activities for the fishers has also made it difficult to change norms regarding “environmentally conscious behavior” in this industry.

Furthermore, the small-scale fishers are unionized, and have organized political opposition to government policies that restrict fishing. They often capture illegal, undeclared fish beyond their allocated quota. [WWF \(2017\)](#) estimates that the amount of hake fished by small-scale artisanal fishers are between 3.8 and 4.5 times the legal quota. As a result, the artisanal sector is responsible for 75% of the hake fish supplied in the market, even though they hold only 40% of the “official” hake quotas.

These fishers go fishing using small boats and nets at night and sell fish after sunrise. They are able to target specific fish types by varying the location and depth at which the nets are dropped. The fish is sold directly at the docks to three types of buyers: (1) fish vendors who

buy the fish to sell them in local markets, (2) intermediaries who supply fish to vendors located in places further from the coast, and (3) households who live close to the *caleta* and buy the fish for their own consumption. There is very little use of ice and refrigeration at this point in the supply chain. The fish that vendors sell in local markets is typically fresh, and captured the night before. Table C.3 in the Appendix describes *caleta* characteristics.

2.2.2 *Ferias*: Outdoor Markets where Hake is Sold

The majority of hake-fish sales to final consumers occur in *ferias*, which are outdoor urban markets organized by municipalities. Each vendor pays a fee every six months to rent a selling spot in the market. *Ferias* are typically navigable only by foot, and each feria serves a limited geographic area of surrounding neighborhoods. To cover more neighborhoods, the vendors rotate between different *ferias* in a pre-set pattern - typically setting up in the same location twice a week. For example, they may sell at a first feria every Sunday and Wednesday and at a second feria every Tuesday and Friday. The group of vendors who move together across neighborhoods is called a *circuit*. A semi-annual fee paid by the vendor to the municipality covers her inclusion in the entire circuit, so the same group of vendors typically rotate across neighborhoods all together. Vendors are not allowed to sell in public places other than *ferias*.

Each municipality typically organizes one circuit of vendors. Large municipalities may have more than one circuit. In such cases, the municipality area is divided in such a way that there is no geographic overlap between circuits. Figures A.2 and A.3 in the Appendix provide visual examples of *ferias* and circuits. Table C.1 describes observable characteristics of fish stalls in *ferias*. Since it is difficult for *Sernapesca* to directly regulate fishers, it chose to enforce the September ban by targeting vendors in *ferias* to prevent them from selling illegal hake fish. Our interventions and our model therefore focus on vendors' decisions to sell hake in September.

3 A Model of Enforcement, Information, and Vendor Adaptation

This section presents a model of vendors’ decisions to sell illegal hake during the September ban. The model is designed to (i) clarify the distinct channels through which the two policy instruments in our experiment—enforcement and a consumer information campaign—affect the incentives to sell, (ii) characterize how these instruments interact with each other in reducing illegal sales, and (iii) explain how vendor learning and defensive adaptation can shape short-run treatment effects and generate predictions on the relative performance of alternative designs that vary enforcement frequency or predictability.

3.1 A General Framework

Time is discrete, $t \in \mathbb{N}$. In each period, a vendor chooses whether to sell hake. Selling yields a private benefit v_t , while selling also exposes the vendor to the risk of detection and a monetary fine $\Omega > 0$. Let $\hat{\theta}_t$ denote the vendor’s *perceived* probability of being detected (or inspected and fined) in period t . The flow utility from selling is

$$u_t = v_t - \Omega \hat{\theta}_t, \tag{1}$$

and the utility from not selling is normalized to zero. Vendors sell if $u_t \geq 0$. To capture underlying heterogeneity and smooth choice probabilities, we write $v_t = \bar{v}_t + \varepsilon_t$, where ε_t is an i.i.d. utility shock with cdf $F(\cdot)$ and pdf $f(\cdot)$. Then the probability of selling is

$$\mathbf{P}_t = \Pr(u_t \geq 0) = F_\varepsilon(\bar{v}_t - \Omega \hat{\theta}_t) \equiv F_\varepsilon(\bar{u}_t), \tag{2}$$

where $\bar{u}_t \equiv \bar{v}_t - \Omega \hat{\theta}_t$ is the mean utility of selling across vendors.

3.1.1 Policy Instruments: Enforcement and Information

The government has two instruments: Enforcement (E) increases the vendor's *perceived* probability of detection. We capture this through a reduced-form expression:

$$\hat{\theta}_t = \hat{\theta}_t(E), \quad \frac{\partial \hat{\theta}_t}{\partial E} \geq 0. \quad (3)$$

The information campaign (I) reduces consumer willingness to pay for hake, thereby lowering the vendor's net rents. We capture this through

$$\bar{v}_t = \bar{v}_t(I), \quad \frac{\partial \bar{v}_t}{\partial I} \leq 0. \quad (4)$$

Equations (2)–(4) imply that, holding everything else constant, either intervention would reduce the probability of selling hake.⁴

Complementarity between Enforcement and Information. A relevant policy question is whether implementing enforcement and the information campaign jointly reduces illegal hake sales by more or less than the sum of their individual effects. In Appendix D.1, we formalize this notion of complementarity and show that it is equivalent to the cross-partial derivative of the probability of selling with respect to the two policy instruments. Using the mean-utility representation of the model, this cross-partial can be written as

$$\frac{\partial^2 \mathbf{P}_t}{\partial I \partial E} = \frac{\partial^2 \mathbf{P}_t}{\partial \bar{v}_t \partial \hat{\theta}_t} \cdot \frac{\partial \hat{\theta}_t}{\partial E} \cdot \frac{\partial \bar{v}_t}{\partial I} = -\Omega f'(\bar{u}_t) \frac{\partial \hat{\theta}_t}{\partial E} \frac{\partial \bar{v}_t}{\partial I}. \quad (5)$$

Since $\frac{\partial \hat{\theta}_t}{\partial E} \geq 0$ and $\frac{\partial \bar{v}_t}{\partial I} \leq 0$, the sign of (5) is governed by $f'(\bar{u}_t)$, i.e., by where the vendor sits in the distribution of utilities *absent any intervention*, which we will refer to as “baseline

⁴The changes in probabilities can be expressed as follows:

$$\begin{aligned} \frac{\partial \mathbf{P}_t}{\partial E} &= \frac{\partial \mathbf{P}_t}{\partial \hat{\theta}_t} \frac{\partial \hat{\theta}_t}{\partial E} = -\Omega f(\bar{u}_t) \frac{\partial \hat{\theta}_t}{\partial E} \leq 0, \\ \frac{\partial \mathbf{P}_t}{\partial I} &= \frac{\partial \mathbf{P}_t}{\partial \bar{v}_t} \frac{\partial \bar{v}_t}{\partial I} = f(\bar{u}_t) \frac{\partial \bar{v}_t}{\partial I} \leq 0. \end{aligned}$$

utility”. The first implication is that complementarity can be either positive or negative. If the mean utility absent any intervention is high (i.e., selling is common in the control group), then additional reductions in $\bar{u}_t$ from introducing an intervention can have large marginal effects, and the joint policy can reduce sales by more than additivity. In other words, enforcement and the information campaign can act as “complements” in reducing illegal sales. Conversely, if baseline mean utility is low (selling is already rare absent any intervention), then there is limited scope to reduce selling further by layering a second intervention on top of the first. In such cases, the joint effects of two interventions will be attenuated relative to additivity. This could produce a small or null interaction effect between the two treatments even if individually each treatment is effective. This can arise merely because vendors are near a region of the utility distribution where $f'(\bar{u}_t)$ implies attenuation, not because either policy is ineffective in isolation.

We have only described a static model thus far, abstracting from any possibility that vendors learn about enforcement and adapt to reduce the probability of getting fined. In the next sub-section, we will allow vendors to learn and adopt defensive actions, which will be central for interpreting short-run dynamics and the effects of variation in the design of enforcement strategies.

3.2 Learning and Defensive Adaptation

Inspections and beliefs –Bayesian updating. Let $y_t \in \{0, 1\}$ indicate whether the vendor experiences an enforcement visit in period t . Visits arrive according to a Bernoulli process with stationary probability θ (which depends on the enforcement regime). Vendors do not know θ and hold a prior $\hat{\theta}_1 \sim \text{Beta}(\alpha_0, \alpha_1)$. Let $Y_t = \sum_{s=1}^{t-1} y_s$ be the number of past visits. Bayesian updating implies

$$\hat{\theta}_t \sim \text{Beta}(\alpha_0 + Y_t, \alpha_1 + t - 1 - Y_t), \quad \mathbb{E}[\hat{\theta}_t] = \frac{\alpha_0 + Y_t}{\alpha_0 + \alpha_1 + t - 1}. \quad (6)$$

Thus, the perceived probability of detection increases in the number and relative frequency of past inspections, with the magnitude of belief updating attenuated by the strength of the vendor’s prior. Appendix E.1 provides further details on how vendors update beliefs and learn in response to the enforcement visits we carry out. Sernapesca’s field inspectors were not empowered to deviate from the standardized protocols they were trained on during the month-long experiment, so we do not consider enforcement strategy adaptations in the model.

3.2.1 Defensive actions and enforcement loopholes

Conditional on selling, the vendor can sell *openly* or deploy a costly defensive action $d_t \in \{0, 1\}$ (e.g., sell “hidden” hake or “frozen” hake), with cost $c \geq 0$. If inspected while selling openly, the vendor is fined with probability 1. If she employs a defensive action, she avoids paying the fine with probability $g(Y_t)$, where $g : \mathbb{N}_0 \rightarrow (0, 1)$ is strictly increasing, and $\lim_{Y \rightarrow \infty} g(Y) = \bar{g} < 1$. g is a function of Y_t because repeated exposure to audits helps vendors learn loopholes and improve evasion, but defensive strategies are never perfectly safe.⁵

Vendor problem and threshold characterization. Let $s_t \in \{0, 1\}$ denote selling. Conditional on Y_t , expected utilities are:

$$\begin{aligned} U[d_t = 0 \mid s_t = 1, Y_t] &= \bar{v}_t - \Omega \mathbb{E}[\hat{\theta}_t], \\ U[d_t = 1 \mid s_t = 1, Y_t] &= \bar{v}_t - \Omega \mathbb{E}[\hat{\theta}_t] (1 - g(Y_t)) - c. \end{aligned}$$

⁵A simple microfoundation for $g(\cdot)$ is “learning about loopholes” as an accumulating stock of enforcement know-how. Suppose there is a large set of potential tactics that reduce the likelihood of being fined when inspected (e.g., learning where inspectors typically look, which containers are rarely checked, or how to time sales or hide inventory). Each inspection provides an opportunity to discover and refine such tactics. Let $K_Y \in [0, 1]$ denote the share of tactics the vendor has learned after Y inspections. If each inspection reveals a new tactic with probability proportional to the remaining unknown share, then $\mathbb{E}[K_Y] = 1 - (1 - \lambda)^Y$ for some $\lambda \in (0, 1)$, which is increasing and concave in Y (diminishing returns to additional inspections). If defensive success scales with this knowledge stock but remains imperfect even with full knowledge, then the probability of avoiding a fine while undertaking a defensive action can be written as $g(Y) = \bar{g} \mathbb{E}[K_Y] = \bar{g}(1 - (1 - \lambda)^Y)$, which is increasing, concave, and converges to $\bar{g} < 1$ as $Y \rightarrow \infty$.

The vendor chooses d_t to maximize expected utility conditional on selling and sells if the resulting value is nonnegative. Define thresholds

$$\underline{\delta}_t \equiv \frac{c}{\Omega g(Y_t)}, \quad \bar{\delta}_t \equiv \frac{\bar{v}_t - c}{\Omega(1 - g(Y_t))}. \quad (7)$$

Proposition 1 (Selling and defense thresholds) *Fix t and Y_t . If $g(Y_t) \leq \frac{c}{\bar{v}_t}$, the vendor never uses defense. She sells openly if $\mathbb{E}[\hat{\theta}_t] \leq \frac{\bar{v}_t}{\Omega}$, and does not sell otherwise. If $g(Y_t) > \frac{c}{\bar{v}_t}$, the vendor sells openly if $\mathbb{E}[\hat{\theta}_t] \leq \underline{\delta}_t$, sells defensively if $\underline{\delta}_t < \mathbb{E}[\hat{\theta}_t] \leq \bar{\delta}_t$, and does not sell if $\mathbb{E}[\hat{\theta}_t] > \bar{\delta}_t$.*

The proof of this proposition is in the Appendix E.2. The threshold $\underline{\delta}_t$ defines the boundary between selling openly versus adopting a costly defensive strategy, while the second threshold $\bar{\delta}_t$ defines the boundary between costly defensive strategy versus not selling at all. As additional visits accumulate, vendors update beliefs about enforcement intensity, shifting both thresholds upward (deterrence), while simultaneously increasing their evasion capabilities $g(Y_t)$, which compresses the interval between them by making defensive selling more effective. The tension between these two forces is the main reason why the short-run effects of enforcement design can be non-monotone.

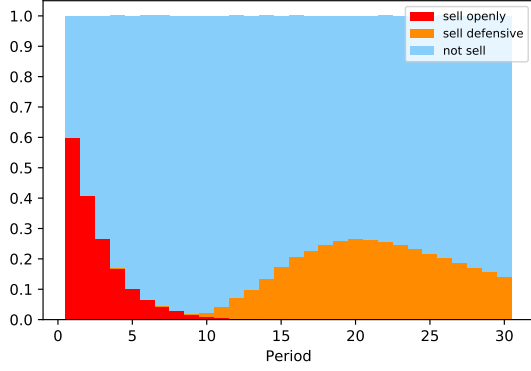
The figure 1(a) provides a numerical example of the timing and scope of adoption of defensive actions. In the early periods, the vendor lacks the knowledge to sell defensively. As the vendor gains experience (Y_t increases), adopting defensive actions becomes more likely, thereby increasing hake sales. So, there comes a point (around the 10th period in 1(a)) where the vendor has experienced enough visits that she has gotten better at circumventing the fine. Beyond that point, selling while undertaking defensive actions would dominate selling openly. The figure 1(b) shows the overall sale of hake fish.

3.3 Implications for Enforcement Design

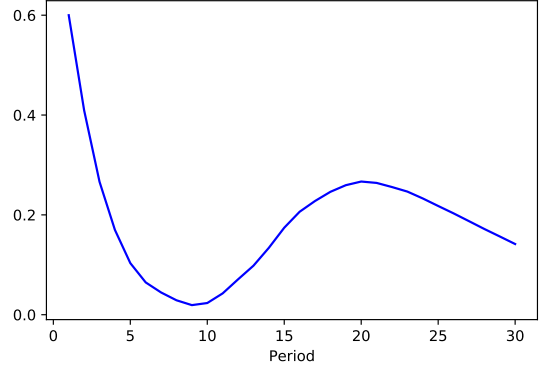
Our experiment varies enforcement along two dimensions: *frequency* (how often vendors are visited) and *predictability* (how concentrated visits are on specific days/markets). The model

Figure 1: Probability of Selling Hake

(a) Adoption of Defensive Actions



(b) Selling Decision



Notes: Figure F.2(a) and F.2(b) describe vendors' decisions on whether and how to sell. This simulation uses the same parameters than previous graph: $\theta = 0.5, v_t = (\bar{v}_t + \varepsilon_t) \sim U(0.5, 1.5), c = 0.1, \Omega = 18, \theta_1 = 0.05, g(Y_t) = 0.7 / (1 + e^{-2 \times Y_t + 12})$, i.e., $\bar{g} = 0.7$. The adoption of defensive strategies starts after a number of periods.

maps these design features into the evolution of $\mathbb{E}[\hat{\theta}_t]$ and $g(Y_t)$.

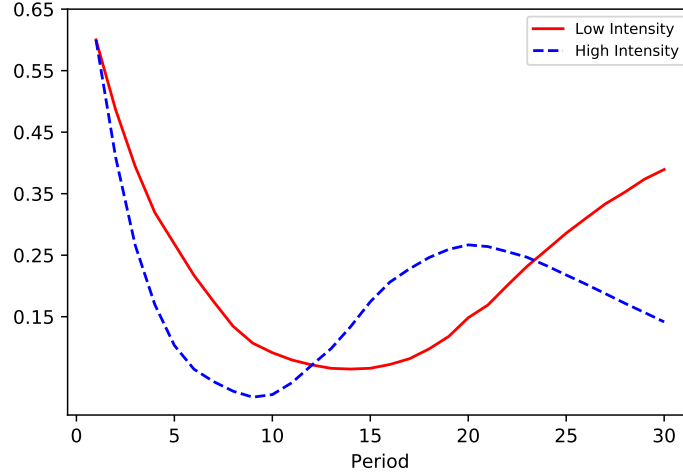
3.3.1 Frequency of visits

Let higher-frequency enforcement increase the true visit probability θ . Higher θ mechanically raises the beliefs $\mathbb{E}[\hat{\theta}_t]$, strengthening deterrence. But higher θ also accelerates the accumulation of experience Y_t , increasing $g(Y_t)$ (successful avoidance of fines) and expanding the region where defensive selling is optimal. Consequently, in the short run, increasing frequency can have *ambiguous* effects on selling: deterrence dominates early when $g(Y_t)$ is low, but defensive adaptation may dominate later once vendors have learned.

Appendix Figures F.2(a) and F.2(b) plot simulations of the vendor's adoption of defensive strategies under those high- and low-frequency enforcement scenarios. More intense enforcement initially reduces hake sales faster (as vendors update more quickly about θ), but vendors also begin adopting defensive actions earlier. This makes high-frequency enforcement relatively less effective in later periods. Figure 2 shows that high-frequency enforcement is less effective at preventing sales between periods 12 and 24, but also that the comparison reverses as we approach the long-run equilibrium. We study short-run dynamics in our experiment (within 30 days of the ban's imposition), and the simulations show that within that period, the regulator

could delay the vendor’s ability to learn and better achieve her policy goal by reducing the frequency of visits.

Figure 2: Probability of Selling Hake



Notes: This figure shows the proportion of times in which a vendor sells hake depending on the frequency of the visits. This graph depicts 1000 simulations using the following model parameters $\theta^{high} = 0.5, \theta^{low} = 0.3, v_t = (\bar{v}_t + \varepsilon_t) \sim U(0.5, 1.5), c = 0.1, \Omega = 18, \theta_1 = 0.05, g(Y_t) = 0.7 / (1 + \exp\{-2 \cdot Y_t + 12\})$, i.e., $\bar{g} = 0.7$. The probability of selling decreases quickly as the enforcement begins, but it increases as vendors learn about enforcement weaknesses. After a number of periods, it converges to the “long-run” equilibrium based on the model’s structural parameters.

3.3.2 Predictability of the schedule

Vendors operate in “circuits” in which the same set of vendors move around to different neighborhoods (setting up a market or feria in each location) on different days of the week. The “Predictable enforcement” arm we implement is equivalent to concentrating audits on a subset of ferias/days, making $\hat{\theta}_t$ high in “targeted” locations and low elsewhere. “Unpredictable enforcement” spreads audits (more) evenly across ferias/days. To formalize, consider two ferias $i \in \{1, 2\}$ visited with probabilities θ^i , with fixed enforcement capacity $\Theta = \theta^1 + \theta^2$. Vendors update beliefs separately by location using the history of visits in each feria. A predictable (targeted) policy sets $\theta^{-i} = 0$ for one feria, whereas an unpredictable policy sets $\theta^1 = \theta^2 = \Theta/2$. With a fixed enforcement capacity, the relative effectiveness of predictable (targeted) versus unpredictable enforcement depends on whether the regulator can generate sufficient overall deterrence to meaningfully affect behavior across all locations.

When enforcement capacity is low, spreading inspections evenly makes perceived detection

probabilities moderate everywhere, so deterrence is weak in all markets. In that case, concentrating enforcement in a subset of locations can be optimal: targeted enforcement sharply raises perceived detection in the targeted feria, quickly deterring sales there, even though it necessarily reduce deterrence away from non-targeted locations.

By contrast, when enforcement capacity is sufficiently high, the regulator can raise perceived detection probabilities across all locations simultaneously. In this case, unpredictable enforcement dominates targeting: spreading inspections prevents vendors from reallocating sales across locations and generates broader deterrence. Appendix E.1 formalizes how vendors form beliefs in a setting with multiple ferias/days with potentially heterogeneous enforcement probabilities.

3.4 Model Predictions

The model delivers testable predictions for “short run” behaviors that will guide our empirical analysis.

[1] **Main effects.** The effects of enforcement and information on the probability of selling are weakly negative: $\partial \mathbf{P} / \partial E \leq 0$ and $\partial \mathbf{P} / \partial I \leq 0$.

[2] **Interaction / complementarity.** Whether the enforcement and information treatments will amplify each other when applied jointly (relative to the sum of their individual effects when applied in isolation) depends on how common illegal sales are absent any intervention – i.e. the distribution of utilities from selling at baseline.

[3] **Dynamics.** Treatment effects will not be stable over time in the enforcement arm. As vendors update beliefs and accumulate experience, the propensity to sell evolves along with the frequency and number of enforcement visits.

[4] **Defensive actions.** Vendors will adopt defensive strategies (e.g., hiding/freezing) in response to enforcement, and their propensity to do so will increase with cumulative audit

experience Y_t .

[5] **Frequency.** Higher-frequency enforcement has ambiguous short-run effects: it increases deterrence through higher perceived detection risk but can accelerate defensive adaptation, potentially weakening effectiveness later in the month.

[6] **Predictability.** When enforcement capacity is low, concentrating inspections in a subset of days or markets can be relatively effective by creating strong local deterrence, albeit at the cost of increasing sales in non-targeted locations. When enforcement capacity is sufficiently high, unpredictable enforcement is more effective in the short run because it deters sales across all days and markets and limits displacement by preventing vendors from reallocating activity to safer locations.

3.4.1 Note on time horizon

The theoretical analysis above deliberately abstracts from long-run effects of enforcement, and instead focuses on short-run dynamics. We make this choice to stay consistent with the design of our experimental intervention, which were implemented only for one month during which vendors repeatedly interacted with inspectors and consumers. Over longer horizons, learning about enforcement intensity and enforcement loopholes converges, beliefs stabilize, and behavior is governed primarily by static deterrence considerations—akin to the standard [Becker \(1968\)](#) model in which agents respond to a known probability of detection and a known fine. In that long-run environment, the problem becomes comparatively simple, and learning no longer plays a central role.

Importantly, the definition of “short run” itself depends on the nature and frequency of interactions. In settings with infrequent interactions—such as annual tax filings — learning about enforcement is slow, and short-run dynamics may persist for years. In contrast, vendors face daily selling opportunities and potentially frequent inspections in our experimental setting, so learning and adaptation can occur rapidly within a month. As a result, short-run

responses are not only empirically relevant but also economically meaningful. In many policy environments including ours, short-run effects are at least as important as long-run outcomes, as they determine immediate compliance during critical enforcement windows and shape the effectiveness of temporary or episodic regulatory interventions.

4 Experimental Design

This study was implemented in close collaboration with the Chilean National Fish Service (Sernapesca). Their goal from this project was to limit hake fishing, sales and consumption during the September ban. It is difficult for them to directly regulate fishers because they operate out of geographically dispersed and politically organized *caletas*, and because their activities occur out in the water at night. *Sernapesca* therefore expressed an interest in exploring options to better regulate the fish sales at *ferias* where hake is commonly sold.

4.1 Sample

We conduct our experiment in the five central regions of Chile, home to 74% of the Chilean population. The caletas located along the coastal villages and cities scattered across these five regions account for 98% of all hake fish harvested in Chile. We conduct our experiment in all ferias in these regions except for those in the city of Santiago.⁶

An important benefit of conducting the experiment at such a large and comprehensive scale is that it allows us to trace the market-level equilibrium effects of our intervention and track any displacement of illegal hake sales towards control markets, because all potential markets (including ones where the interventions were not applied) are in our database. We collected data on the universe of circuits in our sample area, and from every fish vendor operating in those circuits. We mapped all ferias served by each caleta where the fish are caught. The unique long and thin geographic shape of Chile means that ferias are generally located very

⁶Santiago is unique in that there is one big centralized fish market called *Terminal Pesquero Metropolitano* (TPM) where vendors buy from intermediaries to re-sell at neighborhood ferias. TPM is already well-monitored by *Sernapesca*, and our interventions therefore did not need to be implemented there.

close to the caletas from where they source fish (22 miles away on average). This made it relatively easy to connect vendors to the fishers they source from, and trace how the effects of our interventions are transmitted along the supply chain for hake fish.

There are 280 ferias (fish markets) operating in the 70 municipalities in our sample, and these ferias are organized into 106 separate *circuits*. In order to identify and map all existing ferias and circuits, we combined administrative data from multiple sources (Ministry of Economics and Sernapesca) along with information gathered from phone conversations with staff in every municipality. We then used Google Maps to define the consumer “catchment area” for each feria. We identify the neighborhoods which are likely served by each feria, considering the walking distance and road accessibility from the neighborhood to the feria, as well as the residential versus commercial/industrial characteristics of the neighborhoods. The location of the ferias and their organization as circuits informed the design of our enforcement intervention, while the definitions of the residential neighborhoods and their connections to each feria were important for the design of our consumer information campaign.

4.2 Interventions

This study experimentally evaluates the effects of two interventions that aimed to reduce illegal sales of hake during the September ban period. **Enforcement** targeted the **supply** of hake by monitoring vendors (θ in our model) and enforcing penalties on those found to be selling illegal hake. An **Information Campaign** was designed to sensitize consumers about this environmental problem and discourage hake consumption during the ban, in order to lower the **demand** for hake (v in our model).

Design of Enforcement Intervention. The enforcement intervention deployed government officials from *Sernapesca* to periodically visit ferias where fresh hake is usually sold, and levy fines if vendors are caught illegally selling hake during the September 2015 ban period. An enforcement visit consisted of a pair of Sernapesca officials visiting all fish stalls in a market. The officials were instructed to follow the usual Sernapesca protocols to search for illegal fish at each

stall. We negotiated the overall study design at a higher level within Sernapesca, and these field inspectors were not made aware of the existence (or aims) of the study. They merely followed instructions from their supervisors on where and when to visit markets. Inspectors followed standardized Sernapesca inspection protocols throughout the month-long study period and did not receive additional training, tools, or performance-based incentives that would encourage adaptation in response to changes in vendor behavior (like selling “hidden” or “frozen” fish). While vendors may have adapted their selling strategies, Sernapesca’s detection strategies did not change.

To limit the chances of inspector-vendor collusion, Sernapesca rotated their inspectors across markets during the month, so that repeat interactions with vendors were minimized. Appendix Table C.17 documents that even after multiple enforcement rounds, a substantial share of visits involved inspectors visiting a market for the first time. And for the same reason the two inspectors who formed a team were rotated. As a part of this randomized controlled trial, Sernapesca agreed to conduct this monitoring at specific locations and according to schedules defined by the research team and shared the details about their monitoring activities with the research team.

Our conversations with vendors prior to September 2015 suggested that they were already aware of the hake ban. The most important change in 2015 compared to earlier years was that enforcement activities were now applied more consistently and regularly. The punishment for illegal sales was a US \$200 fine plus confiscation of the illegal fish. \$200 is equivalent to two weeks of earnings for the average feria vendor, representing a significant threat.

As our model highlights, we anticipated that fish vendors would react to the enforcement activity by devising new strategies that would help them avoid paying fines. We introduced random variations in enforcement policy design to investigate mechanisms that may be robust to agents’ efforts to circumvent policy:

1. *Predictability*: We randomly varied the ease of predictability of the enforcement. In some areas, Sernapesca monitors followed a consistent schedule (e.g. M,W at 9am) while in

other areas, they were asked to follow a less predictable schedule defined by the research team. The research team randomly varied the day in which the visit is deployed in any given week, keeping constant the total number per week. The latter is a more expensive enforcement strategy because it requires having monitors on-call for longer windows. This strategy was therefore practically more difficult for Sernapesca to implement.

2. *Frequency*: We randomly varied audit frequency at the circuit level, so that some circuits (groups of vendors) only received one visit per week, while others received multiple visits. Increasing the frequency of monitoring visits is more expensive, but it changes vendors' beliefs about the likelihood of visits, θ . It can also affect the speed of learning.

Enforcement activities were randomized at the circuit-level, covering all 106 market-circuits. This randomization was stratified to ensure balance with respect to a few important spatial and market characteristics: Whether the circuit (a) was located in a coastal municipality, (b) was the only one operating in its municipality, and (c) served geographically isolated communities.

Design of Information Campaign. The demand-side intervention was an awareness campaign designed to inform consumers about the September ban on hake sales. Sernapesca distributed letters, flyers and hung posters in the residential neighborhoods randomly assigned to this intervention. The message contained in the flyers and posters was simple: “In September, Respect the Hake Ban.” The letters, signed by the Director of Sernapesca, informed recipients about the hake ban every September, noted the decline in the hake population to a critical level as a result of over-exploitation, and encouraged consumers to not consume hake that month. Appendix A.3 shows samples of flyers and the letter. The primary goal of this intervention was to lower demand, which lowers vendors' benefits from selling hake (v in the model). In previous years, Sernapesca had used a smaller budget to place informational ads in newspapers and highway billboards. So, Information was distributed directly to consumers at a household level for the first time in 2015 as a result of our intervention.

Using our mapping exercise described in section 4.1, and combining it with the location of

major roads and crossings, we define boundaries of neighborhoods and divide the municipality up such that the population-size of neighborhoods would be roughly equal. We conducted this intervention in the 48 most populated municipalities and identified 270 distinct neighborhoods in those municipalities. Figure A.7 provides example maps. The randomization procedure was as follows:

1. Neighborhoods in 35 of the 48 most populated municipalities were assigned to the information treatment, and the remaining 13 municipalities were pure controls where no neighborhood received the letters, flyers or posters.⁷
2. Second, specific neighborhoods within each information treatment area were randomly chosen to receive the treatment.
3. Third, we randomly selected around 200 addresses in each of 102 neighborhoods, and mailed out letters to each of those 20,400 addresses. 200 letters cover roughly 15% of all potential addresses in a representative neighborhood. Based on information from the postal service, we subsequently learnt that at least 13,000 letters were correctly delivered.⁸ 80,000 flyers were distributed by trained field personnel to people walking in the streets, and directly to households within the 102 treated neighborhoods. 3,000 posters were placed around treated neighborhoods where they would be publicly visible, such as at bus stations, community centers, and street intersections.

⁷We also randomly varied the proportion of neighborhoods receiving treatment to examine if there are larger changes in norms around acceptability of socially harmful behavior when households observe many neighbors simultaneously receive the same information about the illegality of hake consumption. 18 of the information treatment municipalities were assigned to “high saturation” where two-thirds of the neighborhood in the feria’s catchment area would receive the letters, flyers, and posters. In 17 other “low saturation” municipalities, only one-third of the neighborhoods received those mailings. This saturation variation made little difference, and is not the focus of this paper.

⁸Although 13,000 were explicitly tracked, it is likely that around 16,500 were actually delivered, because the postal service did not receive any delivery failure notice in those cases. The leading cause of delivery failure was that we were forced to infer and construct addresses using Google maps, and many of those addresses did not actually exist.

4.2.1 Cross-Randomized Experimental Design

The enforcement treatment and the information campaign were cross randomized in a 2x2 experimental design so that we could study potential complementarities between the two approaches. The right column of Figure 3 describes the experimental design; the first panel lists the number of circuits assigned to each of the four treatment cells.

The majority of markets were assigned to Enforcement because that column contains additional sub-treatments with variation in enforcement policy design. Those variations in predictability and frequency of enforcement visits were cross-randomized so that we have sufficient statistical power to study the effect of each variation, one at a time. The second panel in figure 3 shows the number of circuits assigned to each sub-treatment cell. To study the effects of predictability of enforcement, we will compare the 39 circuits where Sernapesca monitored on a predictable schedule against the 44 circuits where they monitored on an unpredictable schedule. Similarly, to study the effects of audit frequency, we will compare the 34 circuits assigned to high-frequency against the 49 circuits assigned to low-frequency.^{9,10}

We control for other dimensions of random assignment whenever we focus on the effects of one particular dimension. Each of our treatments could have spillover effects on control markets, and we analyze spillovers in section 8. Finally, details on the implementation of enforcement activities and inspection outcomes are reported in Appendix Section C.1.4, which documents inspectors’ administrative records, including detection rates of illegal hake.

5 Data

Figure 3 describes the timing of the interventions and data collection activities. We conducted seven different surveys in total. In order to gather information on vendor compliance, we deployed “mystery shoppers” to surreptitiously gather information about hake availability in

⁹The probability of assignment to low-frequency enforcement and to unpredictable schedules was a little higher compared to other cells. In our analysis, we will control for these differences.

¹⁰The randomization of enforcement sub-treatments (predictability and frequency) did not explicitly condition on information campaign assignment, however, the realized allocation of info campaign within enforcement sub-treatment yielded a broadly balanced split across cells.

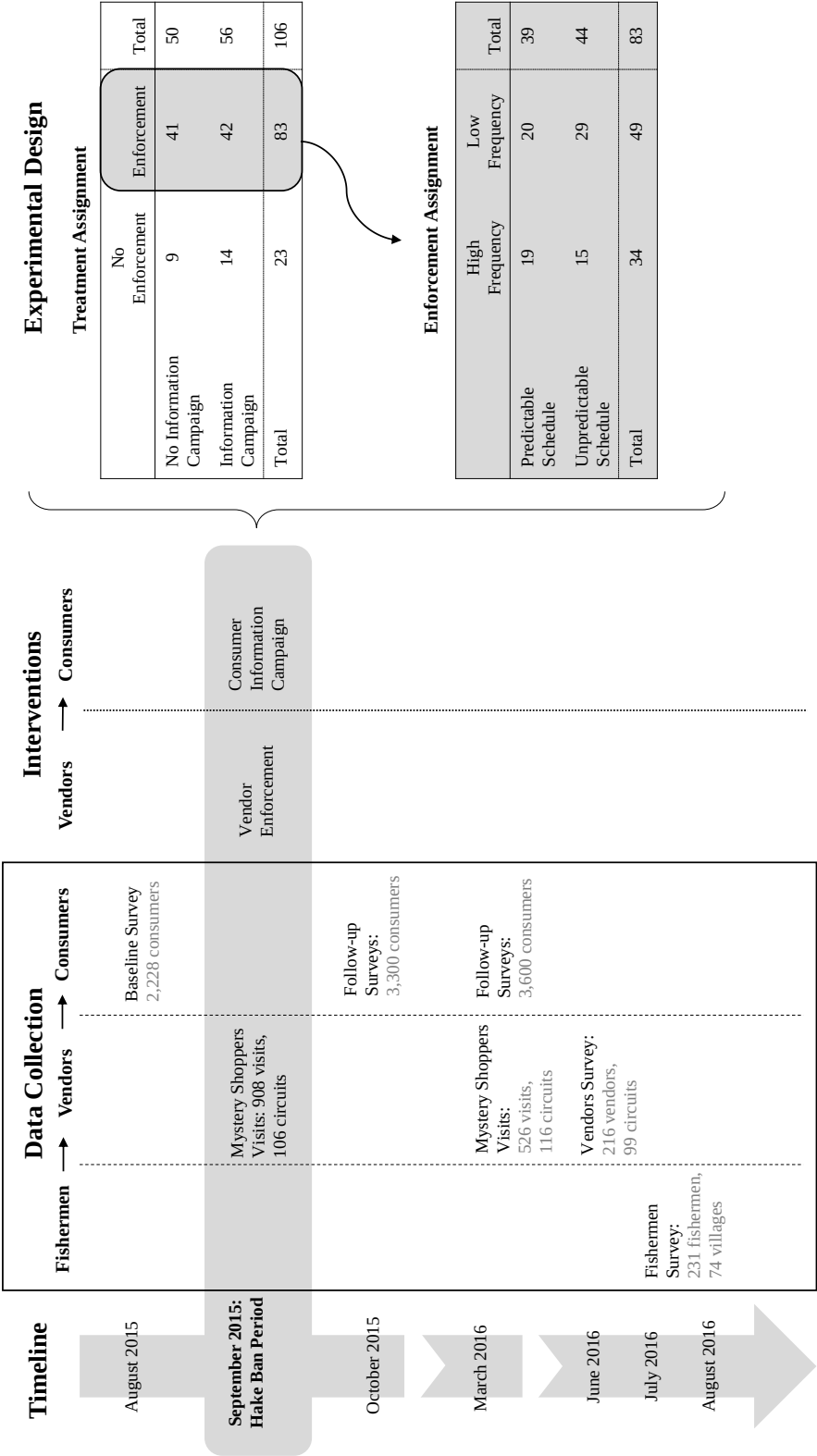
fish markets, once during the ban (September 2015), both before and after our interventions started, and again six months later in March 2016. We conducted two rounds of surveys of consumers during those same two periods. We also surveyed fishers at *caletas* and vendors at *ferias* to map the fish supply chain and investigate spillovers.

5.1 Mystery Shopper Surveys

For us to reliably measure illegal activity, fish vendors cannot know that they are monitored. To develop a strategy to address this challenge, the research team visited dozens of *ferias* before the ban to understand the market structure and relationships between vendors and consumers. We learned that vendors do not know most shoppers, so an unfamiliar face will not necessarily raise suspicion. This made it a good environment to deploy *mystery shoppers*. 29 enumerators were trained to work as mystery shoppers.¹¹ They were trained to look and act like ordinary shoppers, to pose as buyers and (try to) purchase hake fish from the vendors and were not told the treatment status of any market. They visited each circuit three times on average during the September ban. We conducted an additional round of mystery shopper visits in March 2016 to track longer term effects outside the intervention period.

¹¹They were mostly women between the ages of 40 and 50, because this demographic represents the typical feria customer profile.

Figure 3: Timeline of Data Collection and Interventions



These mystery shoppers gathered information on whether it was possible to buy hake, and the market price of the fish. They also collected information on what else was available for sale at the fish stalls and their prices, and noted what was being purchased by other shoppers in their presence. The visit protocol was piloted and refined through multiple pre-period visits to ferias to make sure we elicited the required information without raising suspicion. Given this methodology, we could not collect information about the total quantity of hake being sold, because that would be unnatural for a typical shopper to ask about and would have raised suspicion. The main outcome variable from this survey is therefore an indicator for whether it was possible to buy hake at any particular stall.¹²

5.1.1 Identifying Defensive Actions

Defensive actions are at the heart of our theory of learning and adaptation. These are normally difficult to observe because they are illegal and designed to be hidden. However, our data collection strategy allows us to uncover such hidden actions. Mystery shoppers uncovered two strategies most commonly used by vendors to circumvent enforcement: They hide the hake they sell (instead of displaying it openly), and they put the fish over ice and claim that it was caught legally in August, and frozen since then.¹³ We discuss both these strategies in greater detail below.

Hiding: Mystery shoppers were trained to ask vendors for hake even if it was not visibly on sale in the stall and they noted each occurrence of “hidden hake”. We never shared the specific vendor or feria identity with our government partners, so as to protect vendor privacy and abide by our research ethics protocol. These data were very useful for the evaluation, but were never used to target enforcement.

The hidden hake fish was often stored in a cooler behind the board that displayed the

¹²The mystery shoppers also noted down general characteristics of the stall and vendor. They also noted the behavior of fish vendors and conversations occurring in their presence. This is how we learned about the defensive strategies vendors employed, as described in the next section.

¹³There are other possible illicit reactions that are impossible for mystery shoppers to observe safely, such as bribes paid or threats issued during vendor-inspector interactions.

stall’s fish prices. This is costly for vendors, because displaying the fish available for sale and attracting customers’ attention are the main marketing tools at the vendors’ disposal. Many of our mystery shoppers noted that they observed regular consumers asking vendors for hake when it was not visible. The hiding strategy appears to work because many consumers are willing to partake.

Freezing: On paper, vendors are not allowed to sell hake fish in any form in September. In practice, Sernapesca inspectors were more lenient with vendors who were detected selling “frozen” hake. This is the practice of freezing the fish on ice and claiming that it was harvested in August, before the ban. We had not anticipated this reaction, but mystery shoppers noted the practice for us early enough such that we were able to collect systematic data on it. Many of those vendors admitted to our mystery shoppers that the “frozen” fish was in fact, fresh. Matching our mystery shopper data at the daily level to the administrative data on fines levied (from Sernapesca’s registry of inspector visits) suggests that inspectors were indeed much less likely to levy penalties when the vendor was claiming to sell “frozen” hake.

Selling frozen hake is costly for vendors because consumers prefer fresh fish, and because freezing requires freezers and access to electricity. From other rounds of data, we see that freezing is virtually non-existent during the rest of the year and so, this does appear to be a strategy that vendors use to circumvent the September ban.

5.2 Consumer Surveys at Fish Markets

We also surveyed consumers before and after the ban period. A separate team of enumerators (distinct from our mystery shoppers) stopped consumers close to points of entry and exit of the fish market to administer this surveys. To encourage unbiased responses, enumerators informed consumers that the survey was conducted by university-based researchers, and that it aimed to gather information about food consumption in ferias. They were not asked to provide any personal identifiable information, and were only asked about the list of food purchased in the feria in the past month - avoiding direct questions about the consumption of hake. We also

asked consumers to provide a sense of their home location on a physical map we carried, so that we could match their residence to the neighborhoods assigned to the information treatment.

We conducted three rounds of consumer surveys; we surveyed 2,228 consumers in August 2015 (before the ban) through 50 enumerator visits. After the ban, we surveyed 3,300 consumers in October 2015 through 54 enumerator visits and 3600 in March 2016 through 95 enumerator visits. This produced three rounds of a repeated cross-section. The same consumers were not followed over time.

5.3 Survey of Vendors at Markets and Fishers at Fishing Villages

We surveyed fish vendors in every market in our sample in June 2016 (outside the hake ban period). We asked vendors about the suppliers and intermediaries they source their fish from, so that we could map out the supply chain. We also asked vendors about their contacts with fish vendors who operated in other circuits, in order to study spillover and network effects.

To understand whether the effects of our interventions were transmitted upstream via the supply chain, we conducted a survey of fishers during July-August 2016 in every coastal village in the region where hake fish is caught and distributed. We surveyed 231 fishers from 74 fishing villages (*caletas*). Figure A.4 in Appendix A.2 contains a map of all *caletas* and fish markets.

Surveying fishers was valuable for two reasons. First, the interventions were designed to ultimately reduce illegal fishing, so understanding the activities of the fishers is essential for public policy. Second, the treatment effects may have spilled over to control areas if treatment and control markets are served by the same fishing village. Understanding these supply-chain connections are important for analyzing spillovers. Figure 3 organizes our interventions and data collection activities along the supply chain for fish.

6 Program Evaluation Results

We report experimental treatment effects first, and then use the daily data to test the model’s predictions on learning and adaptation. We registered this trial in the AEA registry before

data collection was completed. Our treatment effects analysis and the outcome variables we focus on closely mirror the project narrative we uploaded before we had access to any data. We highlight notable departures from the pre-analysis plan (PAP) in Appendix section H. Tests of the learning model were not pre-specified.

6.1 Empirical Strategy

Mystery shoppers visited several stalls in each market multiple times in September 2015. These visits created a stall-day level panel dataset of 906 visits. The first enforcement visit to various markets by Sernapesca officers occurred between Sept 4 and 10. Our panel data consists of 242 visits during the pre-enforcement period, plus 664 visits during the post-enforcement period. We use the following regression specification to evaluate the interventions, where each observation refers to a mystery shopper visit at fish stall s , in feria f , from circuit c visited on day t :

$$y_{sft} = \beta_0 Post_t + \beta_1 T_c + \beta_2 T_c \times Post_t + \beta_3 y_{sfc0} + X_{ct}' \beta_4 + \varepsilon_{sft} \quad (8)$$

y_{sft} is an indicator for whether illegal hake fish was available at that stall on that day. The treatment assignment (T_c) varies at the circuit level. The variable $Post_t$ indicates the post-intervention period, September 8-30.¹⁴ We control for weather on each day, whether the inspector visited the market that day, a few socioeconomic covariates (e.g. municipality crime rate), randomization strata fixed effects, and the baseline (pre-intervention) value of the dependent variable. The error term, ε_{sft} , is clustered at the circuit level, which was the unit of randomization. The coefficient of interest for the evaluation is the parameter β_2 , which captures the difference between treatment and control groups during the post-intervention period. In most of our tables, we will only report the β_2 coefficients, and suppress all others.

To study consumer fish purchase behavior, we use the post-intervention surveys of consumers conducted at ferias, because the pre-intervention surveys did not cover all markets, and we therefore do not have a full panel of consumer surveys. We use the following regression

¹⁴Many of the information campaign letters arrived at households even after September 8. There are other reasonable ways to define the post-intervention period, and we make a conservative choice. The Appendix Table C.10 reports the estimated effects varying the starting date of the “Post Intervention” period.

specification to evaluate the effect of interventions, where each observation refers to a to a single consumer i , surveyed in feria f , from circuit c :

$$y_{ifc} = \gamma_1 T_c + X_{ic}' \Gamma + \epsilon_{ifc} \quad (9)$$

Where y_{ifc} measures the number of times the consumer purchased hake fish in the past month. T_c is the treatment status at the circuit level, and X_{ic} represents a set of covariates, including socioeconomic characteristics of the municipality, individual demographics (usual fish consumption, age, gender, and household income) and strata fixed effects. Consumers are assigned treatment status based on the feria where they were interviewed.¹⁵ Standard errors are clustered at the circuit level.

Experimental Balance. Appendix section C.2 discusses experimental balance. Table C.6 presents the balance of relevant market characteristics and weather variables across different treatment arms. The tables C.7 and C.8 present the same estimates, but decomposing by the enforcement variations in predictability and frequency. Overall, treatment arms appear well balanced in the main specification. However, there is a slight imbalance (in terms of the aggregate F-stat) across the enforcement sub-treatments. We verify that our results are robust to controlling for any combination of covariates. All the regressions we report control for the full set of observable market characteristics and weather characteristics. The first three rows of Table C.9 show the differences in the sale of illegal hake in the pre-intervention period. While these are statistically insignificant, the coefficients are positive (corresponding to a smaller proportion of stalls offering hake during the pre-intervention period in control markets). In contrast, statistically significant differences between the treatment and the control group appear in the post-intervention period as a result of the interventions.

¹⁵While that is the only sensible choice for the enforcement treatment, we could have also used the person's address to link them to the information treatment. Results look very similar either way, and we have imperfect information on individual addresses, so we use the feria location.

6.2 Treatment Effects on Hake Sales Observed by Mystery Shoppers

Column 1 of Table 1 shows the effect of the interventions on whether fresh, visible hake was available for sale in that stall, as detected by mystery shoppers. Column 2 shows effects on whether hake in any form (either fresh/visible or hidden in the back, or “frozen” hake that is kept on ice) was available for sale. Each dependent variable is binary, and we report marginal effects from a Probit regression. The table 1 presents the coefficients of interest of regression equation 8, which track the effects of the demand-side information campaign, the supply-side enforcement treatment, or the interaction between the two (ferias where both interventions were simultaneously administered), during the post-intervention period.¹⁶

Table 1: Treatment Effects on Hake Sales

VARIABLES	(1) Fresh, Visible Hake	(2) Any Hake Available (Hidden, Frozen, Visible)
Information Campaign Only	-0.133 (0.066)	-0.131 (0.074)
Enforcement Only	-0.178 (0.082)	-0.130 (0.089)
Info Campaign and Enforcement	-0.179 (0.074)	-0.139 (0.094)
Change in Dep. Var. in Control Group During Intervention Period	-0.21	-0.36
N	901	901

Notes: This table reports the effect of each treatment arm on the availability of illegal hake fish. The variable Fresh Hake indicates when the hake was available fresh. Hake available indicates when was possible to buy fish in any form. The table reports marginal effects from a Probit regression. Other controls are included: municipality characteristics, strata fixed effects and the average level of the outcome variable in pre-intervention period. We control for pre-treatment values for the outcome variables in addition to the treatment indicator, because not all markets were visited in pre-intervention period. Robust standard errors clustered by circuit (the unit of randomization) in parentheses.

In column 1, vendors in markets exposed to the information campaign are 13.3 percentage

¹⁶We randomized the Information Campaign over the subset of the 48 most populous municipalities in our sample (out of 70 total). We control for an indicator for these 48 municipalities in all our regressions. We have also run regressions restricting the analysis sample to these 48 municipalities, and the results look very similar.

points less likely to be selling fresh, visible hake relative to control group vendors.^{17,18} This is quite a large effect, considering that about 43% of vendors in control markets were selling hake before the interventions were launched. Vendors operating in markets where Sernapesca monitors visit to levy penalties become 17.8 percentage points less likely to sell fresh, visible hake. The combination of the two treatments also produces a 17.9 percentage point decrease in hake availability, so there is no evidence that the information campaign complements the enforcement strategy to make it more effective.

When we add “hidden” and “frozen” hake to fresh/visible hake sales in column 2 to create a broader dependent variable that captures any type of hake sales, the enforcement treatment effects become smaller and lose statistical significance. Taken together, the two columns suggest that while the interventions reduced vendors’ propensity to engage in illegal activity that could be easily monitored by regulators (visible sales in column 1), it is not so clear whether it actually reduced the underlying environmental harm that we care about (column 2). Through the lens of our model, the reduction in the size of the treatment effect moving from column 1 to 2 stems from the defensive strategies that vendors adopt in response to the audits.¹⁹

The effects of receiving both the information campaign and enforcement (17.9 percentage points in column 1 and 13.9 pp in column 2) are almost identical to the effects of enforcement alone – i.e., adding the second treatment does not improve effectiveness relative to each treat-

¹⁷The “Information Campaign” group is a marker for circuits located in municipalities assigned to receive the High-Saturation Information Campaign, where the majority of neighborhoods were treated with the campaign. Appendix Table C.4 explains why we made this modeling choice. Our consumer survey data indicates that the majority (69%) of shoppers we found shopping at *ferias* located in “control” neighborhoods in high-saturation treatment municipalities resided in neighborhoods that were treated. It therefore makes more sense to code such *ferias* as ‘treated’ with the information campaign. Appendix C.7 shows the results of re-estimating the results in Tables 1, but reverting to coding *ferias* in control neighborhoods as not treated with information. The results are qualitatively similar. The high-saturation information treatment has significantly larger effects on hake sales than the low-saturation treatment.

¹⁸The p-value for the “Info Campaign Only” arm increases from 0.044 (standard delta method) to 0.105 under the pairs cluster bootstrap (Cameron et al., 2008), which accounts for finite-sample variability not captured by the normal approximation. This suggests that inference for this arm is sensitive to the small-sample correction, and its statistical significance should be interpreted with some caution. The remaining results remain statistically significant under standard thresholds ($p < 0.05$, $p < 0.1$) even under this more conservative procedure.

¹⁹It is curious that the control group experienced larger reductions in “any hake” (column 2) than in “fresh, visible hake” (column 1). This is because a few control group vendors practiced freezing during the pre-intervention period (first week of September), but they stopped doing so after the interventions started. Apparently vendors in the control group learnt that there would *not* be much enforcement in their *ferias*, and reacted accordingly.

ment implemented in isolation. From the perspective of our model, this implies that vendors' utility from selling hake illegally *absent any intervention* must be quite low on average. This is testable by examining control group vendors' propensity to sell hake illegally in September. Indeed, only 14% of control group vendors sell hake in any form during the month of September. The lack of complementarity between the two intervention arms, therefore, appears to arise from the fact that hake sales, absent any intervention, are low enough such that applying one intervention does not leave much of a margin for the second intervention to have any effect in our setting.

6.3 Consumer Behavior

We consider the mystery shopper data to provide the most reliable measure of illegal behavior, but the consumer surveys at markets allows us to report changes in purchase patterns during the ban. The first column of Table 2 shows treatment effects on the number of times that consumers report buying hake fish during the previous month. The reported coefficients are marginal effects from a Poisson regression, evaluated at the mean of all covariates. We see significant decreases in (self-reported) hake purchase across all treatment arms, and so results are generally consistent with the mystery shopper survey. However, in these consumer reports, the treatment effects appear larger in information campaign areas where purchases decrease by 50% compared to the control group. This may be because the direct communication consumers received through the information treatment created some self-reporting bias.

Consumer behavior was also indirectly influenced by the enforcement activity. Not only did self-reported hake purchases decrease there relative to control markets, the third column also shows that consumers were about twice as likely (or 8-11 percentage points more likely) to mention to our enumerators, totally unprompted, that they did not buy hake because there was a September ban in place. Our enumerators did not ask consumers any questions about the ban, but were instructed to note down whenever a consumer spontaneously mentioned the ban. Consumers treated with the information campaign were 15 percentage points more likely

Table 2: Treatment Effects on Fish Consumption

VARIABLES	(1)	(2)
	Num. Times Hake Purchased	Mention Ban (unprompted)
Information Campaign Only	-0.275 (0.071)	0.146 (0.045)
Enforcement Only	-0.111 (0.049)	0.082 (0.047)
Info Campaign and Enforcement	-0.098 (0.046)	0.107 (0.051)
Mean Dep Var Control Group	0.49	0.07
N	3218	3319

Notes: This table presents the effect of different treatments on the reported consumption of hake fish during September 2015. The column 1 shows the marginal effects from a Poisson regression because the dependent variable is count data, the column 2 shows marginal effects from a Probit regression. Consumers were not asked about the ban, but surveyors registered if the ban was mentioned spontaneously. These regressions include socioeconomic characteristics and strata fixed effects. The numbers of observations in columns 1 and 2 differ because some consumers could not recall the number of times they purchased hake in the past month. Both Poisson and Probit are nonlinear models, and the average marginal effects of each treatment depend not only on the coefficients reported in this table, but also on the values of the covariates. Robust standard errors clustered by circuit in parentheses.

to mention the September ban unprompted, so evidently the treatments were at least successful in spreading more information and awareness relative to control areas.

6.4 Variations in the Design of the Enforcement Strategy

We experimentally manipulated the enforcement schedule along two dimensions: Predictability and Frequency. Table 3 uses the mystery shopper data, and repeats the regression setup of Table 1, except that the enforcement treatment is now sub-divided into areas where the monitoring schedule was either predictable or unpredictable (column 1), or sub-divided into areas where monitoring was conducted at high versus low frequency (column 2). These provide direct tests of Predictions [5] and [6] highlighted in Section 3.4.²⁰

The first column shows that the enforcement strategy was more effective when it was unpredictable, in line with model predictions. When enforcement follows a predictable schedule (e.g.

²⁰See Table C.13 in the Appendix for a disaggregated specification that reports treatment effects separately for each experimental arm. The estimates confirm the patterns in the pooled specifications: the largest reductions arise under enforcement with an unpredictable, low-frequency schedule, while other enforcement arms yield smaller and less precise effects, and the information campaign alone also reduces illegal hake availability.

Table 3: Treatment Effect on Hake Sales by Enforcement Strategy

VARIABLES	(1)	(2)
	Any Hake Available (Fresh, Visible, Hidden or Frozen)	
Information Campaign only	-0.134 (0.073)	-0.135 (0.072)
Enforcement on Predictable Schedule	-0.060 (0.083)	
Enforcement on Unpredictable Schedule	-0.192 (0.094)	
High Frequency Enforcement		-0.070 (0.095)
Low Frequency Enforcement		-0.162 (0.090)
p-value of Predictable = Unpredictable Sch.	0.036	
p-value of Low = High Int. Enf.		0.280
Change in Dep Var in Control		
During Intervention	-0.36	-0.36
N	901	901

Notes: This table presents the coefficient corresponding to the interaction term $T_c \times Post_t$ for each treatment. To retain statistical power, the cells “Enforcement only” and “Enforcement + Info Campaign” from Table 1 are combined under “Enforcement” and then sub-divided by schedule predictability (column 1), or intensity (column 2). So these coefficients should be interpreted as the average effects of enforcement when half the sample is also exposed to the information campaign. Note that we previously find evidence of null interaction effect between enforcement and info campaign (Muralidharan et al., 2019). Column 1 includes a dummy for the intensity sub-treatment, and column 2 includes a dummy for the predictability sub-treatment, but those coefficients are not shown. Each regression controls for the dependent variable in pre-intervention period, strata fixed effects and municipality characteristics. Probit regression marginal effects are reported. Robust standard errors clustered by circuit in parentheses..

every Tuesday at 10am), its effect is not statistically different than zero. However, when we make the monitoring visit schedule difficult for vendors to predict, we see that there is a much larger and statistically significant decrease of 19 percentage points in vendors’ propensity to sell hake, even after we account for vendor defensive reactions like hiding and freezing. The effect of the unpredictable schedule is statistically significantly larger than predictable enforcement. The lack of predictability makes it difficult for vendors to anticipate the visit pattern, which according to the model, delays learning and makes it more difficult to deploy effective defense.

The second column shows results separately for the subgroup of vendors who received monitoring visits once a week (low frequency), and other vendors who were visited twice a week, which means that monitors followed a circuit around in the different market locations where those vendors set up stalls on different days of the week (high frequency). The high frequency visits in principle could have lowered illegal hake sales by increasing vendors’ beliefs about the

likelihood of enforcement ($\hat{\theta}$). In practice, high frequency backfired in line with what was predicted by Appendix Figure F.3: vendors learned faster and evidently deployed defensive actions more effectively in the short run. Devoting additional resources to enforce more frequently is 9.2 percentage points less effective. The gap between low and high frequency is meaningful in magnitude, but not statistically distinguishable.²¹

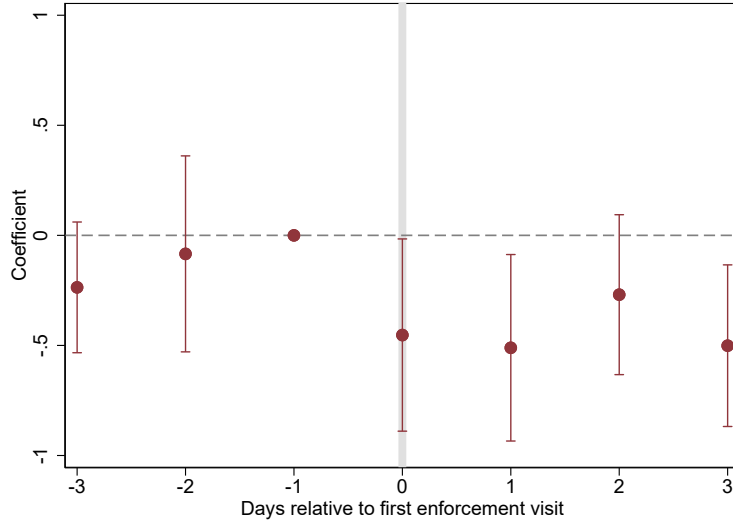
7 Empirical Evidence on Learning and Adaptation

We merge our daily data collected via mystery shoppers with the administrative data from Sernapesca inspectors to evaluate some of the specific theoretical predictions on how vendors learn and adapt to enforcement. In our model the choices vendors make each day is determined by the full history of their experience, so in the data we link observations made by mystery shoppers at a specific feria on a given day to the history of enforcement visits in that feria and circuit. Appendix C.1.4 describes Sernapesca’s enforcement activities in more detail. We organize the data this way to test Prediction [3] from Section 3.4 that vendors’ selling propensity changes as they learn about enforcement and adapt.

Response to the first enforcement visit. A natural way to examine how vendors update their beliefs and adjust their behavior is to focus on their response to the *first* enforcement visit they experience. In the model, vendors begin the season with priors on enforcement intensity and scheduling, which are updated as they observe a series of completed inspections. The first visit is therefore the most salient informational event: it provides direct evidence that enforcement is active. Studying vendor behavior in response to this event allows for an analysis of short-run vendor responses to new information about enforcement risk, a key feature of our model.

²¹Implementation data reveal a modest imbalance in enforcement intensity: circuits assigned to unpredictable schedules received, on average, more visits than those assigned to predictable schedules (see Table C.5). To address the concern that differences in visit frequency may confound estimates of predictability, we control for the number of visits in Appendix Table C.11. The results show that the gap between unpredictable and predictable enforcement, if anything, increases and becomes more precisely estimated, consistent with the additional visits in unpredictable circuits attenuating the baseline estimates.

Figure 4: Hake relative to the first visit



Notes: The figure reports coefficient estimates and confidence intervals from an event-study regression that relates the outcome variable, “Any Hake,” to indicators for days relative to a circuit’s first enforcement visit. Event time is defined as the number of days since the first enforcement day; observations outside the window are pooled into the endpoint bins. The omitted (baseline) period is day -1. The specification includes circuit and calendar-day fixed effects, absorbing all time-invariant circuit characteristics and common daily shocks. Standard errors are clustered at the circuit level.

We estimate an event-study specification that analyzes a vendor’s propensity to sell immediately after Sernapesca inspectors visit a market for the first time:

$$y_{ct} = \sum_{k \neq -1} \beta_k \mathbb{1}\{t - \tau_c = k\} + \alpha_c + \delta_t + \varepsilon_{ct}, \quad (10)$$

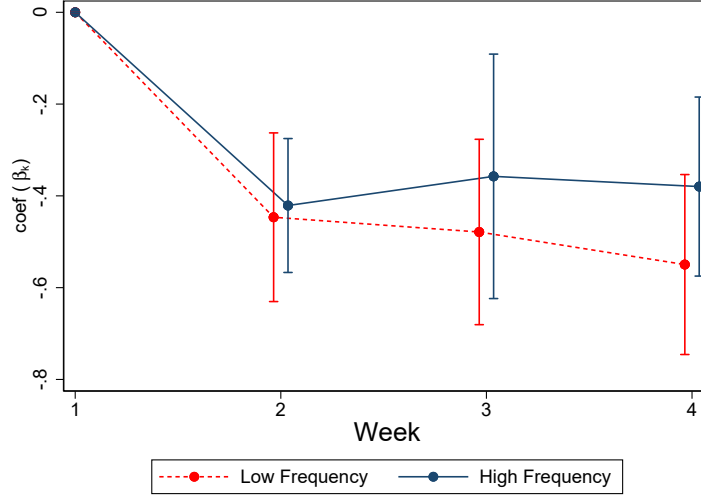
where y_{ct} is an indicator equal to one if vendors in circuit c sell any hake on day t , τ_c denotes the calendar day of the first enforcement visit in circuit c , α_c are circuit fixed effects, and δ_t are calendar-day (September-day) fixed effects. The omitted category is $k = -1$, so coefficients β_k measure changes in selling behavior relative to the day preceding the first visit.²²

Figure 4 plots the estimated coefficients and corresponding confidence intervals. We observe a sharp decline in the probability of selling hake on the day of the first enforcement visit, followed by persistent negative effects in the subsequent three days. As soon as vendors in a market

²²Standard errors are clustered at the circuit level. The sample is restricted to circuits that received at least one enforcement visit.

experience a new enforcement event in early September, there is a persistent deterrence effect. Our model would rationalize this reaction as an upward shift in vendors' perceived inspection risk.

Figure 5: Hake Available



Notes: This figure shows how the sale of hake evolved week by week. The graph plots the coefficients of the treatment-week interactions. Each relevant coefficient is normalized relative to the first week. We exclude the first three days of the month to keep the weeks balanced, i.e., the first week starts on Sept 4th and ends on Sept 10th. Each regression controls for crime rate and strata fixed effects and the average outcome variable before the implementation. We cluster standard errors at the circuit level.

Number of Visits: Figure 5 compares the week-to-week behavior of vendors exposed to different frequencies of enforcement. Consistent with the theory of learning, the two treatments produce similar effects at the beginning of the month, but effects diverge over time. Vendors exposed to higher visit frequency sell *more* hake, not less by the end of the month. This is consistent with the idea from our model that more interactions with auditors allow vendors to learn about enforcement loopholes and adopt defensive strategies.

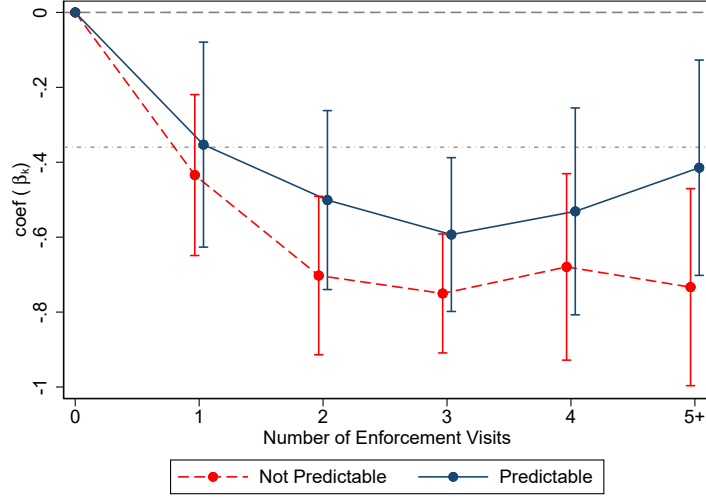
Figure 6 plots the likelihood of selling hake on a given day as a function of the number of inspections received at that *feria* until that day.²³ We see that receiving more visits reduces the probability of selling over time. However, the effect is non-linear: Earlier visits have a

²³The estimates are obtained from the following regression specification:

$$Y_{sfc} = \sum_{n=0}^N (\beta_n^P \times \mathbf{1}(\#Enf_{ct} = n) \times Pred_c + \beta_n^U \times \mathbf{1}(\#Enf_{ct} = n) \times UnPred_c) + X_{ct}'\Gamma + \varepsilon_{sfc} \quad (11)$$

The term $\mathbf{1}(\#Enf_{ct} = i)$ indicates circuits that have been visited n times by Sernapesca officials at the moment the secret shopper collected the data.

Figure 6: Hake Available



Notes: This figure shows how the sale of hake depends on the number of visits received until (including) the day the mystery shopper observed the behavior of the vendor. The horizontal line at -0.36 serves as a reference for the decrease in the probability of selling hake in the control group. This specification controls for crime rate, strata fixed effects, and the average outcome variable before the implementation. We cluster standard errors at the circuit level.

larger effect on reductions in hake sales than subsequent visits. This is especially true in the experimental arm with a predictable visit schedule. This differential effectiveness over time in the predictable arm was also evident in the theoretical simulations [Figures ?? and ??]. The intuition is that vendors learn that one of the ferias where they sell is not being targeted, and continue selling illegally at that location.

Visit Schedule and Targeting. We test this intuition directly in Appendix C.4 using *within-circuit* variation to study whether the same vendor shifts sales to non-targeted days and markets. Each fish vendor rotates between ferias within a circuit on different days of the week in a pre-determined pattern. We see substantial evidence supporting this form of specialization of illegal sales in the non-targeted *feria* (model Prediction [6]).

Appendix Table C.14 shows that, conditioning on the number of visits received, vendors who experienced inspections in various ferias on different days of the week (DOWs) reduced hake sales by an extra 9 percentage points (p-val<0.01) in the second half of the month relative to vendors who were targeted at a single feria always on the same DOW. Furthermore, Tables C.15 and C.16 study vendors' decisions to sell hake in the *non-targeted* feria in the second

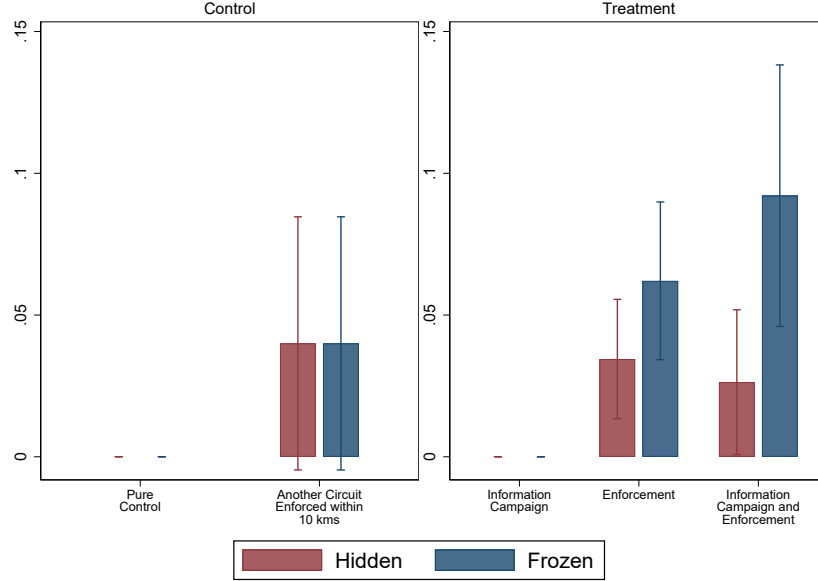
half of the month in circuit-fixed-effects regressions. We see that the same vendor sells more at markets and weekdays where she did *not* experience a visit, relative to another market/weekday where she did. The same vendor is also less likely to adopt a costly defensive action (hiding or freezing) in the “non-targeted” market. Further, the vendor is also more likely to shut down her stall entirely in the “targeted” market during the second half of the month.

This behavior resembles the theoretical simulations displayed in Figure F.3: the learning model predicted that the same vendor would behave differently in the targeted versus non-targeted ferias, as they learn about differences in inspection probabilities. The high levels of illegal sales in non-targeted ferias helps to explain why the predictable schedule of enforcement was less effective overall. While the observed patterns of behavior is very consistent with our model, we relegate this “within-circuit” evidence to the appendix because Sernapesca chose which feria to visit within each circuit (partly based on logistical considerations), and this variation therefore cannot be treated as random.

Defensive Actions. We now study vendors’ adoption of defensive actions (hiding and freezing), to test Prediction [4] from our model. Our mystery shoppers collected systematic data on vendors’ propensity to sell “hidden” fish from the back that was not displayed at the stall, and “frozen” fish that they claimed was caught in August. Hiding was clearly used as a defensive strategy to circumvent the September ban: we conducted another mystery shopper survey six months after the ban, and we did not observe even a single stall selling fish that was not publicly visible at that time. There are several pieces of circumstantial evidence in our data that freezing is also pretense; that fishers and vendors are not actually protecting the environment by catching fish in August and freezing it until September. First, we document more freezing in the second half of September 2015 than during the first half, after vendors have had a chance to learn about the enhanced regulatory activities. Real freezing would have been much less costly to engage in during the first half of the month. Second, we collected data on stall characteristics, and availability of a freezer in a stall is not at all predictive of freezing. If anything, our mystery shoppers find that stalls without freezers are more likely to be selling frozen fish

post-intervention. Third, many secret shoppers noted down that in their conversations with vendors, many vendors admitted (and even insisted) that the fish was fresh even though it was labeled as frozen.

Figure 7: Hidden and Frozen Hake Fish

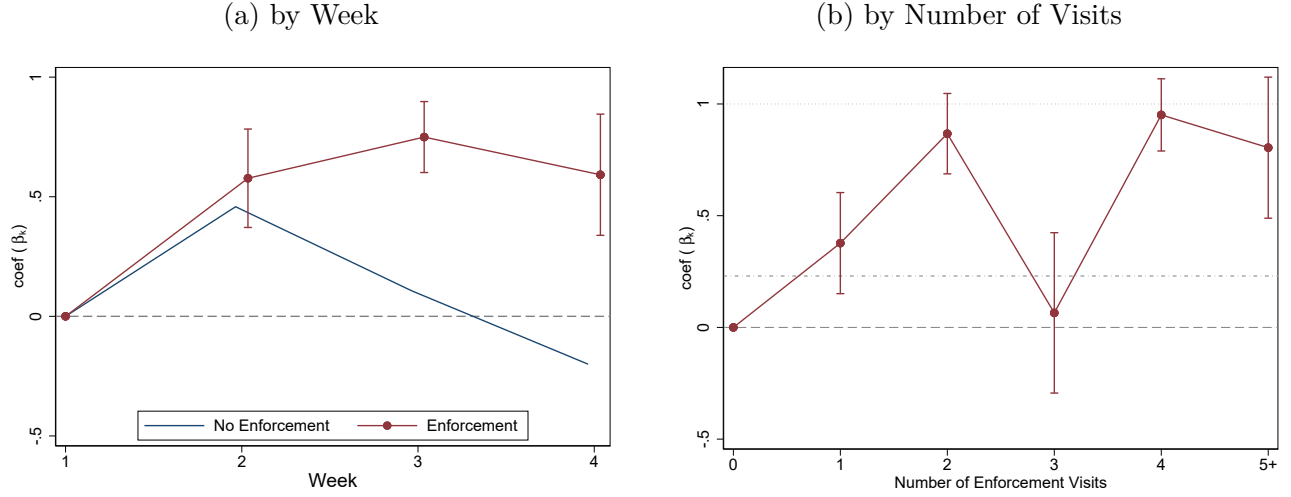


Notes: This figure shows the unconditional mean of hidden hake for different treatment statuses in the post-intervention period. The level of frozen hake is statistically different from zero for markets assigned to Enforcement and Enforcement and Info Campaign. The level of hidden is statistically different from zero for markets with Enforcement and spill-overs.

Figure 7 shows the prevalence of freezing and hiding across treatment groups in the post-intervention period. We divide up the control group into markets that have another circuit that is randomly assigned to enforcement within 10 kilometers (to capture any information spillovers), and *pure control* markets that are more than 10km away from any treated area. Several notable patterns emerge:

1. We do not observe any hiding or freezing at all in pure control markets in the post intervention period. In contrast, 7.2% vendors operating in circuits that received Sernapesca inspector visits sell frozen fish (p-value < 0.01), and 3.2% of those vendors engage in hidden hake sales (p-value 0.01).
2. Vendors operating in circuits exposed only to the information campaign did not engage in any hiding or freezing in the post intervention period. Vendors (sensibly) employ

Figure 8: Proportion of Defensive Hake



Notes: Figure 8 describes the conditional probability of selling hake either frozen or hidden. The coefficients were obtained from an OLS regression over the sample of the stalls selling hake that day. In Panel (a), each treatment assignment interacts with weekly dummies. In Panel (b), treatment interacts with dummies of each number of visits received until (including) the day the mystery shopper observed the behavior of the vendor. The horizontal line at 0.21 serves as a reference for the increase in the conditional probability of selling hake defensively in the control group (that received no enforcement visit). We include strata fixed effects and cluster at the circuit level. The "No Enforcement" category includes observations assigned to the control group and the information campaign. The low level observed at three visits in Panel (b) is partly explained by a small(er) denominator in the fraction.

these defensive strategies only against Sernapesca inspectors, not informed consumers. This implies that the information campaign did not simply signal enhanced government attention to the problem. The consumers are an important independent actor whose knowledge affects vendor behavior. This result also explains why the enforcement strategy appeared to produce larger decreases in fresh, visible hake sales than the information campaign (column 1 of table 1), but not once you take vendor defensive strategies into account (column 2).

3. 4% of vendors who operate in control markets - but located close to treated areas - engage in hiding and freezing, in contrast to 0% in *pure control* markets (p-value 0.02). There appear to be some spatial spillovers in information about Sernapesca visits, and in vendor behavior. We will explore these spillovers at greater depth in Section 8.

Adopting Defensive Actions Over Time. Panel (a) of Figure 8 describes the proportion of all hake sold in frozen or hidden form (as opposed to fresh, visible form), within the set of stalls that sell hake at all. Freezing and hiding were unusual at the beginning of the month. Vendors

exposed to Sernapesca enforcement increasingly adopted these defensive strategies from one week to the next, while vendors not exposed to enforcement did not. By the end of the month, nearly 70% of the stalls selling hake in the enforcement areas hid or froze. Panel (b) shows the conditional probability of hiding or freezing as a function of the number of visits received. After experiencing the 4th enforcement visit, virtually every vendor selling hake chooses to either hide or freeze their produce. There is a curious dip in this panel after visit 3 that breaks the pattern. This appears to be a small numbers issue. When breaking the data down by prior number of enforcement visits, there were only 5 mystery shopper observations between enforcement visit 3 and visit 4, due to idiosyncratic reasons. Overall, these data strongly suggest that vendors indeed adopted defensive strategies in response to enforcement, as stated in model Prediction [4].

Finally, consistent with Prediction [1], the information campaign reduces hake sales even after accounting for defensive strategies (as already shown in Table 1), and consumer surveys conducted again 6 months after the ban ends (see table C.19) shows that the demand-side effects somewhat persist over the long run.

8 Spillovers and Market Level Effects

Interpreting our experiments causally requires a Stable Unit Treatment Value Assumption (SUTVA) whereby targeting enforcement in some treatment markets does not have an indirect effect on the control markets where we collect data. Our interventions may have had spillover effects on control markets through information transmission, or by changing equilibrium prices. It may also have affected the behaviors of other market actors, such as the fishers who supply to vendors. It could have also changed the prices and quantities of other fish that can act as substitutes for hake. In this section we study these spillovers and equilibrium effects, and report experimental results net of spillovers. We collected additional data to track spillovers and equilibrium effects, including a survey of fishers, a survey of vendors to understand their social and supply-chain connections to vendors operating in other markets, GIS data on the

location of all markets, and data on the prices and availability of substitute fish. The vendor and fisher surveys allow us to map the supply chain for each of the ferias in our sample. The geography of Chile (with a very long coast) creates large spatial variation in the locations of *ferias* where vendors sell and *caletas* where the fishers bring in their catch, which in turn produces variation in geographic and social connections between different market actors (see Figure A.4).

We organize the spillovers results as follows. We first re-run our main experimental specification (Table 3, column 1) controlling for spatial, social, and supply-chain spillovers. Second, we show how fishers upstream adjusted their behavior in response to enforcement at markets downstream – which is one channel by which effect spill over to control markets. Third, we re-run our main specifications correcting for the possibility that some vendors may exit the market altogether in response to the treatment. Fourth, we study the market equilibrium effects on prices and consumption of other substitute fish, because some spillovers may operate through a price mechanism.

8.1 Spillovers on Control Markets

We identified three primary channels through which treatment may affect behavior of control markets, and collected data on each channel:

1. *Spatial spillover*: Control markets located geographically close to a treated market may feel the effects of treatment because they share consumers with the treated area.
2. *Social spillover*: If control market vendors are socially connected to vendors operating in treatment areas, they may be more likely to learn about *Sernapesca*'s enforcement activities.
3. *Supply chain spillover*: Treatment and control vendors may source from the same fishers. If a supplier changes fishing behavior due to treatment, that could indirectly affect fish sales in control markets.

Of these different channels, an increase in fish sales in control markets due to indirect effects is of greatest econometric concern. If fishers dump all excess hake in control markets when vendors in treated markets are unwilling to buy hake, then the treatment-control difference will appear to show that the treatment was effective, when in fact hake sales were simply spatially displaced towards the control group. Our regressions would over-estimate the effects of treatment in that scenario. This is why it’s important for us to re-investigate these effects controlling for these sources of spillovers.

In Table 4, we re-estimate the effects of predictable and un-predictable enforcement originally reported in Table 3, but now controlling for potential channels of spillover effects.²⁴ The main treatment effects get a little stronger after controlling for spillovers, but the spillover effects are only suggestive and statistically imprecise.

The first column presents the benchmark: unpredictable enforcement reduces hake availability by 15.7 percentage points in this specification without accounting for any spillover. The second column controls for spatial spillovers, with the indicator “within 10km of Treated Market” turning on for untreated markets that have at least one treated *feria* within a distance of 10 kilometers.²⁵ ²⁶ The coefficient of this variable is negative but small and statistically indistinguishable from zero, suggesting very limited spillovers based on shared consumers due to geographic proximity. The third column includes an indicator for control markets where at

²⁴We follow a procedure similar to Miguel and Kremer (2004) in estimating treatment effects in the presence of spillovers. We divide the control markets into subgroups; (a) Control areas that are more likely to have been affected by treatment due to geographic or social or supply chain connections, which we call “Spillover Group”, and (b) Control areas un-connected to treatment markets, which we call “Pure Control”. Note that sub-dividing the control group this way reduces the number of markets allocated to the omitted category. To retain sufficient statistical power, we therefore focus on re-estimating the effects of enforcement treatment variations only, because spillovers cause the greatest econometric concern (of over-estimating treatment effects) for this particular result. In this setup, some of the markets in the omitted category received the information treatment, so the regression coefficients will look a little smaller in this table compared to Table 3. For the same statistical power reasons, we only study an overall spillover effect of enforcement, and do not try to estimate separate sub-treatment spillovers.

²⁵Using the 10 km radius evenly divides the control group into “pure control” and “spill-over market”, and therefore maximizes statistical power. Alternative definitions produce similar results.

²⁶Vendors connected to a larger number of other circuits are more prone to being exposed to the treatment, and that variation is not random. To control for this, we include a full set of dummy variables for the number of other circuits that each reference circuit is connected to, separately for spatial, social and supply-chain connections. Thus, the variation of exposure to spillovers stems only from the treatment status of other markets, which is exogenous because it was randomly assigned. (Miguel and Kremer, 2004).

Table 4: Treatment Effects on Hake Sales Controlling for Spillovers to Control Markets

VARIABLES	(1)	(2)	(3)	(4)
	Any Hake Available (Fresh, Visible, Hidden or Frozen)			
Enforcement on Predictable Schedule	-0.023 (0.083)	-0.030 (0.069)	-0.076 (0.080)	-0.058 (0.060)
Enforcement on Unpredictable Schedule	-0.157 (0.091)	-0.167 (0.075)	-0.199 (0.084)	-0.177 (0.084)
Spatial Spillover (within 10 km of Treated market)		-0.017 (0.082)		
Social Connection Spill-over (Vendor knows a Treated Vendor)			-0.071 (0.076)	
Supply-Chain Spill-over (Sources from same <i>Caleta</i> as Treated Vendor)				-0.077 (0.081)
Change in Dep Var in Control During Intervention	-0.36	-0.36	-0.36	-0.36
N	901	901	901	901

Notes: This table re-estimates treatment effects controlling for possible spillover effects from treatment to control markets. We focus on enforcement treatments to ensure that the control cell size is large enough to be divided by exposure to spill-overs. We only present the coefficient corresponding to the interaction term $T_c \times Post_t$ for each treatment. Controls for T_c , $Post_t$, covariates, and baseline value of the dependent variable are included, but those coefficients are not shown. The table reports marginal effects from a Probit regression. The dependent variable is an indicator for any type of hake (fresh-visible, hidden or frozen) for sale in the stall. Robust standard errors are clustered by circuit, which was the unit of randomization. .

least one vendor reported that they knew a vendor in a different market that was randomly assigned to the enforcement treatment. The coefficient on this variable suggests that there was a 7 percentage reduction in hake availability in markets experiencing this “social spillover”, but this effect cannot be statistically distinguished from a null effect with any confidence. Controlling for this form of spillover increases the effect of unpredictable enforcement to a 19.9 percentage point reduction in hake availability ($p < 0.05$). Finally, column 4 includes an indicator for control markets who source from fishers operating out of *caletas* that primarily supply to other markets that were assigned to the enforcement treatment. We see a 7.7 percentage point reduction in vendors’ propensity to sell hake in control markets that are connected to treated markets through shared suppliers, but the effect is again not statistically precise.

Importantly, accounting for these spillover effects make the main treatment effects of unpredictable enforcement on enforced areas a little larger and more statistically precise. This is because controlling for spillovers allow us to compare treated areas to the subset of “pure” control areas unaffected by the treatment.

8.2 Treatment Effect Transmission along the Supply Chain

For the supply chain spillover channel to be relevant, the fishers supplying hake to these vendors must have altered their behavior in some way. To understand those changes, we directly survey fishers operating out of every *caleta* (fishing village) that serves the markets in our sample.²⁷ The reactions of fishers are particularly important to track because our interventions conducted at the final point-of-sale has to somehow get transmitted up the supply chain to fishers, for these interventions to ultimately protect the hake population. Only if fishers start perceiving the effects of these interventions on demand conditions will they change fishing behavior in ways that improve environmental outcomes.

Since we did not have baseline data from fishers for years preceding the September 2015 ban, we ask them retrospective questions in 2016, in which the fishers are asked to compare demand and profits during September 2015 (when our interventions were launched) relative to September 2014. To minimize possible response bias given the government fishing ban, we were careful to phrase our questions generically, to cover revenues earned from all types of fish, and not just hake specifically. Retrospective answers may be subject to recall bias, but since these fishers were not directly treated, it is less likely that the recall bias is correlated with treatment assignment. To report treatment effects on fishers, we have to connect each *caleta* to treatment and control markets. We use the vendor survey on the structure of the supply chain -i.e. which caletas each vendor buys from - to link fishers to the randomized treatments.

Table 5 reports results. Column 1 shows that fishers operating out of *caletas* that sell to at least one circuit which had been randomly assigned to enforcement, are 24 percentage points more likely to report that they earned less in September 2015 from all fishing activities compared to September 2014, relative to fishers in caletas that supply to control *ferias*.²⁸ Fishers operating out of caletas that supplied both enforced markets and markets that experienced the information campaign were 36 percentage points more likely to report lower revenues in the

²⁷A few caletas in the regions covered by our sampling frame are only used by divers who harvest seafood, not fish -and we therefore exclude those *caletas*.

²⁸We could instead define exposure based on the proportion of circuits enforced, and results look similar. The “at least one” formulation is attractive because it partitions the sample into equal halves.

Table 5: Treatment Effect Transmission to Fishers in Caletas

VARIABLES	(1) Earned Less in Sept 15 than Sept 14	(2) Feria Vendors buy less Hake in Sep15 compared to Sept 14	(3) Consumers are informed of Hake Ban
At least one circuit Enforced	0.238 (0.105)	0.169 (0.293)	-0.033 (0.147)
Info Campaign	0.043 (0.158)	-0.101 (0.322)	0.343 (0.186)
At least one circuit Enforced and Info Campaign	0.358 (0.128)	0.553 (0.315)	0.173 (0.195)
Mean Dep Var Control Group	0.31	0.40	0.77
N	202	179	217

Notes: This table reports OLS coefficients based on fishers' responses. The variable Information campaign correspond to caletas located in municipalities assigned to receive any level of information campaign. The variable "At least one circuit enforced" considers all circuits located in the same municipality of the caleta. Socioeconomic variables of the caletas are included as covariates. In average, three fishers were surveyed in each caleta. The numbers of observations in columns 1, 2 and 3 differ because some fishers could not recall the earnings and vendor behavior in specific months. The dependent variables of each column are dummy variables. Robust standard errors clustered at caleta level in parentheses.

month of the interventions, compared to the same month in the previous year. Treatment effects were perceived by fishers upstream in the fish supply chain. Column 2 shows that these fishers are more likely to report that vendors were less willing to buy hake in September 2015 compared to the previous year, but this result is marginally significant with ($p < 0.10$). Column 3 shows suggestive evidence ($p < 0.10$) that fishers linked to the information campaign areas are more likely to report that final consumers are aware of the hake ban. The effects on earnings and purchases are not present for fishers linked to the "information campaign only" areas. Somehow, the enforcement activities were more salient upstream in fishers' minds. All these results should be viewed as suggestive, because they are based on a survey conducted 8 months after the intervention, where fishers were asked retrospectively about their experience during the September 2016 ban.

8.3 Change in Number of Stalls Selling Fish

Our intervention may force some fish vendors to exit the market altogether. Appendix Figure B.2 shows that the average number of fish stalls decreases in the markets randomly assigned to the enforcement treatment, especially during the second half of September. This itself is an important effect of the treatment, but it is not captured by the treatment effects reported in

Table 1. Table C.18 in Appendix C.5 describes how we correct our estimates for stalls exiting. The correction makes the effect of enforcement larger, but it does not affect the coefficients for other treatments very much.

8.4 Effects on Fish Substitutes

We collected data on prices and availability of other fish species in the same markets where hake is sold. The September ban is only specific to hake fish, so we might expect consumers to substitute to other fish varieties. This may be because informed consumers choose to avoid hake fish during the ban, or because the enforcement treatment reduces hake availability or increases its price.

The universe of data from all markets suggests that there are seven possible fish substitutes for hake,²⁹ but a typical stall only offers two or three varieties of fish.³⁰ The most common fish substitute is pomfret, which can be found in two-thirds of all markets. Pomfret is larger and (arguably) more tasty than hake fish and is not over-exploited. In Table 6, we study the availability of pomfret (column 1), or any other non-hake fish including pomfret (column 2), as a function of the treatment status of the market where the fish stall is located.

The penultimate row of the table indicates that stalls in control markets are 29 percentage points more likely to start selling pomfret during the September hake ban, so it appears that vendors in general move towards substitutes during the ban. The increase in pomfret sales during September is larger in treated areas (by a further 12-15 percentage points, which results in a 41-44 percentage point increase during the hake ban), but the treatment-control differences are barely statistically significant.³¹ The p-value for only one of the three coefficients (associated with Predictable Enforcement) is below 0.10. Column 2 investigates treatment effects on

²⁹They are pomfret, mackerel, silverside, salmon, sawfish, albacore and southern hake. Of these substitutes, the southern hake is the only one with a similar ban, but in August. The southern hake is considerably larger than the common hake and is harvested in the southern regions of the country, without any geographical overlap with the common hake. More details are available in Subpesca (2015).

³⁰Table C.2 in the Appendix describes the availability and price of different fish species observed by mystery shoppers in ferias during September 2015.

³¹Consumers are more prone to substitute products at similar price levels (see Table C.2). The hake is considerably cheaper than the pomfret and other relevant fish species. This fact may have limited the willingness to substitute for different fish species.

Table 6: Do Vendors Substitute to Selling Other Fish in Response to Treatment?

VARIABLES	(1) Pomfret Available	(2) Any Other Fish Available
Information Campaign Only	0.146 (0.098)	0.004 (0.035)
Enforcement on Predictable Schedule	0.133 (0.079)	0.027 (0.031)
Enforcement on Unpredictable Schedule	0.115 (0.078)	0.065 (0.033)
Change in Dep Var in Control Markets During Intervention	0.29	0.09
N	901	6328

Notes: The table reports marginal effects from a Probit regression. The unit of observation in the first column is stall $\times$ secret shopper visit, and in the second column is stall $\times$ secret shopper visit $\times$ possible substitute fish variety. We only present the coefficient corresponding to the interaction term $T_c \times Post_t$ for each treatment. Controls for T_c , $Post_t$, covariates, and baseline value of the dependent variable are included, but those coefficients are not shown. Robust standard errors are clustered by circuit in parentheses.

the vendor’s decision to offer each of seven different fish substitutes for hake.³² The coefficients indicate that vendors who faced unpredictable enforcement become 15.5 percentage more likely to switch to selling other fish during the hake ban, compared to the 9 percentage point increase in control markets. This 6.5 percentage point treatment-control difference is statistically significant ($p=0.051$).

8.5 Effects on Prices

We collected data on fish prices during all our mystery shopper visits. However, prices are observed only when the fish is available for sale and hake is only available in 26% of markets. Treatment changes the propensity to sell illegal hake fish, so it affects the selection of which prices are observed. There are therefore large sample-selection issues that complicates any analysis of treatment effects on prices, and we refrain from running regressions on the price of hake. The most consumed fish during September (and second most consumed fish during the rest of the year) is Pomfret, which is available in 68% of the stalls (see Appendix Table C.2). Since pomfret is more often available (and not banned), we instead run regressions to study

³²The sample size is larger in this regression because selling each fish variety is treated as a separate decision, but our standard errors are still clustered by the unit of randomization of the treatment (the circuit).

treatment effects on the price of pomfret.

As a descriptive exercise, Figure B.3 shows that the price of hake increased week-to-week in September, over the course of the ban period. Pomfret prices fell by 10% in the second week and that lower price remained stable thereafter. This time-series pattern in prices is consistent with fishers upstream in the supply chain shifting away from hake and towards catching pomfret during our interventions in September 2015. Through conversations with fishers during our survey, we learned that they are able to adjust their fishing strategy to target different species if there are market signals that hake demand is low. To do so, they change the location and depth at which their nets are dropped.

Table 7: Treatment Effect on Fish Prices

VARIABLES	(1) Log Price Pomfret	(2) Log Price Substitute
Information Campaign Only	0.210 (0.109)	0.140 (0.096)
Enforcement Only	-0.017 (0.066)	-0.021 (0.055)
Info Campaign and Enforcement	0.081 (0.065)	0.047 (0.059)
Change in Dep Var in Control		
During Intervention	-0.20	-0.27
N	614	939

Notes: The table reports treatment effects on hake substitutes' price from OLS regressions. The outcome variable is the log of price per kilo. The unit of observation in the first column is *stall with pomfret available* $\times$ *secret shopper visit*, and in the second column is *stall with any substitute available* $\times$ *secret shopper visit* $\times$ *substitute available fish variety*. We only present the coefficient corresponding to the interaction term $T_c \times Post_t$ for each treatment. Controls for T_c , $Post_t$, covariates, and baseline value of the dependent variable are included, but those coefficients are not shown. Robust standard errors are clustered by circuit in parentheses.

Table 7 shows treatment effects on pomfret prices (column 1) and prices of all substitute fish including pomfret (column 2). We find that the price of substitutes weakly increase ($p - value < 0.1$) in markets where the information campaign discouraging hake consumption in surrounding neighborhoods, suggesting that part of the demand for hake shifted towards substitutes. Relative to the control group, markets that received enforcement show small and insignificant price decrease. The fact that we observe these differential price effects suggests that fish markets are at least somewhat segmented.

9 Cost-Effectiveness of Enforcement vs. Information

Given the complications associated with enforcing regulations documented in this paper, and the complexity of designing regulations that are robust to unanticipated defensive reactions from enforced agents, it is useful to determine how cost-effective the enforcement strategies were relative to an information campaign. We collected data from *Sernapesca* on the full administrative costs of implementing each treatment, so that we can report on the relative cost-effectiveness of enforcement and information strategies.

We define effectiveness of our interventions on the basis of our treatment effects on all hake sales (visible, hidden or frozen). Since the fish sold in ferias comes directly from fishing villages and was harvested the same day or the day before, we assume that reduced hake sales is proportional to the decrease in hake fishing. The fisher survey results reported in section 8.2 suggests that fishers did feel the effects of the interventions. Our interventions were conducted at scale covering all major markets where hake is sold, which implies that our data are net of “leakages” of hake from our sampling areas.

In Table 8, we conduct the relative cost-effectiveness analysis by taking our best estimates of the effects of treatments on reduction in hake sales and combining it with an estimate of the number of fish available in the market that we compute using the data we collected from vendors. This allows us to create an estimate of the extra hake fish that are “saved” due to these treatments. Methodological details underlying these calculations are in Appendix G.

We compare this number with the cost of implementing each intervention to compute how much it cost to save each fish under each of the treatment assignments. Overall, the information campaign appears more cost effective than the enforcement strategy. This is partly because enforcement becomes less effective as vendors learn to hide and freeze fish and circumvent regulation. Enforcement costs US\$6.05 per saved fish, compared to \$4.98 under the information campaign.

However, once we examine specific versions of the enforcement strategy that were more successful at curbing hake sales, we see that sending monitors on an unpredictable schedule is

Table 8: Cost-Effectiveness Analysis

	(1)	(2)	(3)	(4)
	Reduction of Hake Sale	Units of Hake Saved	Implementation Costs (USD)	Cost of Saving One Hake (USD)
Enforcement (<i>Overall</i>)	0.13	10,399	\$ 62,900.25	\$ 6.05
<i>Unpredictable</i>	0.192	15,358	\$ 69,190.27	\$ 4.51
<i>Predictable</i>	0.06	4,799	\$ 62,900.25	\$ 13.11
<i>Low Frequency</i>	0.162	12,959	\$ 53,475.84	\$ 4.13
<i>High Frequency</i>	0.07	5,599	\$ 99,613.61	\$ 17.79
Info Campaign	0.13	3,257	\$ 16,213.53	\$ 4.98

Notes: This table shows the benefits and costs of implementing each intervention. Column (1) reports the estimated effects (in percentage points) of treatments in the sale of any type of hake. Column (2) is computed based on the numbers of stall per feria, number of days a week the feria operate and number of fish available in a normal stall. Column (3) is reported by Sernapesca and represents a combination of fixed and variable costs. Finally, column (4) correspond to the ration of (3) over (2). These calculations assume the control group had zero enforcement nor information campaign. As we discussed in section C.1.4, the control group (mistakenly) received a few enforcement visits, the cost is negligible.

a more cost effective way to protect hake, even after accounting for the fact that unpredictable monitoring schedules were more costly for *Sernapesca* to maintain because it required slack personnel capacity. The cost of “saving” a hake via unpredictable enforcement drops to \$4.51. Not surprisingly, less frequent monitoring schedule is most cost-effective (\$4.13 per saved hake) because it was both more effective at reducing hake sale than high-frequency enforcement, and it was obviously also cheaper to implement. Predictable and high-frequency audits were total policy failures in that they were 250-400% too expensive per hake saved, given the subversive adaptation by hake vendors.

These calculations are useful to gauge the *relative* cost-effectiveness of alternative strategies to protect hake, but it does not tell us whether any of these strategies would pass a cost-benefit test.³³ Sophisticated benefit calculation would require us to take a stance on the biology of hake fish (how saving a hake in September 2015 translates into a dynamic effect on the hake population via reproduction), and the ecological value of protecting hake. These considerations are outside the scope of our analysis, but our results can be easily combined with benefit

³³Both enforcement and information provision involve fixed and variable cost components, implying that scaling either intervention would reduce average costs per hake saved. Our cost-effectiveness calculations, therefore, compare interventions implemented at the same experimental scale using realized administrative costs and should be interpreted as relative comparisons across policy designs rather than as estimates of minimum achievable costs under full national rollout.

numbers from ecology studies. The analysis in this paper takes the government’s regulatory goal (“Protect hake fish through a September fishing ban”) as given, and studies the consequences of enforcing that regulation, and analyzes the best ways to achieve that goal.

10 Conclusion

Research in many fields of applied microeconomics evaluate the effects of new regulations, such as anti-corruption campaigns, fines for non-compliance with health, hygiene or environmental standards, or penalties for tax evaders. The effectiveness of such policies depend on the (sometimes unanticipated) reactions of the regulated agents to the new enforcement regime, which is in essence a micro version of the “Lucas critique” (Lucas, 1976). Agents adapt once they have had a chance to learn about the new rules, and may discover new methods to circumvent the rules. This paper presents a formal model to explain this process of learning and adaptation, and puts forth a research strategy - composed of an experimental design and creative data collection - that permits an investigation of adaptation strategies employed by regulated agents as well as the effects of regulation net of such adaptive behaviors.³⁴ This research approach should be broadly useful for policy evaluation whenever agents can adapt to circumvent enforcement. As one important example, such concerns were first-order in the design of the Dodd-Frank Wall Street Reform and Consumer Protection Act in 2010 following the global financial crisis. Smith and Muñiz-Fraticelli (2013) write about this regulatory effort:

“[A] major problem with the new financial legislation is that it is responsive to past market innovations without being sensitive to future innovations (...) The problem is that these actors will not always behave in a predictable way. That is the genius of financial innovation; the market always looks for new opportunities for profit, and, as the dawn follows the dark, mischief may arise.”

³⁴An alternative evaluation strategy would be to collect data in the short run before agents have an opportunity to react to the new regime, and in the long-run after they have reacted. This is more expensive, requires more time, and is fundamentally more difficult, because researchers do not always know when and how agents would learn and adapt.

Our experimental variations coupled with our theory of learning yield novel insights about the adaptive behavior of regulated agents, and how to better design policy accounting for their adaptation. Data collected via “mystery shoppers” help us identify the ways in which agents exploit loopholes to continue selling fish illegally. Implementing a high frequency monitoring schedule produces a counter-intuitive result – but one that economic theory can rationalize – it allows vendors to learn the regulators’ strategies faster, and more effectively cheat, thereby undermining enforcement efforts. Our theoretical and empirical results imply that even if monitoring is cheap, the regulator may do better by holding back some enforcement resources. And when you do monitor, adopting an unpredictable schedule makes it more difficult for agents to circumvent enforcement and proves to be the most cost-effective way to reduce hake sales even though it is more expensive to implement.

Our 2x2 experimental design with an information campaign generates insights on the conditions under which demand- and supply-side strategies aimed to reduce illegal activities can act as substitutes or complements. While our results can be informative about ways to curb undesirable behaviors in other settings, we should be careful not to over-claim the broader implications. Each regulatory setting is unique, and contextual details matter. The hake fishing ban in Chile only lasts a month, and consumers, vendors, and fishers have substitute food sources and livelihood opportunities. Such options may not be present in other regulatory settings, and even instituting bans may be politically complex in those cases.

Ultimately we learn that without sophisticated design-thinking, attempts at enforcement can backfire. Designing and implementing a consumer information campaign is a much less complex task, it leverages consumer ethics (Hainmueller et al., 2015), and many regulators may rationally choose to proceed with such simpler approaches. After observing the results of this evaluation, the Chilean government decided to scale-up the information campaign during the 2016 ban on hake fish sales, and conduct similar information campaigns for fishing bans for three other species.³⁵ While the unpredictable, low-frequency monitoring proved to be

³⁵See https://www.povertyactionlab.org/sites/default/files/documents/creating-a-culture-of-evidence-use-lessons-from-jpal-govt-partnerships-in-latin-america_english.pdf

the single-most cost-effective strategy in our evaluation, the government correctly surmised that vendors may have other second and third order subversive adaptations to audits in the long run. In contrast to an enforcement strategy which may need to be constantly revised in response to regulated agents' adaptation, the information campaign is easier to replicate and scale, especially once the government has already incurred the fixed costs of developing campaign materials.

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Online Appendix. Not for Publication

A Appendix Figures on the Research Context

A.1 Fishermen Villages

Figure A.1: Fishermen Village (Caleta)



A.2 Outdoor Markets

Figure A.2: Examples of Ferias



Figure A.3: Example of a Circuit

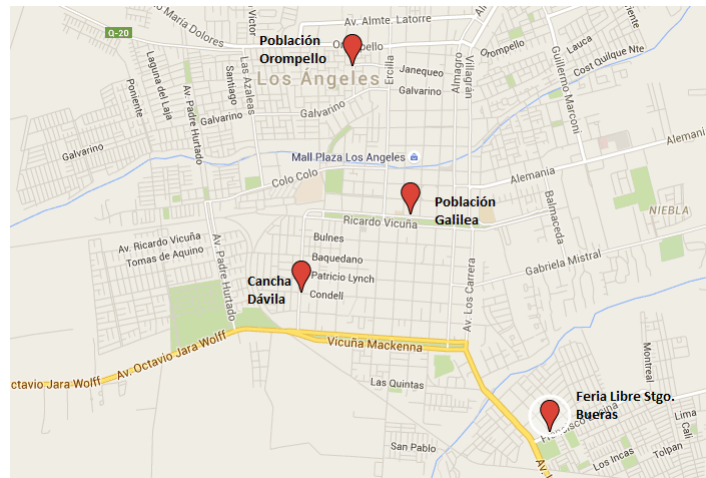
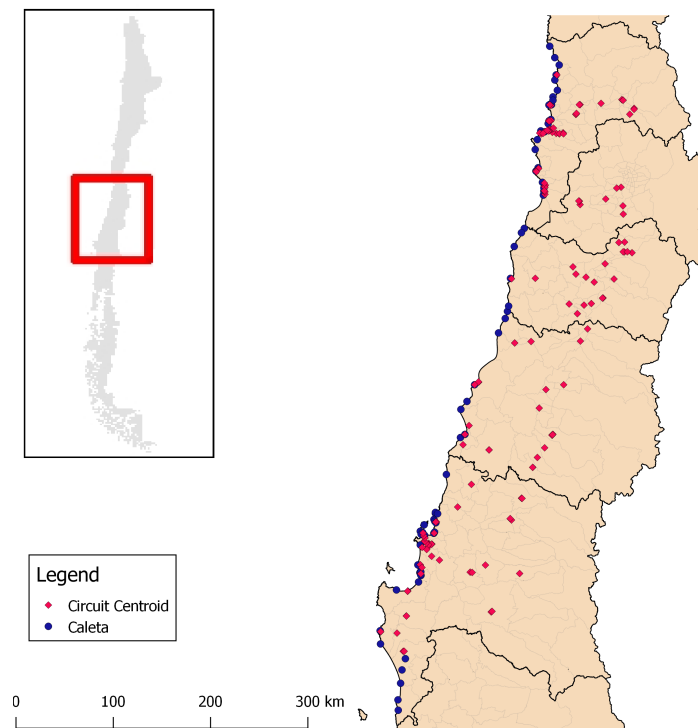


Figure A.3 maps the four ferias that compound one circuit of the city of Los Angeles, VII region.

Figure A.4: Map of Circuits and Caletas



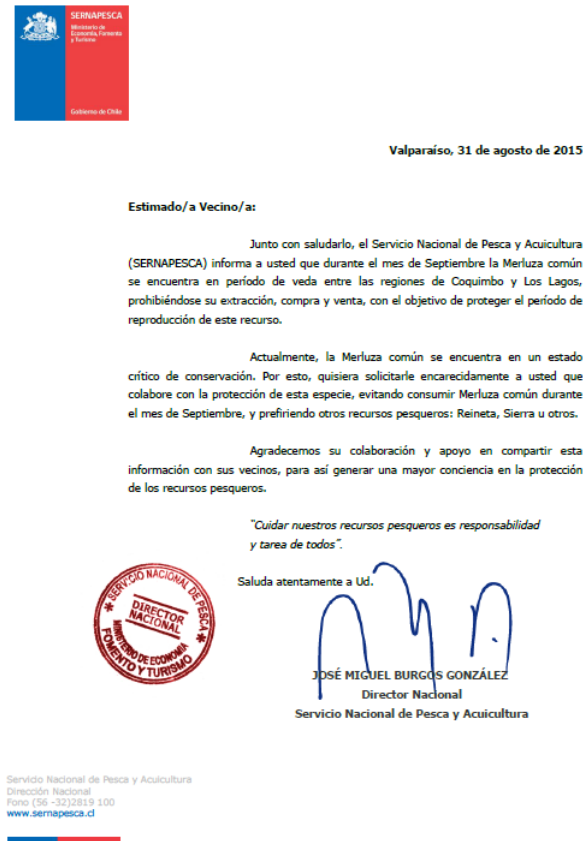
A.3 Interventions

Figure A.5: Flyers



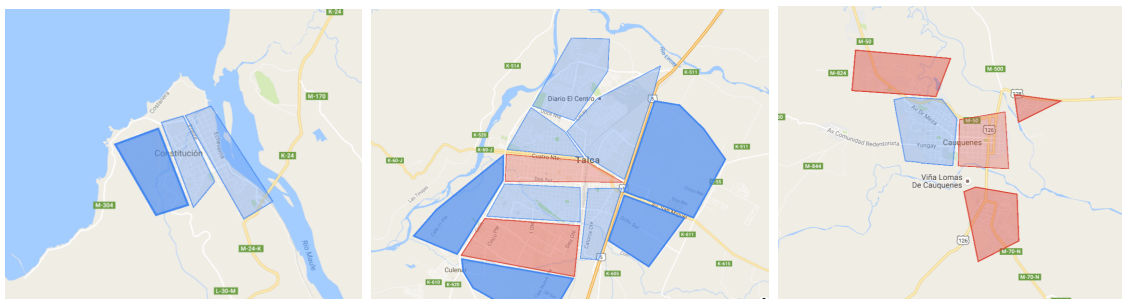
The figure A.5 shows the two types of flyers distributed during the ban period. The message of the one in the right says, “In September respect the Ban”, the one in the right says “This month respect the Ban”.

Figure A.6: Letter to Consumers



The figure A.6 shows the letter distributed to households during September 2015. The letter, signed by Sernapesca's director, informs about the September ban and the fact that hake's conservation is threatened because of overfishing.

Figure A.7: Examples of Neighborhood Treatment Assignment



The 48 most populated comunas were divided randomly into three levels of saturation: high, low and zero. Based on the level of saturation, the information campaign was assigned at the neighborhood level. The figure [A.7](#) shows the map of three different comunas: The comuna in the left didn't receive information campaign, the one in the center received low level of saturation, the one in the right received high level of saturation. In red, those neighborhoods assigned to receive the information campaign.

B Appendix Figures on Results

B.1 Adoption of Defensive Actions

Figure [B.1](#) describes the unconditional probability of selling hake defensively (either frozen or hidden) by week. The probability of selling defensively increases over time, while the overall probability of selling hake decreases substantially along the month. The conditional probability is presented in [B.1](#).

B.2 Number of Stalls

Figure [B.2](#) describes the average number of fish stalls in the first and second half of the month. It shows that the average number of fish stalls does decrease in the markets randomly assigned to the enforcement treatment, especially during the second half of September.

Figure B.1: Adoption of Defensive Actions

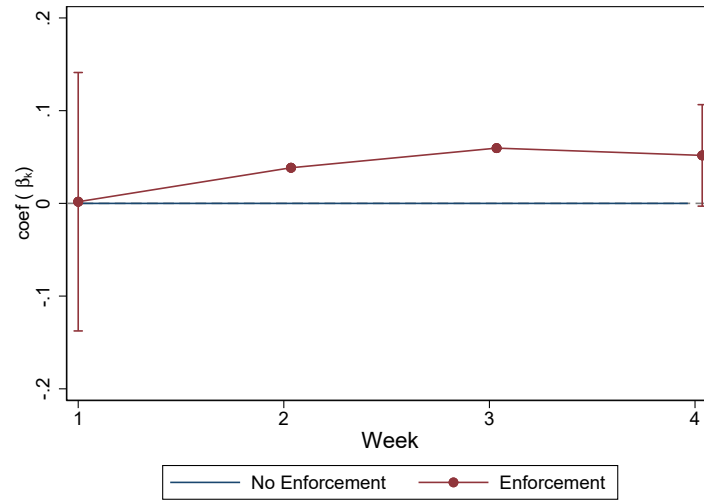
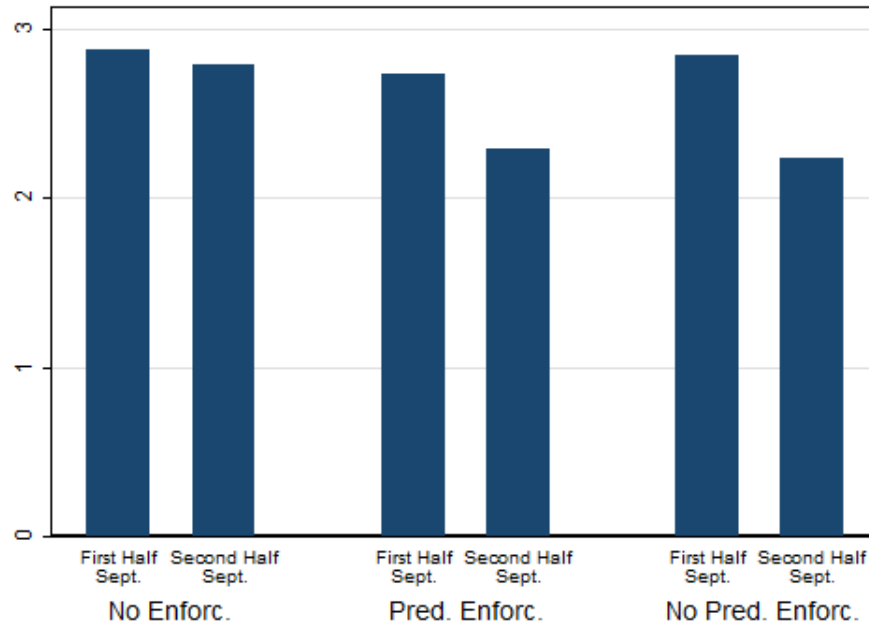


Figure B.1 describes the unconditional probability of selling hake either frozen or hidden. The coefficients were obtained from an OLS regression in which the treatment assignment interacts with weekly dummies. We include strata fixed effects and cluster at the circuit level. The "No Enforcement" category is the omitted category and includes observations assigned to the control group and the information campaign. To facilitate the interpretation, we only present the confidence intervals associated with weeks one and four.

Figure B.2: Number of stalls in Feria by Treatment Assignment

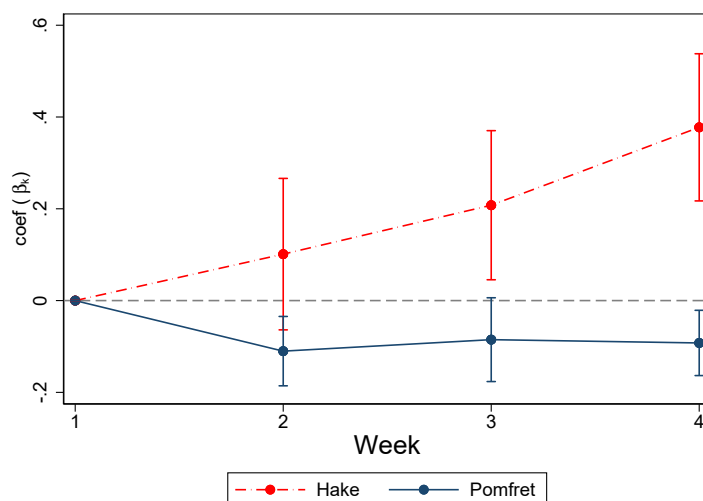


This figure shows the average number of stalls in each feria, separately for the first and the second half of the month. Markets assigned to receive enforcement showed a decrease in the number of stalls between the first and the second half of the month.

B.3 Prices

Figure B.3 describes the week-by-week evolution of (log) prices for hake and pomfret during the ban. Both rates are normalized to their levels in the first week. It shows that the price of hake increased week-to-week, throughout the ban period; the hake price in the fourth week is 40% higher than the first price. Pomfret prices fell by 10% in the second week, and that lower prices remained stable after that.

Figure B.3: Log Prices of Fish During the Ban



This figure shows the evolution of log prices for hake and pomfret, using the first week as a reference. The price of hake continuously increased over the course of September 2015. Hake was 40% more expensive by the fourth week relative to the first week. The price of pomfret decreased around 10% after the first week.

C Appendix Tables

C.1 Descriptive Statistics

C.1.1 Data Collected by Mystery Shoppers

During September 2015, the mystery shoppers interacted with fish-vendors 908 times. The table C.1 describes observable characteristics of the stalls visited and the vendors. In general, each stall was operated by one person. Mostly man, and based on mystery shoppers' guess,

47-year-old. The type of weight used informs about the formality of the stall; digital weights are more precise and expensive.

Table C.1: Fish Vendors in Ferias

Variable	Mean	SD	Min	Max	N
Number of Vendors per Stall	1.089	0.326	1	4	883
Proportion Female Fish Vendor	0.425	0.479	0	1	882
Age Vendor	47.438	10.126	19	75	876
Prices Visibly Listed	0.242	0.429	0	1	908
Type of Weight					
<i>No Weight</i>	0.262	0.440	0	1	848
<i>Mechanical Weight</i>	0.410	0.492	0	1	848
<i>Digital Weight</i>	0.308	0.462	0	1	848

Notes: This table presents observable characteristics of fish stalls visited by mystery shoppers during September 2015. The variable “Age Vendor” was not directly asked but guessed by mystery shoppers. The type of weight used to weight fish is a proxy of the level of formality of the fish stall.

The table C.2 describes the availability of different types of fish in feria stalls during the ban period. During a typical month, hake would be available in roughly 90% of stalls, however, due to the ban period (and our interventions), only 26% of stalls had hake for sale. The fish species offered in markets depend largely on the latitude of the market, i.e., markets located in the southern regions offer slightly different fish species than the stalls located in northern regions.

Table C.2: Fish Availability in Feria Stalls

Fish	Availability	Price/unit (USD)	Price/kg (USD)	Unitary Weight (kg)	N
Hake	0.263	1.08	3.81	0.284	239
Pomfret	0.684	5.27	4.99	1.057	621
Mackerel	0.124	2.05	4.16	0.492	113
Silverside	0.096	0.17	2.92	0.059	87
Salmon	0.139	7.53	9.22	0.816	126
Sawfish	0.057	6.27	6.25	1.003	52
Albacore	0.051	.	9.21	.	46
Southern Hake	0.042	7.30	5.59	1.306	38

Notes: This table presents the availability and average prices of different fish types in feria stalls during September 2015. The mystery shoppers recorded the price for each fish offered for sale in each fish-stall visited. The sale price in each stall was based on units, kilos or both. The unitary weight is estimated using the ratio of these two prices. The albacore is a considerably larger fish type (over 20 kgs) and is only sold in pieces (by kg).

C.1.2 Data Collected in the Fisherman Survey

A round of surveys to Fishermen was collected in August 2016. In total, 231 fishermen were surveyed and asked about their work, typical buyers and fishing behavior. The table C.3 describes the main variables collected in the survey.

Table C.3: Fishermen Characteristics

Variable	Mean	SD	Min	Max	N
<i>Fisherman Boat characteristics:</i>					
Boat Length (mts)	8.52	3.11	6	24	227
Boat Powered by a Motor	0.88	0.33	0	1	231
Fiberglass Boat	0.57	0.50	0	1	228
Wooden Boat	0.39	0.49	0	1	228
<i>Union Participation:</i>					
Number of Unions in the Caleta	1.67	0.90	0	3	230
Fisherman Member of a Union	0.82	0.38	0	1	230
<i>Number of Days that Goes Fishing Every Week:</i>					
Summer	5.01	1.47	1	7	226
Winter	2.25	1.14	0	7	227
<i>Number of Boats in the Caleta:</i>					
Less than 10	0.24	0.43	0	1	189
Between 10 and 30	0.22	0.41	0	1	189
Between 31 and 60	0.24	0.43	0	1	189
Between 61 and 100	0.12	0.32	0	1	189
More than 100	0.19	0.39	0	1	189
<i>Top 3 Most Captured Fishes in the Caleta:</i>					
Hake	0.56	0.50	0	1	230
Sawfish	0.24	0.43	0	1	230
Cuttlefish	0.24	0.43	0	1	230
Pomfret	0.13	0.34	0	1	230
Bass	0.10	0.30	0	1	230
<i>Usual Buyer of the Fish at the Dock:</i>					
Final Consumer	0.58	0.49	0	1	230
Feria Vendor	0.27	0.45	0	1	228
Intermediary	0.60	0.49	0	1	227

Notes: This table describes the responses to the Fishermen Survey carried out in August 2016 to 231 fishermen. On average, three fishermen were surveyed in each of the 74 caletas that operate in the four coastal regions included in our sample. The last section of the table represents the proportion that responded that *Always* or *Most of the Time* the fish was sold to these type of buyers.

C.1.3 Consumer Mobility Between Neighborhoods

The table C.4 shows the proportion of consumers treated by the information campaign depending on the location of the feria where they are surveyed. The striking fact in this table is that in high-saturation municipalities, the proportion of consumers treated with the information campaign is high, regardless of whether we found that person shopping in a feria located in a treatment neighborhood (78%) or in a control neighborhood (69%). High Information campaign saturation is therefore the effective treatment variable, and conditional on that, the specific location of the feria does not matter too much.

Table C.4: Proportion of Consumers located in Treated Neighborhoods

	Survey in Feria located in Treated Neigh		Survey in Feria located in Control Neigh	
	Prop	N	Prop	N
High Saturation Municipality	0.78	1114	0.69	389
Low Saturation Municipality	0.57	559	0.17	825
Zero Saturation Municipality	0	0	0.00	1014
Overall	0.71	1673	0.18	2228

Notes: This table shows the proportion of consumers whose households are located in neighborhoods assigned to receive the information campaign. These statistics are based on households' location reported by surveyed consumers. 71% of consumers surveyed in a feria located in a treated neighborhood live in a household located in a treated neighborhood; the remaining 29% are consumers who live in a household located in a control neighborhood. This table informs about the high consumers' mobility between neighborhoods. In fact, in high-saturation municipalities, the proportion of treated consumers is higher in both, ferias located in treatment and control neighborhoods.

C.1.4 Enforcement Implementation

This section describes the implementation of enforcement activities by Sernapesca officials. The research team planned the schedule of visits to different circuits. The execution was carried out by Sernapesca inspectors, as part of their usual tasks. The information about the actual visits was collected from the reports written by inspectors on a daily basis.³⁶ In total, 230 visits were carried out, equivalent to 659 stall-inspections in 62 circuits. Based on the inspectors' reports,

³⁶These reports contain information on the identity of the inspectors, the ferias visited that day, the number of fish stalls inspected, and whether illegal fish were detected. Importantly, the inspectors' performance does not depend on the information collected by these reports, but they rather work as a logbook of their activities. The research team periodically accessed, systematized and digitized this information.

illegal hake was detected in 11% of inspected stalls. This number is three times smaller than what our secret shoppers observed in markets.

Table C.5 describes the implementation of enforcement visits relative to the treatment assignment. The average number of visits in different treatment arms is slightly smaller than the original plan, this gap is explained by “contingencies” that obstructed the expected routine, and possibly, some under-reporting on behalf of inspectors. Also, a few visits were noted in Control group markets; these were generally markets located near Sernapesca regional offices, that officials unpromptedly visited.

Table C.5: Implementation of the Interventions

Treatment Assignment	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Visits	Number of Different Days of the Week Visited	Circuit Visited at Least Once	At Least One Visit		N
				Number of Visits	Number of Different Days of the Week Visited	
No Enforcement	0.39	0.30	0.22	1.80	1.40	23
Enforcement	2.53	1.49	0.69	3.68	2.18	83
Low Freq. and Unpredictable Sch.	1.48	1.14	0.62	2.39	1.83	29
High Freq. and Unpredictable Sch.	5.00	2.80	0.87	5.77	3.23	15
Low Freq. and Predictable Sch.	1.30	0.85	0.65	2.00	1.31	20
High Freq. and Predictable Sch.	3.47	1.68	0.68	5.08	2.46	19

This table reports the unconditional mean of visits to circuits in different treatment arms. Column 4 presents the average number of visits conditional on receiving at least one visit. The difference between the number of visits in circuits assigned to High-Intensity Enforcement relative to Low Intensity is statistically significant at 1%.

The experimental design varied two margins of enforcement deployment; the frequency, and the predictability of the visits. The implementation of frequency variations can be evaluated based on the number of visits to each circuit. The predictability variation can be assessed based on the number of different days of the week in which visits were carried out. If predictable enforcement was implemented correctly, it should repeat the days of visits every week, so we should expect fewer days of the week visited. On average, circuits assigned to enforcement received 2.53 visits in 1.49 different days of the week. Both treatment variations generate significant differences in the relevant margins: High-frequency circuits received substantially more visits than low-frequency circuits. Unpredictable circuits were visited in more days of the

week than predictable enforcement. The columns 4 and 5 compare these margins conditioning on receiving at least one visit.

C.2 Balance

We did not conduct a full baseline survey, but had access to municipality administrative data and weather data with which we could check balance across treatment arms. The table C.6 shows balance tests across the main treatment arms. Tables C.7 and C.8 also show balance tests with respect to the enforcement predictability and frequency sub-treatments.

Overall, the various treatment arms appear well balanced in terms of important socio-economic and weather characteristics (e.g. poverty rate, rainfall). The joint test F-statistics of all variables are insignificant for different treatment arms. The delinquency rate (i.e., per-capita police cases for major offenses) is lower in municipalities assigned to receive the information campaign relative to the control group. The regressions reported below control for this variable, but we have verified that the reported treatment effects are not sensitive to adding this control.

C.2.1 Balance Tables

The table C.6 presents the balance of relevant characteristics across different treatment arms. These variables include market's observable characteristics, socioeconomic characteristics of the municipality and weather information of the day of the visit by a mystery shopper. The columns 1, 2, 4 and 6 present the mean and SD of these variables for different treatment arms. The columns 3, 5 and 7 compare the difference relative to the control group as well as its p-value. Finally, joint significance tests are also reported in the last two columns. The tables C.7 and C.8 present the same estimates but decomposing by the enforcement variations: predictability and frequency.

Table C.6: Randomization Balance on Observables

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Control	Info Campaign		Enforcement		Enforc. and Info Camp.	
	Mean	Mean	Diff	Mean	Diff	Mean	Diff
Indicator Fixed Stalls	0.573 (0.497)	0.644 (0.484)	0.083 [0.740]	0.489 (0.501)	-0.069 [0.574]	0.509 (0.501)	0.013 [0.920]
Distance to Closest Caleta (kms)	16.507 (25.082)	11.572 (9.503)	3.465 [0.516]	14.863 (22.626)	-3.606 [0.386]	26.425 (29.037)	2.245 [0.682]
Poverty Rate Municipality	19.006 (4.780)	17.567 (3.313)	-2.148 [0.244]	18.026 (5.483)	-1.079 [0.412]	16.734 (7.549)	-0.316 [0.851]
Av. Monthly Income Municipality (USD)	791.767 (149.808)	875.475 (172.858)	18.446 [0.846]	790.514 (140.334)	-2.506 [0.953]	830.683 (139.251)	20.464 [0.673]
Delinquency Rate Municipality	0.038 (0.015)	0.029 (0.002)	-0.013 [0.016]	0.036 (0.015)	-0.001 [0.835]	0.034 (0.009)	-0.004 [0.480]
Rain Indicator	0.290 (0.456)	0.133 (0.344)	-0.124 [0.455]	0.178 (0.383)	-0.114 [0.318]	0.142 (0.349)	-0.122 [0.301]
Average Temperature (Celsius)	12.200 (2.281)	12.126 (2.087)	0.081 [0.942]	11.993 (2.021)	-0.192 [0.797]	11.936 (2.196)	-0.688 [0.346]
Joint Significance							
<i>F statistic</i>			0.609		1.094		0.816
<i>p-value</i>			0.747		0.371		0.575

Notes: This table reports characteristics of circuits included in our sample across treatment arms. The columns (1), (2), (4) and (6) show the mean and the standard deviation for the control and treatment groups. The columns (3), (5) and (7) show the coefficient on treatments from regressions of each characteristic on treatments and strata fixed effects, clustering standard errors at the circuit level. The p-values are reported in brackets. The socio-economic characteristics are aggregated at Municipality level. These variables should be interpreted as the characteristics of the Municipality where the circuit is located. Also, this table reports weather information of the day that different circuits were visited by mystery shoppers. Finally, joint significance test statistics: F statistic and p-values, for all variables on each treatment arm are reported in the last two rows of the table.

Table C.7: Randomization Balance: Enforcement Predictability

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Control	Info Campaign		Enforc: Predictable Schedule		Enforc: Unpredictable Schedule	
	Mean	Mean	Diff	Mean	Diff	Mean	Diff
Indicator Fixed Stalls	0.573 (0.497)	0.644 (0.484)	0.080 [0.749]	0.396 (0.490)	-0.148 [0.270]	0.575 (0.495)	0.045 [0.710]
Distance to Closest Caleta (kms)	16.507 (25.082)	11.572 (9.503)	3.822 [0.479]	17.286 (23.417)	1.704 [0.678]	20.487 (27.384)	-4.926 [0.333]
Poverty Rate Municipality	19.006 (4.780)	17.567 (3.313)	-2.173 [0.240]	17.130 (6.185)	-1.739 [0.212]	17.890 (6.450)	-0.075 [0.960]
Av. Monthly Income Municipality (USD)	791.767 (149.808)	875.475 (172.858)	18.828 [0.843]	809.457 (153.747)	-1.068 [0.981]	801.805 (130.506)	9.235 [0.834]
Delinquency Rate Municipality	0.038 (0.015)	0.029 (0.002)	-0.013 [0.015]	0.036 (0.015)	-0.002 [0.648]	0.035 (0.012)	-0.001 [0.782]
Rain Indicator	0.290 (0.456)	0.133 (0.344)	-0.126 [0.446]	0.117 (0.322)	-0.158 [0.162]	0.202 (0.402)	-0.080 [0.488]
Average Temperature (Celsius)	12.200 (2.281)	12.126 (2.087)	0.076 [0.946]	12.058 (2.057)	-0.170 [0.824]	11.904 (2.107)	-0.491 [0.498]
Joint Significance							
<i>F statistic</i>			0.609		1.954		1.717
<i>p-value</i>			0.747		0.067		0.111

Notes: This table reports characteristics of circuits included in our sample across treatment arms. The columns (1), (2), (4) and (6) show the mean and the standard deviation for the control and treatment groups. The columns (3), (5) and (7) show the coefficient on treatments from regressions of each characteristic on treatments and strata fixed effects, clustering standard errors at the circuit level. The p-values are reported in brackets. The socio-economic characteristics are aggregated at Municipality level. These variables should be interpreted as the characteristics of the Municipality where the circuit is located. Also, this table reports weather information of the day that different circuits were visited by mystery shoppers. Finally, joint significance test statistics: F statistic and p-values, for all variables on each treatment arm are reported in the last two rows of the table.

Table C.8: Randomization Balance: Enforcement Frequency

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variable	Control	Info Campaign		Enforc: High Frequency		Enforc: Low Frequency	
	Mean	Mean	Diff	Mean	Diff	Mean	Diff
Indicator Fixed Stalls	0.573 (0.497)	0.644 (0.484)	0.098 [0.696]	0.372 (0.484)	-0.164 [0.211]	0.591 (0.492)	0.039 [0.753]
Distance to Closest Caleta (kms)	16.507 (25.082)	11.572 (9.503)	4.009 [0.451]	16.224 (23.309)	-5.302 [0.330]	21.269 (27.292)	0.600 [0.885]
Poverty Rate Municipality	19.006 (4.780)	17.567 (3.313)	-1.975 [0.279]	16.242 (7.011)	-2.256 [0.101]	18.564 (5.576)	0.140 [0.922]
Av. Monthly Income Municipality (USD)	791.767 (149.808)	875.475 (172.858)	17.128 [0.857]	821.315 (142.660)	23.237 [0.587]	792.779 (138.930)	-8.734 [0.840]
Delinquency Rate Municipality	0.038 (0.015)	0.029 (0.002)	-0.013 [0.014]	0.036 (0.013)	-0.000 [0.973]	0.035 (0.014)	-0.003 [0.547]
Rain Indicator	0.290 (0.456)	0.133 (0.344)	-0.128 [0.439]	0.193 (0.396)	-0.077 [0.491]	0.143 (0.350)	-0.143 [0.215]
Average Temperature (Celsius)	12.200 (2.281)	12.126 (2.087)	0.050 [0.964]	12.127 (2.204)	-0.192 [0.799]	11.853 (1.984)	-0.447 [0.545]
Joint Significance							
<i>F statistic</i>			0.609		2.016		2.090
<i>p-value</i>			0.747		0.058		0.049

Notes: This table reports characteristics of circuits included in our sample across treatment arms. The columns (1), (2), (4) and (6) show the mean and the standard deviation for the control and treatment groups. The columns (3), (5) and (7) show the coefficient on treatments from regressions of each characteristic on treatments and strata fixed effects, clustering standard errors at the circuit level. The p-values are reported in brackets. The socio-economic characteristics are aggregated at Municipality level. These variables should be interpreted as the characteristics of the Municipality where the circuit is located. Also, this table reports weather information of the day that different circuits were visited by mystery shoppers. Finally, joint significance test statistics: F statistic and p-values, for all variables on each treatment arm are reported in the last two rows of the table.

C.3 Additional Tables

Table C.9 shows all coefficients from estimating regression 8, including coefficients for the pre-intervention period in the first three rows. The bottom three rows that add an interaction with “ $\times$ Post” capture the effect of the interventions. While the markets randomly assigned to various arms appear balanced based on pre-existing administrative data, the first three rows allow us to explore whether there were statistical differences in vendor behavior between treatment and control groups in the first week of September, immediately before our interventions were launched. While these effects are not statistically significant, the coefficients are sizable, indi-

cating some imbalance in that first week. So our main results rely on a difference-in-difference estimate (the interactions with “ $\times$ Post”), and significant differences between markets emerge after the interventions are launched. But the reliance on a difference-in-difference estimate implies that the exact definition of the “post” period becomes critical. Our main specification uses September 8 because this was the first market day after the first Monday of the month (Sept 7). Ferias do not operate on Mondays, so September 8 is a clear and natural pre/post boundary. A second reason we chose September 8 is that exactly half the ferias received their first enforcement visit before September 8, and half the ferias received it after. Choosing the “median date” of the first enforcement visit felt like a natural cutoff.

Table C.10 explores the sensitivity of our estimates to changing the starting date of the “Post Intervention” period. The results are relatively stable across specifications, and the effect sizes based on our chosen Sept 8 cutoff falls between the effect sizes from a Sept 6 vs a Sept 9 cutoff (Monday, Sept 7 is omitted because ferias don’t operate that day).

Table C.9: Treatment Effects on Hake Sales

VARIABLES	(1)	(2)
	Fresh, Visible Hake	Any Hake Available (Hidden, Frozen, Visible)
Information Campaign Only	0.080 (0.056)	0.029 (0.058)
Enforcement Only	0.114 (0.070)	0.092 (0.060)
Information Campaign and Enforcement	0.078 (0.070)	0.100 (0.065)
Information Campaign Only $\times$ Post	-0.133 (0.066)	-0.131 (0.074)
Enforcement Only $\times$ Post	-0.178 (0.082)	-0.130 (0.089)
Info Campaign and Enforcement $\times$ Post	-0.179 (0.074)	-0.139 (0.094)
Change in Dep. Var. in Control Group During Intervention Period	-0.21	-0.36
N	901	901

Notes: This table reports the effect of each treatment arm on the availability of illegal hake fish. The variable Fresh Hake indicates when the hake was available fresh. Hake available indicates when was possible to buy fish in any form. The table reports marginal effects from a Probit regression. Other controls are included: municipality characteristics, strata fixed effects and the average level of the outcome variable in pre-intervention period. We control for pre-treatment values for the outcome variables in addition to the treatment indicator, because not all markets were visited in pre-intervention period. Robust standard errors clustered by circuit (the unit of randomization) in parentheses.

Table C.10: Effects varying the post-intervention period

	Any Hake Available		
	Post ≥ 6	Post ≥ 8 (current)	Post ≥ 9
Info Campaign Only	-0.050 (0.117)	-0.131 (0.074)	-0.169 (0.086)
Enforcement Only	-0.048 (0.093)	-0.130 (0.089)	-0.065 (0.089)
Info Campaign and Enforcement	-0.058 (0.102)	-0.139 (0.094)	-0.189 (0.093)
Change in Dep Var in Control Group			
During Intervention Period	-0.78	-0.36	-0.29
N	901	901	901

Notes: This table reports the effect of each treatment arm on the availability of illegal hake, using different definitions of the “Post Intervention” period. The “Post ≥ 7 ” was omitted because Sept 7th was a Monday and ferias usually don’t operate on Mondays. The variable “Any Hake Available” indicates whether it was possible to buy fish in any form. The table reports marginal effects from a Probit regression of the “ $\times$ Post” terms in the main regression equation. Other controls are included: municipality characteristics, strata fixed effects, and the average level of the outcome variable in the pre-intervention period. Robust standard errors clustered by circuit (the unit of randomization) in parentheses.

The table below reproduces Table 3 in the paper but includes the control “total number of visits the circuit has received until the day before the data was collected by the mystery shopper.”

The coefficient of the added variable (not presented in the table) is positive, although small and insignificant (coef= 0.04, SE=0.07), i.e., increasing the frequency of enforcement leads to higher sales in the short run due to learning and adaptation. Including this control, if any, exacerbates the estimated gap between Unpredictable and Predictable enforcement (and tightens the standard errors) reinforcing the paper’s main findings. This is not surprising: the Unpredictable enforcement group received more visits, so controlling for the number of enforcement visits removes the attenuation effect of receiving more visits.

C.3.1 Robustness: Cluster Bootstrap p-values

Table C.12 reports the coefficients from Table 1 together with “delta-method” standard errors and p-values, alongside bootstrap p-values computed using 1,000 replications. We implement this bootstrap exercise to assess the robustness of our inference in a setting with a relatively

Table C.11: Treatment Effects Controlling for the Number of Past Visits

VARIABLES	(1)	(2)
	Any Hake Available (Fresh, Visible, Hidden or Frozen)	
Info campaign only	-0.134 (0.069)	-0.134 (0.069)
Enforcement on Predictable Sch.	-0.069 (0.077)	
Enforcement on Unpredictable Sch.	-0.209 (0.090)	
High Frequency Enforcement		-0.085 (0.090)
Low Frequency Enforcement		-0.166 (0.087)
Change in Dep Var in Control		
During Intervention	-0.36	-0.36
N	901	901

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the coefficient corresponding to the interaction term $T_c \times Post_t$ for each treatment. To retain statistical power, the cells “Enforcement only” and “Enforcement + Info Campaign” from Table 1 are combined under “Enforcement” and then sub-divided by schedule predictability (column 1), or intensity (column 2). So these coefficients should be interpreted as the average effects of enforcement when half the sample is also exposed to the information campaign. Note that we previously find evidence of null interaction effect between enforcement and info campaign (Muralidharan et al., 2019). Column 1 includes a dummy for the intensity sub-treatment, and column 2 includes a dummy for the predictability sub-treatment, but those coefficients are not shown. Each regression controls for the dependent variable in pre-intervention period, strata fixed effects and municipality characteristics. Probit regression marginal effects are reported. Robust standard errors clustered by circuit in parentheses..

small number of clusters, where asymptotic approximations may be less reliable. Bootstrap p-values are generally somewhat larger than their delta-method counterparts, consistent with more conservative inference that accounts for finite-sample variability. Importantly, these differences are modest and do not affect the substantive conclusions. For the “fresh, visible hake” outcome (Table 1, column 1), the “Info Campaign and Enforcement” arm remains statistically significant under the bootstrap ($p = 0.036$), while the Enforcement-only arm yields a bootstrap p-value of 0.066 (vs. 0.028), and the Info Campaign-only arm (the smallest treatment group) yields 0.105 (vs. 0.044). For the “Any Hake” outcome (Table 1, column 2), bootstrap and delta-method p-values are closely aligned across all treatment arms.

Table C.12: Robustness: Bootstrap p-values

Outcome	Treatment arm	N	AME	SE	<i>p</i> Delta	<i>p</i> Bootstrap
Fresh Hake	Info campaign only	901	-0.132	(0.066)	0.044	0.105
Fresh Hake	Enforcement only	901	-0.179	(0.081)	0.028	0.066
Fresh Hake	Info Campaign and Enforc.	901	-0.179	(0.073)	0.014	0.036
Any Hake	Info campaign only	901	-0.131	(0.074)	0.075	0.117
Any Hake	Enforcement only	901	-0.131	(0.088)	0.139	0.138
Any Hake	Info Campaign and Enforc.	901	-0.139	(0.094)	0.138	0.146

Notes: All regressions include baseline controls, covariates, strata fixed effects, and treatment indicators. AME = average marginal effect coefficient. SE in parentheses, computed via the delta method. *p-Delta*: two-sided p-value from delta-method z-statistic. *p-Bootstrap*: pairs cluster bootstrap p-value (Cameron et al., 2008), percentile-t method, applied to the marginal effect difference directly.

C.3.2 Disaggregation of treatment effects

We complement the pooled specifications presented in Table 3 with a disaggregated specification, reported in Table C.13, which provides the complete set of treatment effects across all six experimental groups. These estimates allow for an assessment of heterogeneity across enforcement designs and the information campaign. In line with the main results, the largest reductions in illegal hake availability arise under enforcement with an unpredictable, low-frequency schedule, while other enforcement arms yield smaller estimated effects, and the information campaign alone also reduces illegal hake availability. As expected, given the smaller number of circuits per cell, the estimates are less precise than the pooled specifications, but the overall patterns are consistent with those emphasized in the main text.

C.4 Adaptation to the Schedule of Visits

Our model suggests that vendors would learn and adapt to the pattern of visits. We use the daily data over the course of the September ban to study how the vendors adjust to the visit patterns they observe. Table C.14 shows how selling decisions differ in the second half of the month, depending on how concentrated the earlier inspections were in specific ferias and on specific days of the week (DOWs). We control for the total number of visits in studying the effects of “targeting” only one feria or day-of-week. We find that auditing *different* ferias on *different* DOW reduces hake sales by an extra 9 percentage points (p-val<0.01) in the second

Table C.13: Disaggregated treatment effects

	(1) Any Hake Available
Information Campaign only	-0.135 (0.072)
Enforcement: Predictable, Low Frequency	-0.099 (0.090)
Enforcement: Predictable, High Frequency	-0.019 (0.081)
Enforcement: Unpredictable, Low Frequency	-0.198 (0.086)
Enforcement: Unpredictable, High Frequency	-0.140 (0.114)
Change in Dependent Variable in Control	
During Intervention	-0.36
N	901

Notes: This table presents the coefficients corresponding to the interaction term $T_c \times Post_t$ for each treatment arm, without pooling across sub-treatments. Each coefficient should be interpreted as the treatment effect for a specific combination of enforcement predictability and frequency (and the information campaign). Compared with Table 3, this specification provides the fully disaggregated set of effects across all experimental cells. Given the smaller number of circuits per cell, these estimates are less precise, which motivates the pooled specifications in the main text. Each regression controls for the dependent variable in the pre-intervention period, strata fixed effects, and municipality characteristics. Probit regression marginal effects are reported. Robust standard errors clustered by circuit in parentheses.

half of the month, relative to targeting all visits at the same feria.

Table C.15 studies vendors’ decisions to sell hake in the *non-targeted* feria in the second half of the month. This is a circuit-fixed effects regression, so the coefficient “DOW not visited $\times$ 2nd half” shows the same vendor’s decision to sell on a weekday in which he did not experience a visit relative to another weekday when he did. We see that the hake selling in the second half of the month was higher in ferias and DOWs that did *not* receive enforcement relative to the ones that did.

We consider this evidence as only suggestive and placed it in the appendix, because Ser-napesca chose which feria to visit within each circuit partly based on logistical considerations, and this cannot be treated as random. Indeed, the un-interacted terms in the regression show some differences (in the opposite direction!) across ferias within the circuit in the first part of the month.

Table C.16 shows that this same effect is not only seen in the propensity to sell, but also in

Table C.14: Hake Available based on the Number of different ferias and Days of the Week visited

VARIABLES	(1)	(2)	(3)
	Any Hake Available		
N Ferias Visited	0.041 (0.030)		0.039 (0.033)
N Ferias Visited $\times$ Second Half	-0.091 (0.023)		-0.081 (0.035)
N DOWs Visited		0.030 (0.055)	0.014 (0.056)
N DOWs Visited $\times$ Second Half		-0.098 (0.073)	-0.037 (0.077)
Change Dep Var First - Second Half	-0.31	-0.31	-0.31
N	906	906	906

Notes: This table studies how the probability of selling hake depends on the number of different days of the week (DOWs) and the number of different ferias that a circuit got visited during the ban. The observations are divided between the first and second half of the month to retain enough statistical power; other pre-post decompositions produce similar results. The table presents OLS coefficients of the relevant variables. Since DOWs and N Ferias are positively correlated, the columns 1 and 2 run them separately. Column 3 includes both variables and interactions. Each regression controls for “Second Half”, the total number of visits, and the interaction of the two variables. Also, they control for the dependent variable in the pre-intervention period, and strata fixed effects and municipality characteristics. Also, each regression controls for treatment assignment. Robust standard errors clustered by circuit in parentheses.

the number of stalls that vendors choose to continue to operate in the second half of the month. The interacted coefficients “... $\times$ second half” show that more stalls disappear entirely in the second half of the month in the targeted ferias operating on targeted DOWs, relative to the non-targeted. The effect is larger when the vendors operate in more than two ferias, because those are the circuits where vendors have more options to adjust and displace sales across days of week.

The preceding tables explain why predictable enforcement is less effective. As our theoretical model lays out; vendors learn from the pattern of targeted ferias and targeted days of week, and adjust to sell more on non-targeted days.

Table C.15: Hake Sale across DOWs and Ferias within Circuit

	(1)	(2)	(3)	(4)	(5)	(6)
	Any Hake Available		Circuits that Rotate between More Than 2 Ferias		Any Defensive Action	
VARIABLES	Full Sample		Full Sample		Full Sample	
Feria Not Visited	-0.209		-0.299		-0.053	
	(0.157)		(0.182)		(0.106)	
Feria Not Visited $\times$ Second Half	0.058		0.267		-0.028	
	(0.086)		(0.130)		(0.046)	
DOW Not Visited		-0.252		-0.303		-0.034
		(0.131)		(0.161)		(0.084)
DOW Not Visited $\times$ Second Half		0.192		0.286		-0.046
		(0.104)		(0.139)		(0.072)
Change Dep Var First - Second Half	-0.31	-0.31	-0.39	-0.39	-0.01	-0.01
N	906	906	218	218	906	906

Notes: This table examines whether the behavior of the vendors varied across days of the week or ferias. It shows the OLS coefficient of dummy variables indicating whether the observation was collected in a feria or day that was not visited by Sernapesca officials during the ban. The observations are divided between the first and second half of the month to retain enough statistical power. Other pre-post decompositions produce similar results. These regressions include circuit fixed effects, so the coefficients capture within circuit variation. The columns (3) and (4) restrict the analysis only to circuits that rotate between more than two ferias. It controls for “Second Half” and weather covariates. Robust standard errors clustered by circuit in parentheses.

Table C.16: Number of Stalls across DOWs and Ferias within Circuit

	(1)	(2)	(3)	(4)
	Number of Stalls		Circuits that Rotate between More Than 2 Ferias	
VARIABLES	Full Sample		Full Sample	
Feria Not Visited	-0.510		-0.956	
	(0.320)		(0.368)	
Feria Not Visited $\times$ Second Half	0.354		1.331	
	(0.464)		(0.771)	
DOW Not Visited		-0.024		-0.241
		(0.220)		(0.312)
DOW Not Visited $\times$ Second Half		0.317		0.995
		(0.330)		(0.450)
Change Dep Var First - Second Half	0.03	0.03	0.20	0.20
N	374	374	104	104

Notes: This table examines whether the number of stalls selling fish varied across days of the week or ferias. It shows the OLS coefficient of dummy variables indicating whether the observation was collected in a feria or day that was not visited by Sernapesca officials during the ban. Every observation correspond to a feria and are divided between the first and second half of the month to retain enough statistical power. Other pre-post decompositions produce similar results. These regressions include circuit fixed effects, so the coefficients capture within circuit variation. The columns (3) and (4) restrict the analysis only to circuits that rotate between more than two ferias. It controls for “Second Half” and weather covariates. Robust standard errors clustered by circuit in parentheses.

Table C.17: Inspector’s Past Experience in Enforced Circuits

	Neither Inspector visited this circuit before	At Least 1 Inspector had not visited this circuit before	N
Visit 1	1.00	1.00	63
Visit 2	0.47	0.78	55
Visit 3	0.30	0.68	37
Visit 4	0.15	0.62	26
Visit 5	0.00	0.50	20
Visit 6	0.17	0.17	12
Visit 7	0.20	0.20	5
Visit 8	0.00	0.25	4
Visit 9	0.00	0.50	2
Visit 10+	0.00	0.00	6

Notes: This table describes the proportion of cases in which both or one inspector is “unknown” to the circuit-vendors (i.e., is the first time they visit that circuit), depending on the order of the visit. Mechanically, visit 1 (first visit) only contains “unknown” inspectors, since the pool of inspectors is limited, this proportion decreases in the 2nd, 3rd, and 4rd visit

C.5 Exit of Stalls Correction

Our main results are based on the information gathered by mystery shoppers from the operative stalls at the moment of the visit, which does not capture the fact that the “missing” stalls are not selling hake. To correct for this issue we identify the average number of stalls per circuit/visit before and after the interventions. The comparison between these two averages informs about the number of “missing” stalls per circuit.³⁷ The number of stalls observed by mystery shoppers in every visit in the post treatment period is increased by computed number of missing stalls. The added observations have zero fish available.^{38 39}

³⁷We allow the number of missing stalls to be non-integer, and negative if the number of stalls increased.

³⁸If the number “missing” stalls is negative: the number of stalls observed by mystery shoppers in every visit in the pre-treatment period is increased by that number.

³⁹Since we allow the number “missing” stalls to be non-integer, we add a random noise that distributes uniform between -0.5 and 0.5, and then, the sum of the “missing” number and the noise is rounded to the closest integer. This correction makes the expansion more representative of the right (possibly non-integer) number.

Table C.18: Treatment Effect on Hake Availability Correcting for the Exit of Stalls

VARIABLES	(1) Fresh, Visible Hake	(2) Any Hake Available (Fresh-Visible, Hidden or Frozen)
Panel A: Main Specification		
Info Campaign Only	-0.118 (0.060)	-0.115 (0.065)
Enforcement Only	-0.190 (0.082)	-0.141 (0.091)
Info Campaign and Enforcement	-0.156 (0.084)	-0.130 (0.104)
Panel B: Variation in Predictability of Enforcement		
Info Campaign Only	-0.111 (0.062)	-0.121 (0.064)
Enforcement on Predictable Schedule	-0.091 (0.073)	-0.061 (0.087)
Enforcement on Unpredictable Schedule	-0.246 (0.089)	-0.197 (0.100)
Panel C: Variation in Frequency of Enforcement		
Info Campaign Only	-0.113 (0.062)	-0.121 (0.064)
High Frequency Enforcement	-0.086 (0.092)	-0.092 (0.101)
Low Frequency Enforcement	-0.184 (0.086)	-0.148 (0.095)
Change in Dep Var in Control During Intervention	-0.17	-0.28
Covariates	Yes	Yes
Baseline Control	Yes	Yes
N	1014	1014

Notes: This table presents the coefficient corresponding to the interaction term $T_c \times Post_t$ for each treatment correcting for the exit of stalls. The increase in the number of observations relative to results presented earlier is due to the fact that the correction is done by adding the “missing” stalls (calculated comparing the number of stalls per circuit before and after the interventions). The panel A describes the same specification presented in Table 1. Panels B and C follow the same specification than Table 3. Probit regression marginal effects are reported. Robust standard errors clustered by circuit in parentheses.

C.6 Treatment Effects Six Months After the Ban Period

Table C.19 describe the answers to the consumer survey carried out in March 2016. The survey had the same format as previous surveys; it asked about general consumption behavior based on a list of items, including hake. Even though the survey was carried out in an off-ban period, consumers assigned to the information campaign tend to report less hake consumption.

Table C.19: Hake Purchases Reported by Consumers in March 2016 (Outside Ban Period)

VARIABLES	(1) Purchased Hake last month	(2) Number Times Hake Purchased	(3) Usually Purchase Hake
Information Campaign Only	-0.133 (0.106)	-0.419 (0.234)	-0.121 (0.100)
Enforcement Only	0.018 (0.078)	-0.106 (0.205)	0.093 (0.069)
Info Campaign and Enforcement	-0.017 (0.081)	-0.197 (0.246)	0.017 (0.077)
Mean Dep Var Control	0.59	1.19	0.58
N	3652	3630	3652

Notes: This table presents the effect of different treatment arms on the reported consumption of hake fish by consumers based on the round of surveys collected in March 2016. Columns 1 and 3 show marginal effects from Probit regressions. Column 2 shows the marginal effects from a Poisson regression because the dependent variable is a count data. These regressions control for propensity to purchase other types of fishes and other covariates. Standard errors are clustered based on the circuit where the survey was collected.

C.7 Alternative Definition Information Campaign Treatment

Tables C.20 and C.21 present the main results using a different definition of the Information Campaign treatment: The variable “Information campaign” indicates whether the observations were collected by mystery shoppers in ferias located in treated neighborhoods - regardless of the level of saturation of the municipality. This definition does not include possible information spill-overs between neighborhoods within municipalities assigned to receive information.

Table C.20: Treatment Effect on Hake Availability

VARIABLES	(1)	(2)
	Fresh, Visible Hake	Any Hake Available (Fresh-Visible, Hidden or Frozen)
Information Campaign Only	-0.082 (0.064)	-0.070 (0.071)
Enforcement Only	-0.157 (0.079)	-0.101 (0.094)
Info Campaign and Enforcement	-0.169 (0.079)	-0.121 (0.094)
Change in Dep Var in Control Markets During Intervention	-0.21	-0.36
N	901	901

Notes: This table reports the effect of each treatment arm on the availability of illegal hake fish. The variable “Info Campaign” indicates if the feria where the data was collected is located in a neighborhood assigned to receive the information campaign. Probit Marginal effects of the interactions $T_c \times Post_t$ are reported. Robust standard errors are clustered by circuit and presented in parentheses.

Table C.21: Treatment Effect on Hake Sales by Enforcement Strategy

VARIABLES	(1)	(2)
	Any Hake	Available
Info Campaign Only	-0.073 (0.071)	-0.073 (0.071)
Enforcement on Predictable Schedule	-0.036 (0.089)	
Enforcement on Unpredictable Schedule	-0.169 (0.099)	
High Freq. Enforcement		-0.049 (0.101)
Low Freq Enforcement		-0.140 (0.095)
Change in Dep Var in control Markets During Intervention	-0.36	-0.36
N	901	901

Notes: This table reports the effect of each treatment arm on the availability of illegal hake fish. The first column includes compares the effectiveness of predictable vs unpredictable enforcement. The second column divides enforcement depending on its intensity. Each regression controls for the dependent variable in pre-intervention period, strata fixed effects and municipality characteristics. The variable “Info Campaign” indicates if the feria where the data was collected is located in a neighborhood assigned to receive the information campaign. Probit Marginal effects of the interactions $T_c \times Post_t$ are reported. Robust standard errors are clustered by circuit and presented in parentheses.

D Appendix: Effectiveness and Complementarities Between Demand- and Supply-Side Policies

In this section, we present a general model that helps clarify the forces that shape both the magnitude of each policy’s effect and the extent to which demand- and supply-side interventions reinforce—or attenuate—one another.

D.1 General Model

At each period, a vendor decides whether to sell hake. The government has two policy instruments to curb the sale of hake: intensifying enforcement ($E = 1$) and informing consumers about the ban ($I = 1$). Enforcement (E) increases the probability of detection. The information campaign (I) reduces consumer valuation for hake, hence the rents the vendors get from selling.

We are interested in the extent to which carrying out the two interventions jointly creates a larger (or smaller) effect than the sum of the two interventions. To do so, we introduce some notation:

$$\mathbf{P}_E = \mathbf{P}_0 + \frac{d\mathbf{P}}{dE} \quad (12)$$

$$\mathbf{P}_I = \mathbf{P}_0 + \frac{d\mathbf{P}}{dI} \quad (13)$$

$$\mathbf{P}_{EI} = \mathbf{P}_0 + \frac{d\mathbf{P}}{dE} + \frac{d\mathbf{P}}{dI} + \Gamma \quad (14)$$

where $\mathbf{P}_0$, $\mathbf{P}_E$ and $\mathbf{P}_I$ are the probabilities of selling hake after no, E and I , respectively. The term $\mathbf{P}_{EI}$ denotes the probability of selling after E and I are carried out jointly. The terms $\frac{d\mathbf{P}}{dE}$ and $\frac{d\mathbf{P}}{dI}$ are the effects of E and I on the probability of selling, respectively. The parameter Γ captures the extent to which the *added* effect of receiving E changes if I is received as well. We can rearrange terms and express Γ as follows:

$$\Gamma = (\mathbf{P}_{EI} - \mathbf{P}_E) - (\mathbf{P}_I - \mathbf{P}_0) \quad (15)$$

Note that this is the discrete analog of the cross-partial of $\mathbf{P}$ with respect to E and I , $\frac{d^2\mathbf{P}}{dIdE}$. Also note that these interventions are aimed at reducing the probability of selling hake, hence we expect $\frac{d\mathbf{P}}{dE}$ and $\frac{d\mathbf{P}}{dI}$ to be negative, and so, if $\Gamma > 0$ it means that the negative effects of E and I get *attenuated* when the interventions are carried out jointly, i.e., $\mathbf{P}_{\mathbf{EI}} > \mathbf{P}_E + \mathbf{P}_I$. Conversely, if $\Gamma < 0$, the negative effect on reducing the sale of hake is *exacerbated* when the interventions are carried out jointly, in which case E and I **complement** each other in reducing the sale of hake.

Vendor's Utility Function. The utility of selling hake depends on the value obtained from sales and the probability of receiving a fine. We assume these two terms are separable:

$$u = v - \hat{\theta}\Omega$$

v is the net income from selling hake, and $\hat{\theta}$ is the perceived probability of detection and Ω the fine if detected. The enforcement intervention affects the utility of selling hake through the perceived probability of detection $\hat{\theta}$, while the information campaign enters into the utility via the value v .

We assume $v = \bar{v} + \varepsilon$, where ε is a mean-zero utility shock that captures unobserved factors that affect the utility, and distributes F_ε . We normalize the utility of not selling hake to zero, $u_0 = 0$. Thus, the vendor would sell if the utility of doing so is greater than zero:

$$\begin{aligned}\bar{v} - \Omega\hat{\theta} + \varepsilon &> 0 \\ \bar{v} - \Omega\hat{\theta} &> -\varepsilon \\ \bar{u} &> -\varepsilon\end{aligned}$$

where $\bar{u}$ is the mean level of utility of selling hake. Without loss of generality we assume F_ε is symmetric, $1 - F_\varepsilon(-\bar{u}) = F_{-\varepsilon}(\bar{u}) = F_\varepsilon(\bar{u})$. The probability that a seller decides to sell hake is:

$$\mathbf{P} = F_\varepsilon(\bar{v} - \Omega\hat{\theta}) = F_\varepsilon(\bar{u})$$

Now, using the functional form of probabilities as a function of the distribution of $F_\varepsilon(\cdot)$ and vendor's utility function, we can express each of the terms above:

$$\frac{d\mathbf{P}}{dE} = \frac{d\mathbf{P}}{d\hat{\theta}} \frac{d\hat{\theta}}{dE} = -\Omega f(\bar{u}) \frac{d\hat{\theta}}{dE} \quad (16)$$

$$\frac{d\mathbf{P}}{dI} = \frac{d\mathbf{P}}{d\bar{v}} \frac{d\bar{v}}{dI} = f(\bar{u}) \frac{d\bar{v}}{dI} \quad (17)$$

$$\Gamma = \frac{d^2\mathbf{P}}{d\bar{v}d\hat{\theta}} \frac{d\hat{\theta}}{dE} \frac{d\bar{v}}{dI} = -\Omega f'(\bar{u}) \frac{d\hat{\theta}}{dE} \frac{d\bar{v}}{dI} \quad (18)$$

where $\frac{d\hat{\theta}}{dE} \geq 0$ and $\frac{d\bar{v}}{dI} \leq 0$ capture the channels through which E and I impact vendor's utility function (first stage). Since $\Omega > 0$ and $f(\bar{u}) \geq 0$, from equations (16) and (17), it is easy to note that we should expect negative effects of E and I on the sale of hake; $\frac{d\mathbf{P}}{dE} \leq 0$ and $\frac{d\mathbf{P}}{dI} \leq 0$.

The sign of Γ depends on the sign of $f'(\bar{u})$. Γ is positive if $f'(\bar{u}) > 0$, and negative otherwise. For standard probability distributions, $f(x)$, extreme values of x are less common than values around the mean; i.e., starting from a low value of x , the derivative of $f(x)$ is positive, while for high values of x , the derivative of x is negative as greater (and more extreme) realizations are less likely. Thus, the value of Γ crucially depends on the baseline value of $\bar{u}$:

- If $\bar{u}$ is large, either due to high $\bar{v}$ or low $\Omega\hat{\theta}$, vendors derive high utility from selling hake, and thus $\mathbf{P}_0$ is high. To reduce the probability of selling, carrying the two interventions jointly has a greater effect than the sum of the two interventions in isolation ($\Gamma < 0$).
- Conversely, if $\bar{u}$ is small, (i.e., low $\mathbf{P}_0$), there's little room to reduce further the probability, hence the marginal contribution of performing both interventions jointly is small and less than the sum of the effects of each of them in isolation ($\Gamma > 0$). One extreme scenario would be such that one of the two interventions is enough to reduce the probability to zero, then adding a second intervention would not contribute to reducing the probability of carrying them jointly

D.2 Calibration in mean utility space

The baseline level of $\bar{u}$ is unobservable; however, the experiment exogenously varied enforcement (E) and information (I), and we measured their effects on the probability of selling hake. Now, our model links these policy levers to the vendor's decision to sell hake through the *mean utility* of selling. In particular, by observing how the probability of selling responds to each treatment arm in the RCT, we can recover the underlying shifts in mean utility induced by E and I , as well as the baseline level of mean utility.

This section formalizes that identification and implements a simple calibration.

Mean utility and selling probability. Let the utility of selling hake be $u = \bar{v} - \Omega\hat{\theta} + \varepsilon$, with $\varepsilon \sim N(0, 1)$, where $\bar{v}$ is the deterministic net value of selling, Ω is the fine, and $\hat{\theta}$ is the perceived probability of detection. The *mean utility* of selling as $\bar{u} \equiv \bar{v} - \Omega\hat{\theta}$. A vendor sells hake if $u > 0$, so the selling probability is $\mathbf{P} = \Pr(u > 0) = \Pr(\bar{u} + \varepsilon > 0) = F_\varepsilon(\bar{u}) = \Phi(\bar{u})$, where $\Phi(\cdot)$ is the standard normal cdf.

First-stage in mean utility. We model the impact of enforcement E and information I on vendors' beliefs and valuations through linear first stages:

$$\begin{aligned}\hat{\theta}(E) &= \hat{\theta}_0 + \beta_E E, \\ \bar{v}(I) &= v_0 + \beta_I I.\end{aligned}$$

Hence, the mean utility is

$$\begin{aligned}\bar{u}(E, I) &= \bar{v}(I) - \Omega\hat{\theta}(E) \\ &= v_0 - \Omega\hat{\theta}_0 + \beta_I I - \Omega\beta_E E. \\ &= \bar{u}_0 + \beta_I I - \Omega\beta_E E.\end{aligned}$$

where $\bar{u}_0 = v_0 - \Omega\hat{\theta}_0$ is the baseline mean utility of selling. Given this structure we can write

the mean utilities of each treatment arm is:

$$\bar{u}_I = \bar{u}_0 + \beta_I, \quad \bar{u}_E = \bar{u}_0 - \Omega\beta_E E, \quad \bar{u}_{EI} = \bar{u}_0 + \beta_I - \Omega\beta_E E,$$

with corresponding selling probabilities:

$$\mathbf{P}_0 = \Phi(\bar{u}_0), \quad \mathbf{P}_I = \Phi(\bar{u}_I), \quad \mathbf{P}_E = \Phi(\bar{u}_E), \quad \mathbf{P}_{EI} = \Phi(\bar{u}_{EI}).$$

Experimental moments. The treatment effects estimated from the RCT, generate the following moment conditions:

$$\mathbf{P}_I - \mathbf{P}_0 = -0.133 = \Phi(\bar{u}_0 + \beta_I) - \Phi(\bar{u}_0), \quad (19)$$

$$\mathbf{P}_E - \mathbf{P}_0 = -0.178 = \Phi(\bar{u}_0 - \Omega\beta_E E) - \Phi(\bar{u}_0), \quad (20)$$

$$\mathbf{P}_{EI} - \mathbf{P}_0 = -0.179 = \Phi(\bar{u}_0 + \beta_I - \Omega\beta_E E) - \Phi(\bar{u}_0). \quad (21)$$

The model unknowns are $(\bar{u}_0, \beta_I, \beta_E)$. For any given $\bar{u}_0$, equations (19) and (20) each define a unique solution for $\beta_I(\bar{u}_0)$ and $\beta_E(\bar{u}_0)$ because the probit cdf is strictly increasing. Substituting these reduced-form solutions into (21) yields a single scalar equation in $\bar{u}_0$, which we solve numerically.

Calibration results. Solving the system (19)–(21) via bisection, we obtain

$$\bar{u}_0 \approx -0.9189, \quad \beta_I \approx -0.7654, \quad -\Omega\beta_E \approx -2.1533.$$

In words, the information campaign reduces mean utility by about 0.44 units, and enforcement reduces mean utility by about 1.04 units. The fine is $US\$200$, i.e., $\beta_E = 2.1533/200 = 0.0108$. The baseline mean utility $\bar{u}_0 = -0.9189$ implies a baseline selling probability $P_0 = \Phi(\bar{u}_0) \approx \Phi(-0.92) \approx 0.18$, which coincides with the level we observe in the control RCT control group in the post-intervention period.

D.2.1 The role of baseline utility $\bar{u}_0$ on intervention effects and their complementarity

We can combine the model calibration (based on RCT effects), to provide a broader understanding of how intervention complementarities vary as a function of the baseline level of the utility of selling hake. We treat (β_I, β_E) as constant shifts in the mean utility and evaluate the model at different baseline mean utilities, $\bar{u}_0$. We denote the treatment effects relative to baseline utilities as

$$\Delta_I(\bar{u}_0) \equiv \mathbf{P}_I(\bar{u}_0) - \mathbf{P}_0(\bar{u}_0), \quad \Delta_E(\bar{u}_0) \equiv \mathbf{P}_E(\bar{u}_0) - \mathbf{P}_0(\bar{u}_0), \quad \Delta_{EI}(\bar{u}_0) \equiv \mathbf{P}_{EI}(\bar{u}_0) - \mathbf{P}_0(\bar{u}_0),$$

to gauge $\Gamma(\bar{u}_0)$, we need to introduce an additional term called “additive benchmark,” $\Delta_{\text{add}}(\bar{u}_0) \equiv \Delta_I(\bar{u}_0) + \Delta_E(\bar{u}_0)$, that simply adds the standalone effects of enforcement and information campaign. Thus, we can write equation 15;

$$\Gamma(\bar{u}_0) \equiv \left(\mathbf{P}_{EI}(\bar{u}_0) - \mathbf{P}_E(\bar{u}_0) \right) - \left(\mathbf{P}_I(\bar{u}_0) - \mathbf{P}_0(\bar{u}_0) \right) = \Delta_{EI}(\bar{u}_0) - \Delta_{\text{add}}(\bar{u}_0)$$

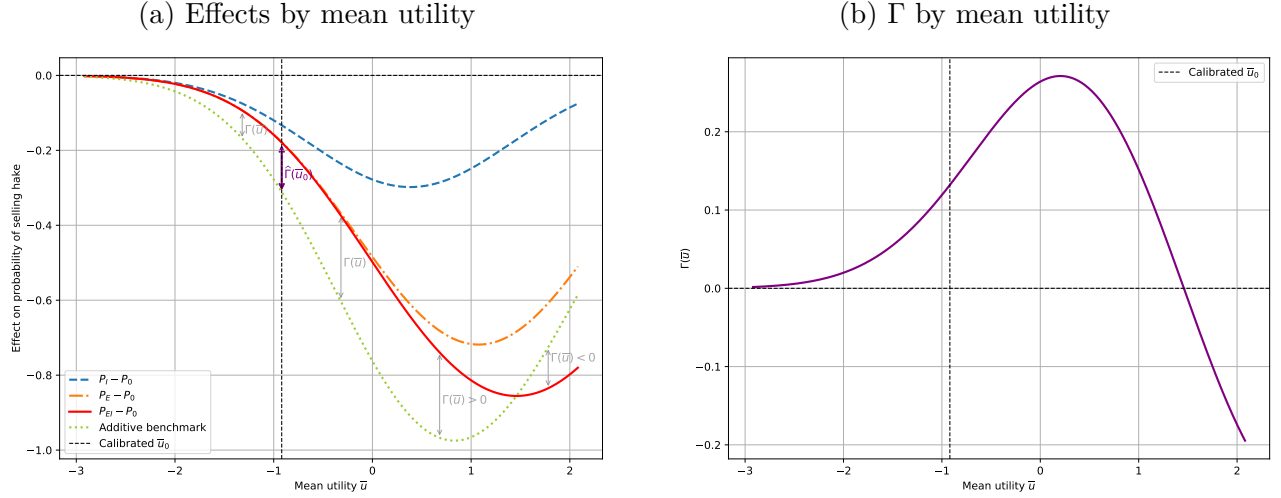
Again, if $\Gamma(\bar{u}_0) < 0$, the joint treatment reduces selling more than the sum of the two single-treatment effects, so enforcement and information are *complements* at that mean utility. If $\Gamma(\bar{u}_0) > 0$, the joint effect is attenuated relative to additivity.

Figure D.1 shows how the effects of the information and enforcement interventions vary with the baseline mean utility of selling hake, $\bar{u}$. The figure highlights that treatment effects are not constant: they depend on where vendors lie on the underlying utility distribution. When $\bar{u}_0$ is low, and baseline selling probabilities are already small, both interventions have limited scope to further reduce sales, and their joint effect is smaller than the sum of the individual effects. As $\bar{u}$ increases, the marginal impact of each intervention grows, and the joint effect begins to deviate from the additive benchmark, ultimately reversing sign as we move across the support of $\bar{u}$.

Panel (b) makes this pattern transparent by plotting $\Gamma(\bar{u})$. The function peaks around

$\bar{u}_0 \approx 0.3$, where the interventions act as substitutes to the greatest degree, and turns negative near $\bar{u} \approx 1.4$, where complementarities emerge. The calibrated baseline value $\bar{u}_0 \approx -0.91$ lies in a region where attenuation dominates, yielding a small positive $\hat{\Gamma}(\bar{u}_0)$ and indicating that, for the baseline vendor type, the two interventions slightly weaken rather than reinforce each other.

Figure D.1: Intervention Effects and Complementarity



Notes: Figure D.1 illustrates the model-implied treatment effects as a function of the baseline mean utility of selling hake, $\bar{u}$. Panel (a) reports the single-treatment effects $P_I - P_0$ and $P_E - P_0$, the joint effect $P_{EI} - P_0$, and the additive benchmark $(P_I - P_0) + (P_E - P_0)$ under the probit specification $P = \Phi(\bar{u})$. The vertical brackets indicate the implied complementarity term $\Gamma(\bar{u}) \equiv (P_{EI} - P_0) - [(P_I - P_0) + (P_E - P_0)]$. Panel (b) plots $\Gamma(\bar{u})$ directly. The vertical line at $\bar{u}_0 = -0.91$ marks the calibrated baseline mean utility, obtained by solving the system of moment conditions that equate the observed experimental effects $(P_I - P_0, P_E - P_0, P_{EI} - P_0) = (-0.133, -0.178, -0.179)$ to their model-implied counterparts. Given the monotonicity of the probit link, these moments uniquely identify the triple $(\bar{u}_0, \gamma_I, \gamma_E)$ governing the reduced-form shifts in mean utility. At $\bar{u}_0$, the distance between the $P_{EI} - P_0$ curve and the additive benchmark corresponds to the estimated complementarity $\hat{\Gamma}(\bar{u}_0)$.

The exercise shows that joint policy effectiveness depends not on the interventions per se, but on where vendors lie along the utility distribution. Vendors with low baseline incentives to sell hake are in a flat region of the choice curve, where additional reductions in utility have limited impact, and the interventions yield diminishing returns. Vendors with higher baseline utilities lie on steeper or more curved regions, where combining interventions can generate more-than-additive effects. The calibration thus highlights that complementarities are heterogeneous and state-dependent, helping explain why agents exposed to these interventions in other settings may respond differently to the same policy bundle.

E Appendix: Model of Belief Formation

E.1 Belief Formation and Enforcement Design

This appendix section expands on the belief-formation mechanism embedded in the model and clarifies how it links to two enforcement design margins studied in the experiment: (i) the *frequency* (intensity) of enforcement visits and (ii) the *predictability* of the enforcement schedule across ferias/days. The main text summarizes these mechanisms; here we provide additional detail and intuition that can be referenced by readers interested in the learning channel.

E.1.1 Beliefs about detection risk: what vendors learn from inspections

The model distinguishes between the *true* enforcement process and vendors' *perceived* detection risk. Let $y_t \in \{0, 1\}$ indicate whether the vendor experiences an enforcement visit at time t . Under a given enforcement regime, visits arrive according to a stationary Bernoulli process with probability θ (or, in the multi-feria case below, a vector of probabilities θ^i). Vendors do not observe θ directly; instead, they infer it from the history of realized visits.

In the single-location benchmark, a conjugate Beta prior and Bernoulli likelihood imply a Beta posterior. If the vendor's prior is $\theta \sim \text{Beta}(\alpha_0, \alpha_1)$ and $Y_t = \sum_{s=1}^{t-1} y_s$ is the number of visits observed before period t , then Bayesian updating yields

$$\theta \mid (y_1, \dots, y_{t-1}) \sim \text{Beta}(\alpha_0 + Y_t, \alpha_1 + t - 1 - Y_t), \quad \mathbb{E}[\theta \mid y_1, \dots, y_{t-1}] = \frac{\alpha_0 + Y_t}{\alpha_0 + \alpha_1 + t - 1}.$$

Hence, perceived probability increases with accumulated experience (the number of visits) and with the *relative frequency* of visits (the share of periods with a visit), while the magnitude of updating is attenuated by prior strength. This learning channel is central in the short run because vendors face repeated selling opportunities during the ban, so beliefs can evolve within the one-month intervention window.

E.1.2 Enforcement frequency: deterrence versus adaptation through learning

A key implication of Bayesian updating is that enforcement intensity affects behavior not only mechanically (through more visits) but also dynamically (through faster learning). In our model, increasing enforcement frequency (a higher visit probability θ) operates through two distinct channels:

- **Deterrence via beliefs.** More frequent visits raise the likelihood that vendors observe visits early in the intervention, which increases $\mathbb{E}[\hat{\theta}_t]$ more quickly. This lowers the expected payoff from selling and tends to reduce illegal sales.
- **Adaptation via experience.** The same increase in enforcement frequency also increases the speed at which vendors accumulate experience Y_t . In the model, experience can raise the effectiveness of defensive actions (captured by an increasing $g(Y_t)$), making it more attractive to continue selling while mitigating the expected fine. This channel can partially offset (or in some periods dominate) pure deterrence.

The tension between these channels makes short-run comparative statics ambiguous. Early in the intervention, when vendors have limited experience and defensive actions are less effective, additional visits primarily update beliefs upward and discourage selling. As Y_t accumulates, additional visits can instead accelerate the transition toward defensive selling (or other forms of evasion), which can attenuate the net effect of enforcement in later periods. This is the core mechanism behind potentially non-monotone dynamics in short-run treatment effects when enforcement intensity is increased.

Importantly, this ambiguity is a *short-run* phenomenon. Over longer horizons, beliefs converge and the learning dynamics fade: $\mathbb{E}[\hat{\theta}_t] \rightarrow \theta$ and $g(Y_t) \rightarrow \bar{g}$. In that long-run environment, behavior depends primarily on static deterrence forces (in the spirit of Becker-style models), and changes in enforcement intensity have the standard monotone effect of reducing illegal activity through higher detection risk. Our experiment focuses on the one-month window, where the learning and adaptation channels are empirically relevant.

E.1.3 Predictability across ferias: belief formation with multiple locations

Vendors in our setting operate across circuits that rotate through different ferias/days. To capture predictability, we allow the probability of enforcement visits to vary across locations. We describe how a vendor forms beliefs when there are two ferias, which is sufficient to capture the logic of targeting and displacement.

Let $y_t^i \in \{0, 1\}$ indicate whether feria $i \in \{1, 2\}$ is visited at time t . Define the multinomial variable $z_t = y_t^1 + 2y_t^2 + 1$, which summarizes the inspection profile in period t .⁴⁰ We assume that z_t has a stationary distribution characterized by $p = (p^j)_{j=1}^4$, where $p^j = \Pr(z_t = j)$. The vendor has a Dirichlet prior over p , $\hat{p}_0 \sim \text{Dirichlet}((\beta_j)_{j=1}^4)$, and updates it using the realized history of z_t .

The implied *visit policy* is $\theta = (\theta^1, \theta^2)$, where $\theta^1 = p^2 + p^4$ and $\theta^2 = p^3 + p^4$. The next lemma shows that, despite the joint multinomial structure, the vendor can update beliefs about each feria's visit probability using only the history of visits in that feria.

Lemma 1 *The vendor's belief about the probability of a visit at feria i at time t satisfies*

$$\begin{aligned}\hat{\theta}_t^1 &\sim \text{Beta}(\alpha_2 + \alpha_4 + Y_t^1; \alpha_1 + \alpha_3 + t - 1 - Y_t^1) \\ \hat{\theta}_t^2 &\sim \text{Beta}(\alpha_3 + \alpha_4 + Y_t^2; \alpha_1 + \alpha_2 + t - 1 - Y_t^2),\end{aligned}$$

where $Y_t^i = \sum_{s=1}^{t-1} y_s^i$ is the number of times feria i has been visited prior to period t .

Lemma 1 implies that the perceived detection risk in each feria is shaped locally: vendors form $\hat{\theta}_t^i$ by combining their prior with the realized frequency of visits in feria i . This provides a microfoundation for how predictability matters in the model: concentrating enforcement in one feria makes beliefs in that feria rise quickly, while beliefs remain low in non-targeted ferias, creating scope for *asymmetries* of illegal sales across days/locations.

⁴⁰Note that z_t takes value 1 if no feria was inspected at time t , value 2 if only feria 1 was inspected, value 3 if only feria 2 was inspected, and value 4 if both ferias were inspected in that period.

E.1.4 Predictable versus unpredictable enforcement under fixed capacity

To connect belief formation to enforcement design, define total enforcement capacity as

$$\Theta \equiv \theta^1 + \theta^2.$$

A *predictable/targeted* policy concentrates enforcement in one feria (e.g., $\theta^1 = \Theta$, $\theta^2 = 0$), whereas an *unpredictable* policy spreads enforcement evenly (e.g., $\theta^1 = \theta^2 = \Theta/2$). Holding Θ fixed, these designs generate different paths for $(\hat{\theta}_t^1, \hat{\theta}_t^2)$ and therefore different incentives to sell across ferias.

The central trade-off is:

- **Targeting raises local deterrence but induces displacement.** Under a targeted policy, beliefs in the targeted feria rise quickly, often inducing sharp reductions in sales there. However, because enforcement capacity is shifted away from the non-targeted feria, beliefs there remain low, and vendors can respond by allocating sales to the less-monitored location.
- **Unpredictability reduces displacement but requires sufficient capacity.** Under an evenly spread policy, beliefs rise in both ferias, reducing the scope for allocating sales to a “safe” location. This is most effective when capacity is large enough to generate meaningful deterrence in *both* locations. When capacity is low, spreading visits thinly may leave beliefs below the relevant deterrence threshold everywhere, in which case targeting can be relatively more effective by at least shutting down sales in one feria.

In the short run, the effectiveness of predictability depends on whether the regulator can raise perceived detection probabilities sufficiently broadly to deter selling across all markets, or whether concentrating enforcement is necessary to generate any meaningful deterrence at all. Finally, note that the multi-feria belief formation channel interacts naturally with vendor adaptation. These dynamics reinforce the importance of distinguishing short-run (learning-driven) from long-run (static) effects when evaluating enforcement design.

E.2 Proofs

[Proof of Proposition 1] First we analyze the vendor's the different options. To avoid unnecessary notation we omit the subindex t and write $g = g(Y_t)$.

- She sells and does not defend if and only if $U[d = 0|s = 1, Y] \geq 0$ and $U[d = 0|s = 1, Y] \geq U[h = 1|s = 1, Y]$. These restrictions together imply that

$$\mathbb{E}[\hat{\theta}] \leq \frac{1}{\Omega} \min \left\{ v; \frac{c}{g} \right\}.$$

- She sells and defends if and only if $U[d = 1|s = 1, Y] \geq 0$ and $U[d^i = 1|s^i = 1, Y] > U[d^i = 0|s^i = 1, Y]$. These restrictions together imply that⁴¹

$$\underline{\delta} < \mathbb{E}[\hat{\theta}] \leq \bar{\delta}.$$

- The vendor does not sell if and only if $\max_{d \in \{0,1\}} U[d|s = 1, Y] < 0$. These restrictions together imply that

$$\mathbb{E}[\hat{\theta}] > \frac{1}{\Omega} \max \left\{ v; \frac{v - c}{1 - g} \right\}.$$

First, note that the conditions $\underline{\delta} < \bar{\delta}$, $v > \frac{c}{g}$, and $v < \frac{v-c}{1-g}$ are equivalent. Therefore, there is a set of beliefs for which the vendor's optimal choice is to sell and defend the hake if and only if $v > \frac{c}{g}$.

If $v \leq \frac{c}{g}$, then $\underline{\delta} \geq \bar{\delta}$ and the vendor never sells and defends. Moreover, as $\min \left\{ v; \frac{c}{g} \right\} = v$ in this case she sells and does not defend if $\mathbb{E}[\hat{\theta}] \leq \frac{v}{\Omega}$ and does not sell otherwise.

Finally, if $v \leq \frac{c}{g}$, the characterization follows directly from the comparison of the three options.

[Long-run Comparative Statics] Define $\underline{\delta}_\infty = \frac{c}{\Omega \bar{g}}$ and $\bar{\delta}_\infty = \frac{v-c}{\Omega(1-\bar{g})}$. First, note that Assumption $\bar{g} > c/v$ implies that $\underline{\delta}_\infty < \bar{\delta}_\infty$. As (a.s.) $\underline{\delta}_t \rightarrow \underline{\delta}_\infty$ and $\bar{\delta}_t \rightarrow \bar{\delta}_\infty$, we have that in the long

⁴¹Recall that $\underline{\delta} = \frac{c}{\Omega g}$, and $\bar{\delta} = \frac{v-c}{\Omega(1-g)}$

run there is a set of beliefs for which the vendor sells and defends. Furthermore, Proposition 1 implies that the vendor sells if and only if

$$\theta \leq \bar{\delta}_\infty = \frac{v - c}{\Omega(1 - \bar{g})}.$$

The comparative statics results follow from analyzing the effect of changes in θ , v , and $1 - \bar{g}$ in the previous inequality.

[Proof of Lemma 1] For any t we define the number of periods *before* t that the vendor has seen $z = j$ (for $j = 1, 2, 3, 4$) by

$$Z_t^j = \sum_{s=1}^{t-1} \mathbb{1}_{\{z_s=j\}}$$

Given $Z_t = (Z_t^j)_{j=1}^4$, Bayesian updating implies that

$$\hat{p}_t \sim \text{Dir}(\alpha + Z_t)$$

As $Z_t^2 + Z_t^4 = Y_t^1$, the previous distribution implies that the vector

$$(\hat{p}_t^1, \hat{p}_t^2 + \hat{p}_t^4, \hat{p}_t^3) = (\hat{p}_t^1, \hat{\theta}_t^1, \hat{p}_t^3) \sim \text{Dir}(\alpha^1 + Z_t^1, \alpha^2 + \alpha^4 + Y_t^1, \alpha^3 + x_t^3).$$

As $\sum_{j=1}^4 \hat{p}_t^j = 1$ and $\sum_{j=1}^4 Z_t^j = t - 1$, the marginal distribution of the probability of being inspected at feria 1 at time $t + 1$ is

$$\hat{\theta}_t^1 \sim \text{Beta}(\alpha^2 + \alpha^4 + Y_t^1, \alpha^1 + \alpha^3 + t - 1 - Y_t^1).$$

The characterization of the distribution of $\hat{\theta}_t^2$ is completely analogous.

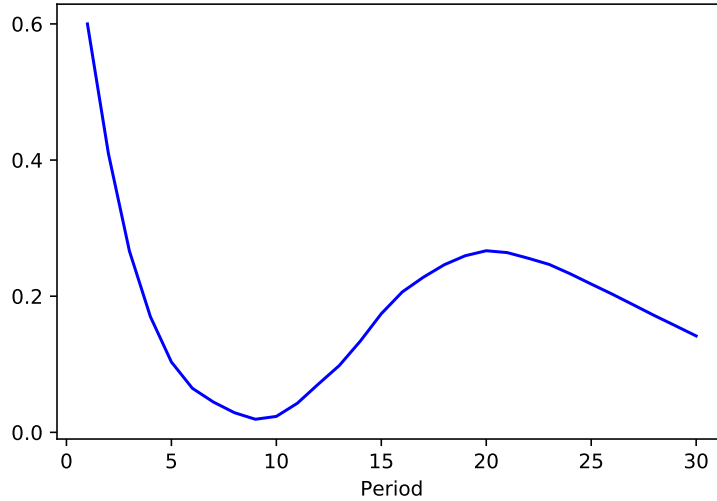
F Appendix: Numerical Simulations

We numerically simulate the behavior of a representative vendor exposed to different levels and schemes of enforcement. These simulations shed light on how the optimal choice evolves as

vendors acquire more information about the pattern of visits and inspection loopholes.

Vendors' Behavior Over Time Vendors decide whether and how to sell hake in every period t . The decision to sell in t is static but incorporates the information collected until $t - 1$. Thus it may vary as more information is incorporated. In particular, vendors continuously update their probability of receiving an enforcement visit as well as the effectiveness of defensive strategies reducing the probability of a fine. Figure F.1 describes how the optimal decision in different periods. It shows that the likelihood of selling is not stable. In this case, it decreases quickly once the enforcement is introduced, and increases as the vendor learn about enforcement weaknesses. After a few periods, it converges to the long-run equilibrium. One direct takeaway of F.1 is that vendors' behavior varies over time; the same policy evaluated in different moments may yield different results.

Figure F.1: Vendor's Decision



This figure shows the proportion of times in which a vendor sells hake in different periods. This graph depicts 1000 simulations using the following parameters $\theta = 0.4, v \sim U(0.5, 1.5), c = 0.1, \Omega = 18, \theta_1 = 0.05, g(Y) = 0.7 / (1 + e^{-8 \times Y + 28})$, i.e., $\bar{g} = 0.7$. The probability of selling decreases quickly as the enforcement begins, however it increases as vendors learn about enforcement weaknesses. After a number of periods, it converges to the “long-run” equilibrium based on model’s structural parameters.

Enforcement Intensity Vendors adapt their behavior according to the pattern of visits they receive. We compare the behavior of vendors exposed to different frequencies of visits. Figure 2 shows that vendors exposed to more intense enforcement tend to decrease the proba-

bility of selling quickly. However, they learn faster about enforcement weaknesses. Thus, after a few periods, the latter effect may counterbalance the higher intensity effect, which makes high-intensity enforcement less effective. As the number of periods increases, the selling decision converges to the long-run optimal. This result has relevant implications for enforcement evaluation and design.

The figures F.2(a) and F.2(b) describe the timing and scope of adoption of defensive actions depending on the frequency of the enforcement. Vendors exposed to more intense enforcement learn quickly about loopholes, so they start adopting these actions earlier.

Figure F.2: Probability of Selling Hake

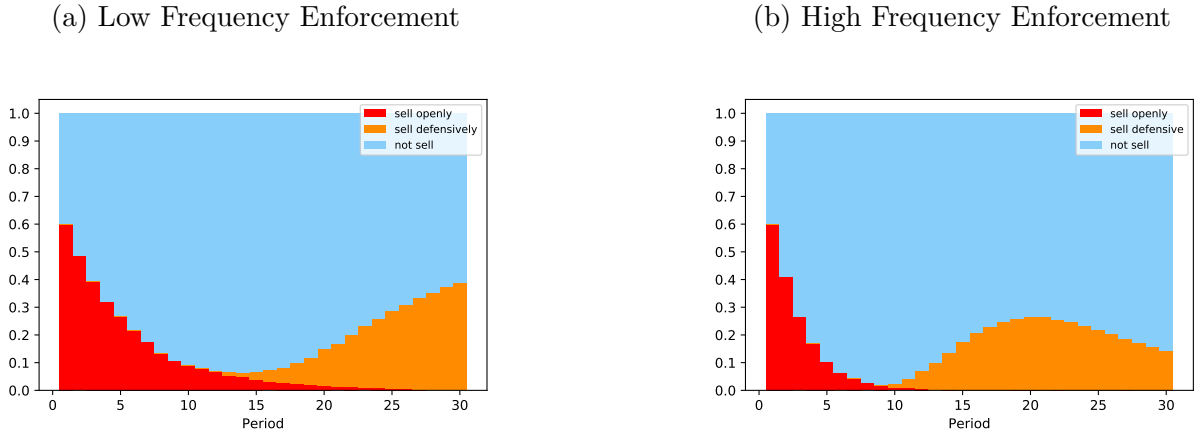
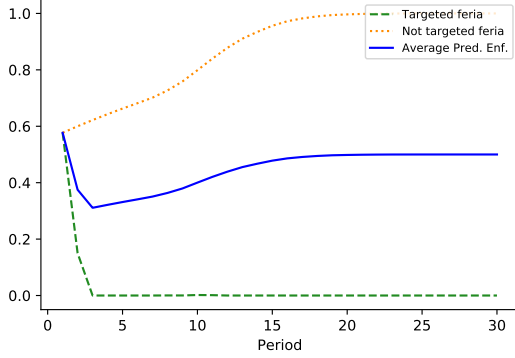


Figure F.2(a) and F.2(b) describe vendors' decision on whether and how to sell. This simulation uses the same parameters than previous graph: $\theta^{high} = 0.5, \theta^{low} = 0.3, v \sim U(0.5, 1.5), c = 0.1, \Omega = 18, \theta_1 = 0.05, g(Y) = 0.7 / (1 + e^{-2 \times Y + 12})$, i.e., $\bar{g} = 0.7$. The adoption of defensive strategies starts after a number of periods.

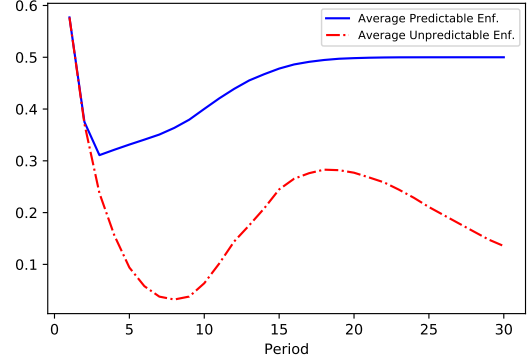
Enforcement Predictability We study the consequences of varying the predictability of the enforcement visits. In particular, we study vendors' behavior, assuming that every circuit has two ferias and that the vendor alternates between them. If enforcement is predictable, one of the ferias receives enforcement more intensely than the other. In our analysis, the probability of receiving a visit in a non-targeted feria is zero. Conversely, under unpredictable enforcement, both ferias have the same likelihood of receiving a visit.

Figure F.3: Probability of Selling Hake

(a) Targeted vs. Non Targeted Ferias



(b) Predictable vs. Unpredictable Enforcement



Notes: Figure F.3(a) compares vendors' decision in targeted and non targeted ferias, assuming that vendors alternate between these ferias. The dashed line correspond to the average probability of sale, which is calculated assuming that in every period there's one half of the vendors in each type of feria. Figure F.3(b) compares the average probability of selling under predictable vs. unpredictable enforcement. These simulations use the following model parameters: $\theta = 0.5, v \sim U(1/2, 3/2), c = 0.1, \Omega = 18, \theta_1 = 0.05, g(Y) = 0.7 / (1 + \exp\{-2 \cdot Y + 12\})$, i.e., $\bar{g} = 0.7$.

With fixed enforcement capacity, learning occurs at the same rate under either a targeted or an unpredictable policy. However, vendors are more likely to adopt defensive actions (or simply not sell) in the targeted feria under the predictable policy, while the probability of selling in the non-targeted feria inevitably will tend to one. We simulate the effects of predictable and unpredictable policies on hake sales in each feria in Figure F.3. Under most functional forms, the unpredictable policy is more effective on average in the short run. Under predictable enforcement, vendors stop selling in the targeted feria almost immediately, but they increase sales in the non-targeted feria. Average sales are higher than under unpredictable policy in the short-run.

Note about Agents Heterogeneity We introduce agents' heterogeneity by agents that differ in their valuation v . Specifically, suppose v is distributed according to the CDF F , whose support is $[\underline{v}, \bar{v}]$. Assume $c < \underline{v} \leq \bar{v} < \Omega$. The applying 1 we get the following result

Corollary 1 *In the case with heterogeneous agents let α_{NS} , α_{SO} , and α_{SD} be the share of*

agents not selling, selling openly, and selling defensively, respectively. These shares are given by

$$\begin{aligned}\alpha_{NS} &= \begin{cases} F(\Omega\mathbb{E}[\hat{\theta}_t]) & \text{if } \Omega\mathbb{E}[\hat{\theta}_t] \leq c/g(Y_t) \\ F(\Omega\mathbb{E}[\hat{\theta}_t](1 - g(Y_t)) + c) & \text{if } \Omega\mathbb{E}[\hat{\theta}_t] > c/g(Y_t); \end{cases} \\ \alpha_{SO} &= \begin{cases} 1 - F(\Omega\mathbb{E}[\hat{\theta}_t]) & \text{if } \Omega\mathbb{E}[\hat{\theta}_t] \leq c/g(Y_t) \\ 0 & \text{if } \Omega\mathbb{E}[\hat{\theta}_t] > c/g(Y_t); \text{ and} \end{cases} \\ \alpha_{SD} &= \begin{cases} 0 & \text{if } \Omega\mathbb{E}[\hat{\theta}_t] \leq c/g(Y_t) \\ 1 - F(\Omega\mathbb{E}[\hat{\theta}_t](1 - g(Y_t)) + c) & \text{if } \Omega\mathbb{E}[\hat{\theta}_t] > c/g(Y_t). \end{cases}\end{aligned}$$

Assumptions The numerical simulations presented above assume a functional form for the learning curve $g(Y_t)$ and a set (and strength) of priors. In particular, we assume: $g(Y_t) = \frac{\bar{g}}{1 + \exp\{-a \times Y + b\}}$. This functional form is handy because, $\lim_{x \rightarrow \infty} g(x) = \bar{g}$, and the parameters a and b dictate the speed of convergence of the function. Other functional forms yield the same qualitative results. The figures F.4(a) and F.4(b) describe how the ability and the beliefs evolve over time for two different levels of enforcement. The idea is that vendors exposed to more intense enforcement develop an ability to circumvent the fine faster, this effect counterbalances the increase probability of a visit.

Figure F.4: Beliefs Updating and Learning Curve

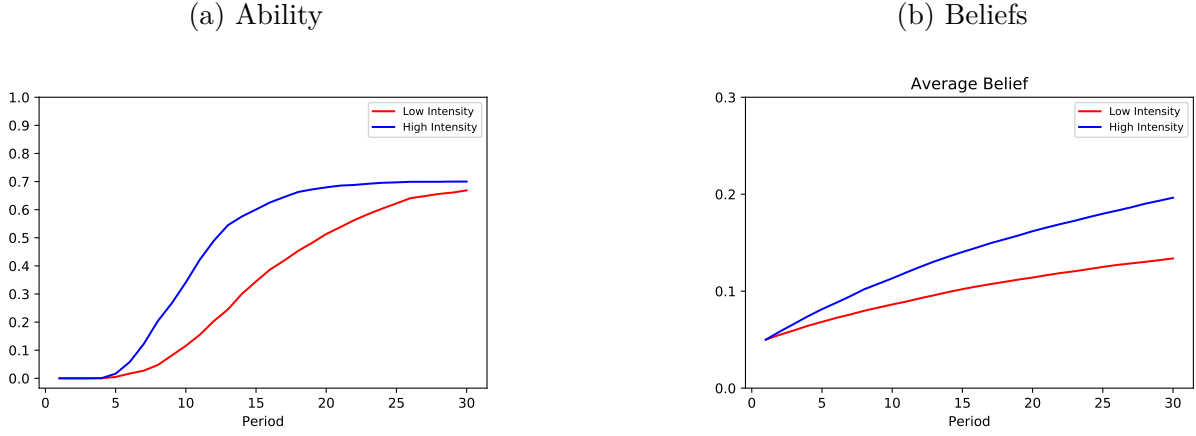


Figure F.4(a) and F.4(b) how vendors learn about loopholes and update the probability of a visit. These figures use the same inputs than other simulations, i.e., $\bar{g} = 0.7, a = 8, b = 28, \theta_1 = 0.05 (\alpha_0 + \alpha_1 = 40)$

G Appendix: Further Details on the Cost-Effectiveness Analysis

The section 9 describes the cost-effectiveness analysis of each treatment. These calculations are based on the following parameters:

- Costs: The total cost of implementing enforcement was \$ 62,900.25, which is divided into fixed costs \$ 7,338.06, and variable costs: \$ 55,562.19. The fixed costs include administrative staff salaries, central office coordination, etc. The variable costs include the specific costs incurred to implement the enforcement (i.e., financial compensation of inspectors, gasoline, etc.). Based on Sernapesca information, deploying enforcement in an unpredictable way is 10% more costly regarding staff availability. The cost of implementing enforcement at low frequency is obtained by calculating the (variable) cost of each visit and multiplying by the number of visits under this new regime, adding the fixed costs.

The total cost of implementing the information campaign was \$ 16,213.53, which includes the printing and distribution of flyers, posters, and letters in treated neighborhoods.

- Reduction of fish sales: The estimated effects of selling hake during the ban presented in section 6 are translated into numbers of fishes “saved.” This exercise takes into account that every stall has 25 hake fishes available, there are 2.57 fish stalls in each feria. Each circuit operates 5 days a week, and the effects consider the three last weeks of September. The enforcement treatment contemplated 83 circuits, whereas 26 circuits are located in municipalities assigned to receive information campaign with a high level of saturation. The information gathered from the vendors surveyed provided useful information to define the right parameters regarding the likely reduction on fish sales as a result of our interventions.

H Appendix: Departures from the Pre-Analysis Plan

We registered this trial on September 15, 2015 (before the data collection was completed) in the AEA registry. Our approach to analysis and the outcome variables we focus on in this paper closely mirror the project narrative we uploaded before we had access to any data. We highlight the most notable departures from the pre-analysis plan (PAP) here:

1. The experimental design section of the PAP mentions that the enforcement group would be divided into two sub-groups: One in which vendors would receive only a warning letter about illegal behavior, and one in which we would follow that up with inspections and fines. In practice, *Sernapesca* officials did not implement the treatments any differently across these two sub-groups. So we do not report this sub-sample analysis. Our data show that vendor behavior was not statistically distinguishable across these sub-groups.
2. The PAP mentions our sample size as 153 circuits, based on information we had collected on the existence of fish markets by calling municipalities before launching the project. During data collection we learnt that 40 of those circuits did not have any fish-stalls. Mystery shoppers could not visit another 7 circuits for logistical reasons. Our final analysis sample therefore contains only 106 circuits. These two sources of attrition are not

correlated with any observable characteristics, nor with the treatment assignment.

3. We had not anticipated that vendors would try to cheat by claiming that the fish was caught in August. This is something we learnt from our mystery shoppers soon after we started data collection. In the PAP, we mention only that we will track vendor reactions to enforcement activity, but do not mention ‘freezing’ specifically.
4. The PAP does not delve into the level of detail that this paper does. For example, we did not know exactly which fish were close substitutes for hake. We learnt from our data that pomfret was the other fish most commonly sold by hake vendors, and we therefore analyze effects on the price of pomfret. This price analysis could therefore be viewed as “exploratory” even though we had pre-specified our interest in studying price effects.